

AUTOMATED INTERPRETABILITY METRICS DO NOT DISTINGUISH TRAINED AND RANDOM TRANSFORMERS

Anonymous authors
Anonymous submission

ABSTRACT

Sparse autoencoders (SAEs) are widely used to recover a sparse, overcomplete basis of latent features from the residual-stream activations of transformer language models. We evaluate whether the SAE quality metrics and automated explanation pipelines in common use admit a discriminative test between trained transformers and randomly initialized ones, i.e., transformers whose parameters are resampled i.i.d. from a Gaussian. Across the Pythia suite (70M–6.9B parameters) and several distinct randomization schemes, we observe that, under a broad range of conditions, SAEs fit to randomly initialized transformers yield auto-interpretability scores and reconstruction-based metrics that are statistically comparable to those obtained from the corresponding trained models. This equivalence establishes that high aggregate auto-interpretability scores are not, in themselves, sufficient evidence that an SAE has recovered features attributable to learned, computationally relevant structure. We accordingly characterize the prevailing SAE metrics as informative but non-sufficient proxies for mechanistic interpretability, and we advocate for the systematic inclusion of randomized null baselines together with targeted quantitative measures of feature ‘abstractness.’

1 INTRODUCTION

Sparse autoencoders (SAEs) are a widely adopted tool in mechanistic interpretability, employed to disentangle the internal representations of neural networks by learning a sparse, approximately interpretable feature basis from network activations (Elhage et al., 2022; Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). Concretely, an autoencoder with a high-dimensional hidden layer is optimized to reconstruct a target activation vector subject to an explicit sparsity constraint on its hidden units (Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024), under the hypothesis that the recovered basis vectors correspond to the underlying concepts, or ‘features,’ encoded by the network (Park et al., 2023; Wattenberg and Viégas, 2024). The iterative refinement of SAE architectures and training objectives is driven by quantitative evaluation metrics, most notably auto-interpretability scores, which quantify the degree of agreement between a generated natural-language explanation and the empirical activation pattern of a latent (Bills et al., 2023; Paulo et al., 2024; Karvonen et al., 2024a).

A necessary condition for an interpretability method to be regarded as robust is that its evaluation metrics separate structure acquired through training from artifacts induced by the input distribution or the model architecture alone. The corresponding diagnostic is a controlled comparison of the method’s output on a trained model against a strong null model, canonically one whose weights are randomly initialized (Adebayo et al., 2020). We instantiate this diagnostic for SAEs and establish that several metrics in standard use fail to yield a reliable separation between the trained and randomized regimes. Specifically, SAEs fit to transformers with random parameters produce latents whose auto-interpretability scores (Bills et al., 2023; Paulo et al., 2024) are, in many configurations, statistically indistinguishable from those of a fully trained model.

This result constrains the inferences that may legitimately be drawn from aggregate SAE quality metrics. A high auto-interpretability score is not sufficient to certify that an SAE has isolated complex, learned computation. Such a score may instead be attributable to lower-order statistical regularities of the training corpus (Dooms and Wilhelm, 2024) or to architectural inductive biases that persist in the absence of any optimization. Indeed, a randomly initialized network still realizes a well-defined,

if elementary, computation—for instance, the preservation or amplification of the sparse structure present in its inputs (Section 4). Under this interpretation, the SAE faithfully recovers this simple, architecture-induced computation rather than any learned structure.

Although a subset of SAE features obtained from trained models demonstrably originates in learned computation, the aggregate metrics in common use are frequently inadequate to determine, for a given SAE, whether such higher-order features have been recovered. These findings carry direct methodological implications for mechanistic interpretability research. We argue, in particular, for the development of more discriminative procedures that separate artifacts from genuinely learned computation, and for the systematic validation of interpretability techniques against appropriately specified null models.

Finally, we advance two candidate mechanisms that could account for these observations. We formulate them as competing hypotheses: (1) the input distribution already exhibits superposition, and randomly initialized networks act approximately as identity maps on this superposed structure, preserving it; and (2) randomly initialized networks amplify existing superposition, or induce superposed structure *de novo*, even when the input is dense and generated i.i.d. from a Gaussian. In Section 4 we construct toy models that establish the plausibility of both hypotheses, while deferring a determination of which mechanism predominates to future work.

2 RELATED WORK

Sparse dictionary learning The phenomenon later termed ‘superposition’ in visual data is among the foundational observations of computational neuroscience. Olshausen and Field (1996; 1997) demonstrated that the receptive fields of simple cells in the mammalian visual cortex are accounted for by sparse coding, i.e., the encoding of a large ensemble of signals (sensory information) via the simultaneous activation of only a small subset of elements (neurons). Coding theory thereby furnishes a formal framework for the problem of recovering the ‘underlying signals’ that give rise to neural network activations (Marshall and Kirchner, 2024).

Sparse dictionary learning (SDL) approximates a set of input vectors by sparse linear combinations of a comparatively small number of learned basis vectors. The learned basis is typically overcomplete, i.e., its cardinality exceeds the input dimensionality. Instances of SDL include Independent Component Analysis (ICA), which recovers a linear representation whose components are maximally statistically independent (Bell and Sejnowski, 1995; Hyvärinen and Oja, 2000). Sparse autoencoders (SAEs) constitute a neural-network formulation of the same objective (Lee et al., 2006; Ng, 2011; Makhzani and Frey, 2014). In the canonical setting, an autoencoder with a single hidden layer whose width substantially exceeds the input activation dimensionality is trained under an objective that imposes or incentivizes sparsity on its hidden activations, so as to recover this latent structure. We use the term **latent** to denote a single hidden unit (dimension) of the autoencoder.

Mechanistic interpretability A now-standard modeling assumption identifies ‘features,’ or concepts, in language models with low-dimensional linear subspaces of the internal activation space (Park et al., 2023; Wattenberg and Viégas, 2024; Engels et al., 2024). Conditional on the existence of such sparse or ‘superposed’ structure, one predicts that targeted interventions on the activations—modifying or substituting them to instantiate different concepts—should causally alter model behavior, a prediction operationalized by ‘steering’ experiments (Meng et al., 2022; Zhang and Nanda, 2023; Heimersheim and Nanda, 2024; Makelov, 2024; O’Brien et al., 2024).

SAEs are a prevalent method for feature discovery: one typically fits a single autoencoder to reconstruct the activations of a single network layer, most commonly the transformer residual stream (Sharkey et al., 2022; Cunningham et al., 2023; Bricken et al., 2023). The proposed architectural variants differ principally in the nonlinearity applied downstream of the linear encoder (Makhzani and Frey, 2014; Gao et al., 2024; Rajamanoharan et al., 2024b; Lieberum et al., 2024). SAEs have additionally been trained under alternative objectives (Braun et al., 2024; Farnik et al., 2025) and fit jointly across multiple layers (Yun et al., 2021; Lawson et al., 2024; Lindsey et al., 2024).

Beyond reconstruction error and preservation of the underlying model’s task performance, SAEs have been assessed by whether they isolate designated concepts (Gurnee et al., 2023; Gao et al., 2024) or

factual knowledge (Huang et al., 2024; Chaudhary and Geiger, 2024), and by whether the recovered features support targeted concept ‘unlearning’ (Karvonen et al., 2024b).

Automatic neuron description A trained SAE commonly comprises tens of thousands of latents, a scale at which manual description of each latent is infeasible. Yun et al. (2021) identify, for each dictionary element, the tokens in a text corpus that maximally activate it, and inspect the resulting activation patterns manually. The prevailing alternative collects per-latent activation patterns over a text corpus and prompts a large language model to produce a natural-language explanation of them (e.g. Bills et al., 2023; Foote et al., 2023). Such automated procedures have been broadly adopted (e.g. Cunningham et al., 2023; Bricken et al., 2023; Gao et al., 2024; Templeton et al., 2024; Lieberum et al., 2024).

Bills et al. (2023) formalize the procedure as follows: generate an explanation of a neuron’s activation pattern from a set of dataset examples, simulate the activation pattern conditional on that explanation, and score the explanation by the correlation between the observed and simulated activations. This procedure is referred to as auto-interpretability (in the sense of self-interpreting). Paulo et al. (2024) introduce classification-based fidelity measures for automatic descriptions that are substantially cheaper to compute than simulation, together with an open-source pipeline implementing them. Choi et al. (2024) apply best-of- k sampling to generate multiple candidate explanations, each conditioned on a distinct subset of the maximally activating examples for a neuron. Notably, they fine-tune Llama-3.1-8B-Instruct on the highest-scoring explanations to obtain computationally inexpensive ‘explainer’ and ‘simulator’ models.

Polysemanticity Lecomte et al. (2024) observed that polysemanticity in neurons can arise incidentally, absent any optimization pressure toward it. A polysemantic neuron (basis dimension) of a layer encodes multiple interpretable concepts (Elhage et al., 2022; Scherlis et al., 2023); individual neurons in a randomly initialized network are, accordingly, expected to be polysemantic. Our analysis instead concerns *superposition* (Elhage et al., 2022; Chan, 2024), a property of the representation carried by an entire layer rather than of any single neuron. Superposition permits a layer to represent a number of (sparse) features that exceeds its (dense) neuron count via sparse coding, on the condition that only a small subset of concepts is active at any given input position, e.g., a given token position.

Training only the embeddings Zhong and Andreas (2024) demonstrated that transformers acquire nontrivial algorithmic capabilities when optimization is restricted to the embeddings alone, with all remaining parameters held at initialization. This result indicates that the behavior of a randomly initialized transformer can be substantially determined by optimizing only a small subset of its parameters. Our setting differs materially: in addition to introducing SAEs, we randomize *all* parameters, embeddings included, in the ‘Step-0’ and ‘Re-randomized incl. embeddings’ variants. The ‘Re-randomized excl. embeddings’ variant retains the pre-trained embeddings but does not optimize them against fixed randomized weights; rather, we freeze the pre-trained embeddings and randomize all remaining weights (Section 3).

Random transformers for board games Karvonen et al. (2024c) reported that SAEs extracted meaningful structure from chess games substantially more effectively when fit to pre-trained transformers than to random-weight transformers. The distributional properties of board-game data, however, differ markedly from those of natural language. In particular, natural language is plausibly sparse—a given concept such as ‘serendipitous’ occurs infrequently—and this sparse structure is ‘aligned’ with conceptual meaning. This alignment need not hold for board games: a functionally salient concept such as a knight fork does not necessarily occur sparsely across game states.

Random one-layer transformers Bricken et al. (2023) found that auto-interpretability scores discriminated reliably between random and trained one-layer transformers. Consistently, we observe that auto-interpretability scores in the randomized setting are comparatively low for smaller models (e.g., Pythia-70m) but that the gap to the trained model contracts as scale increases (e.g., Pythia-6.9b).

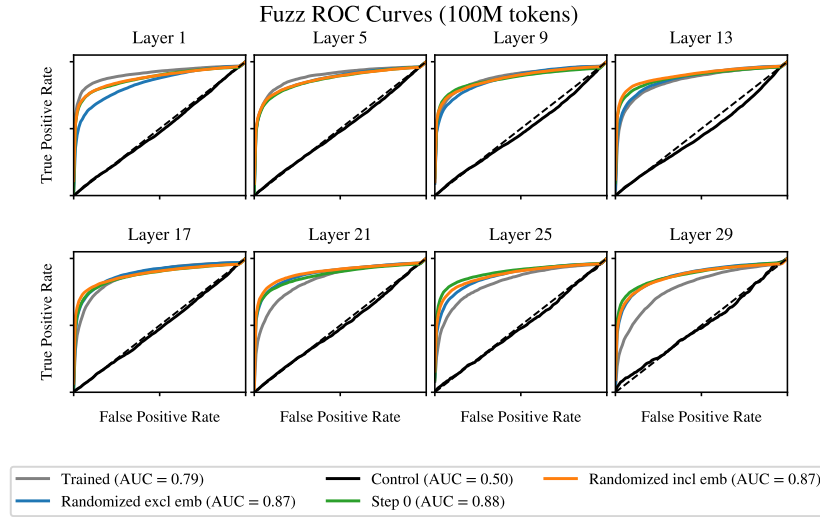


Figure 1: ‘Fuzzing’ ROC curves as a function of layer index for Pythia-6.9b, computed over 100 latents sampled per SAE. The curves for the trained model (gray) and the randomized variants (colored) are mutually overlapping, whereas the control (black) is near the chance diagonal (dotted). Consequently, aggregate AUROC alone is insufficient to attribute a latent to learned computation. Figure 2 reports the remaining metrics across model sizes; Appendix E reports the corresponding variability across random seeds.

3 RESULTS

We fit per-layer SAEs to the residual-stream activation vectors of transformer language models drawn from the Pythia suite, spanning 70M to 7B parameters (Biderman et al., 2023). We compare SAEs fit to the following variants of the underlying transformer, which differ only in how the weights are initialized or perturbed:

- **Trained:** The unmodified, trained model.
- **Re-randomized incl. embeddings:** Every parameter, embeddings included, is resampled i.i.d. from a Gaussian whose mean and variance match those of the corresponding original trained weight matrix, thereby preserving each matrix’s first two moments.
- **Re-randomized excl. embeddings:** Identical to the preceding variant, except that the embedding and unembedding matrices are retained unchanged from the trained model.
- **Step-0:** For each Pythia model, the step0 revision provides the weights at initialization, prior to any optimization (Biderman et al., 2023).
- **Control:** The unmodified trained model, except that at inference time each input token embedding is replaced by an i.i.d. standard Gaussian sample drawn independently per occurrence, so that a token is not assigned a consistent embedding vector. This variant destroys the token-embedding correspondence, and we accordingly expect auto-interpretability to attain chance-level performance.

In the primary experiments, each SAE is trained on 100M tokens from the RedPajama dataset (Weber et al., 2024) with an activation buffer of 10M tokens; Appendix C reports a subset of experiments confirming that the conclusions are unchanged when the token budget is increased to one billion. The layer sampling is dictated by available compute: for models below 410M parameters we fit an SAE at every layer; for Pythia-1b at every second layer; and for Pythia-6.9b at every fourth layer.

Unless stated otherwise, we train k -sparse autoencoders (equivalently, TopK SAEs; Makhzani and Frey 2014; Gao et al. 2024) with expansion factor $R = 64$ and sparsity level $k = 32$. Robustness to this hyperparameter choice is established by a sweep on Pythia-160m over expansion factors

$R \in \{16, 32, 64, 128\}$ and sparsity levels $k \in \{16, 32\}$ (Figure 18). The training code follows Belrose et al. (2025); the evaluation code follows Caden et al. (2025) and Karvonen et al. (2024a).

Auto-interpretability A feature explanation that names a concept can be supplied to a classifier tasked with predicting whether that concept is present in a given text input. The resulting classifier is scored by standard measures, in particular the area under the receiver operating characteristic (ROC) curve (AUROC). Paulo et al. (2024) defined two such classification tasks for evaluating feature explanations, termed ‘fuzzing’ and ‘detection.’ Under ‘fuzzing’ scoring, the positive and negative token examples for a latent (those with non-zero and zero activation values, respectively) are delimited with special characters, and a language model is prompted to identify, conditional on the latent’s explanation, which examples have been correctly delimited. Under ‘detection,’ a language model is prompted to identify which examples contain activating tokens for the feature. Bills et al. (2023) originally proposed ‘simulation’ scoring, defined via the correlation between predicted and observed activations, but its computational cost is prohibitive at scale.

Except where noted, we adopt ‘fuzzing’ scores as our auto-interpretability measure, on the basis that they have been shown to correlate with simulation scores (Paulo et al., 2024). The corresponding AUROC curves for ‘detection’ scoring are provided in Appendix B. For each fitted SAE—indexed by underlying model, variant, and layer—we draw a uniform random sample of 100 latents and compute their auto-interpretability scores. The implementation follows Paulo et al. (2024). Explanation generation and prediction both use Meta-Llama-3.1-70B-Instruct-AWQ-INT4, which is larger than the 8B models of Choi et al. (2024) and, unlike Bills et al. 2023, open-source.

We find that the auto-interpretability scores are markedly closer between the trained and randomized models than between either and the control (Figures 1 and 2). This coincidence of the ROC scores for trained and randomized transformers establishes that ‘fuzzing’ auto-interpretability applied to SAE latent explanations does not, on its own, constitute a discriminative test between the two underlying models.

Evaluation We evaluate a set of standard SAE quality metrics jointly with the auto-interpretability AUROC across Pythia models of 70M to 6.9B parameters. Consistent with the auto-interpretability results, the randomly initialized and re-randomized models (Figure 2; blue, green, orange) lie systematically closer to the trained model (gray) than to the control (black) across metrics.

Two lower-order regularities merit comment. First, both the cosine similarity between the original activation and its SAE reconstruction and the explained variance are typically far lower for the control than for the remaining variants, and the control’s reconstruction error increases with layer index whereas that of the other variants decreases. For the control, we attribute this to the maximum-entropy property of the Gaussian among distributions with fixed mean and variance (Jaynes, 2003): the resampled activations are, in this sense, maximally unstructured and therefore the least compressible by an SAE. As these Gaussian-distributed activations propagate through successive layers, we expect them to depart from Gaussianity and acquire ‘sparser’ structure, hence to become more readily reconstructible (Section 4).

Second, the re-randomized variants (blue and orange) track the trained model more closely than does the variant at initialization (green), an effect most pronounced in the L^1 norm at larger scales. We attribute this to the substantial divergence between the parameter norms of a trained model and those at initialization; by construction, our re-randomization procedure preserves the per-matrix parameter norms of the trained model. Per-layer parameter scale is plausibly consequential, e.g., in regulating the growth of activations along the residual stream (Liu et al., 2020). In the AUROC panels, every variant except the control exhibits AUROC increasing monotonically with model size. We conjecture that latents become more selective as SAE width grows: in narrower SAEs each latent must account for a larger share of the input, so the induced classification task is easier for wider SAEs.

The fifth row of Figure 2 reports the cross-entropy (CE) loss score (loss recovered) against layer index, defined as the increase in loss when the original activations are substituted by their SAE reconstructions, normalized by the increase when the activations are zero-‘ablated.’ The ‘trained’ variant attains CE loss scores comparable to those reported elsewhere (e.g., Kissane et al., 2024; Rajamanoharan et al., 2024a; Mudide et al., 2024). Crucially, the CE loss score is well-defined only for the trained variant: under any randomized variant the model loss is uniformly poor irrespective of whether the original or reconstructed activations are substituted.

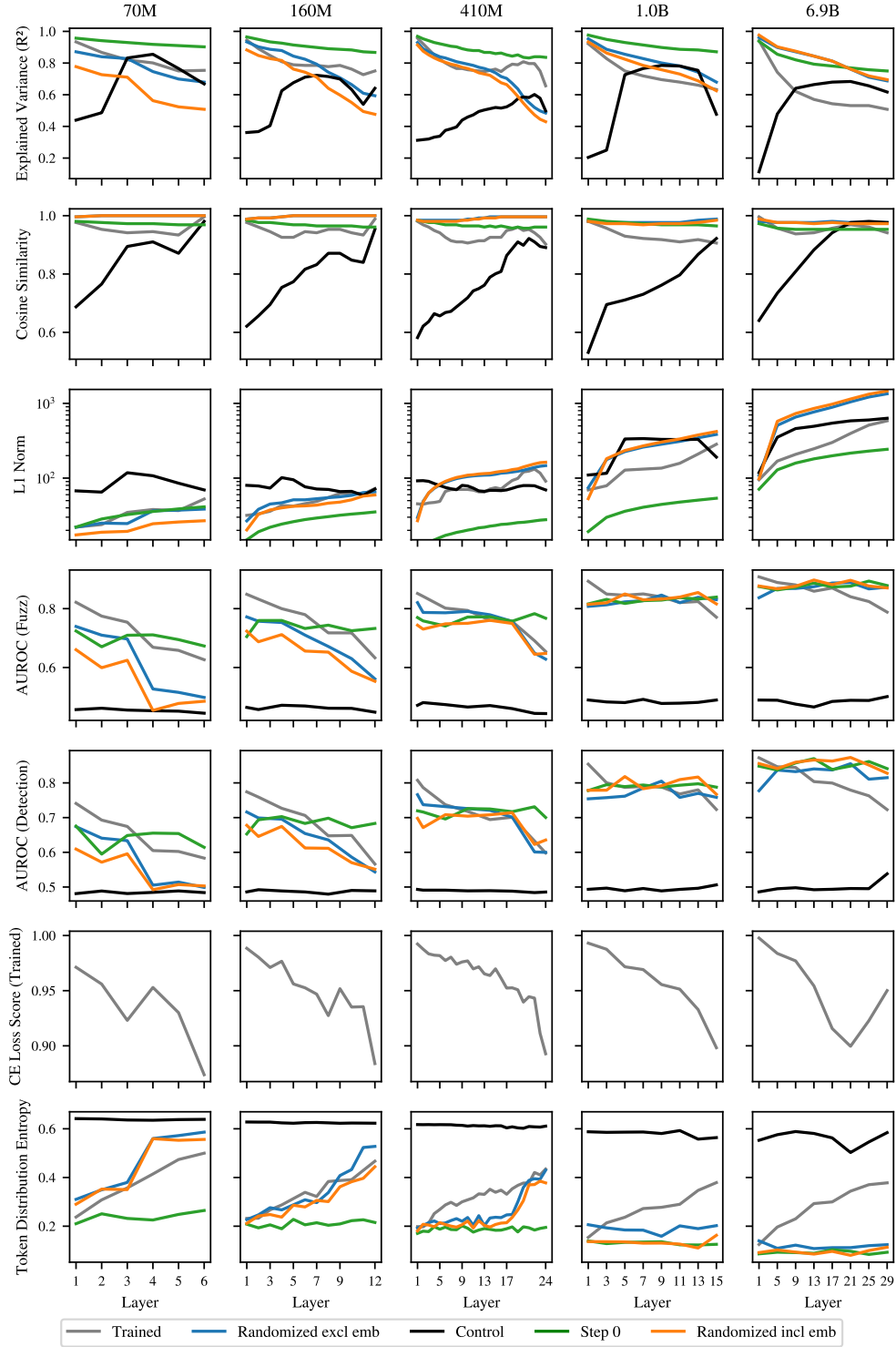


Figure 2: SAE evaluation metrics across the Pythia suite (70M to 6.9B parameters). The metric trajectories as a function of layer index are closely matched across variants at every scale, and the inter-layer variability contracts with increasing model size. Every variant other than the control attains comparable metric values notwithstanding the qualitatively distinct weight-initialization procedures.

Latent explanation complexity Notwithstanding the frequent parity in auto-interpretability scores and evaluation metrics, we hypothesized a priori that SAEs fit to trained versus randomized transformers would recover qualitatively distinct feature sets. Specifically, we predicted that SAEs fit to the randomized variants would recover comparatively simple features keyed to surface characteristics of the input text, but not the more complex, abstract features characteristic of trained transformers (Templeton et al., 2024). As qualitative evidence, Appendix J presents a random sample of latents together with their maximally activating dataset examples for each variant of Pythia-6.9b, and Appendix L provides more detailed per-latent diagnostics.

We have observed that a substantial fraction of SAE latents attain non-zero activation on only a single token, or on a small set of distinct tokens, across a text corpus (e.g., Lin and Bloom, 2024; Doods and Wilhelm, 2024). This motivates a simple complexity measure for a latent, given its maximally activating examples: the extent to which its activation is concentrated on a single token ID as opposed to being distributed over many distinct IDs. We operationalize this measure as the entropy of the empirical distribution of the latent’s activation over token IDs; higher entropy corresponds to activation that is more ‘spread out’ across tokens and hence to a less token-specific latent. This distribution is defined as the total latent activation accumulated per token ID over the set of maximally activating examples used to generate auto-interpretability explanations. The per-latent relationship between this entropy and the ‘fuzzing’ AUROC score is reported in Appendix H.

The last row of Figure 2 reports the entropy of the empirical latent-activation distributions over token IDs. The control variant exhibits uniformly high entropy, as expected: since each occurrence of a token ID draws an independent Gaussian embedding, the token possesses no consistent embedding vector (Section 3). For the trained variant, entropy increases monotonically with layer index, i.e., deeper latents are progressively less likely to have maximally activating examples concentrated on a single token. This too is expected: later layers should encode more abstract features, which are correspondingly less aligned with individual token embeddings. The randomized variants, by contrast, exhibit lower entropy than either the trained or control variants, indicating that their latents activate selectively on one or a few token IDs.

Taken together with the preceding results, this indicates that standard SAE quality and auto-interpretability metrics fail to capture a salient property of SAE features, namely their ‘abstractness.’ Although token-distribution entropy is not a direct measure of abstractness, the analysis shows that the randomized variants—given their auto-interpretability parity with the trained variant—remain capable of recovering simple, single-token features. Critically, and unlike the trained variant, the features of the randomized variants do not increase in complexity with layer index.

4 A TOY MODEL OF SUPERPOSITION IN RANDOM NETWORKS

In Section 1 we posited two mechanisms for the apparent sparsity and interpretability of randomized-transformer activations: either the input distribution itself exhibits superposition that the network preserves, or the network amplifies or induces superposition in the input. We now examine both hypotheses within controlled toy models. We obtain evidence consistent with each mechanism but defer to future work the determination of which predominates for the randomized transformers of the main-text experiments.

4.1 MATRIX MULTIPLICATIONS PRESERVE SUPERPOSITION

We first analyze a simplified generative model to establish that left-multiplication by a weight matrix W preserves superposition. Suppose the superposed input x is generated by first drawing n_s i.i.d. ‘sparse’ features z from a heavy-tailed Lomax distribution, $z \sim \text{Lomax}(\alpha, \lambda)$. The high-dimensional sparse z is projected to the lower-dimensional dense x by a matrix D , followed by additive Gaussian noise of small covariance Σ , so that $x \sim \mathcal{N}(x; Dz, \Sigma)$. The key observation is that the image $x' = Wx$ under any matrix W is *again* superposed: it is generated by the same model as x , with the projection map and noise covariance transformed to WD and $W\Sigma W^T$ respectively, i.e., $x' \sim \mathcal{N}(x'; WDz, W\Sigma W^T)$.

This argument extends heuristically to networks with nonlinearities, as can be seen by visualizing the result of passing the dense activations through a simple feed-forward network (MLP). Figure 3 depicts a configuration with $n_s = 3$ sparse features projected to $n_d = 2$ dense features, in which the

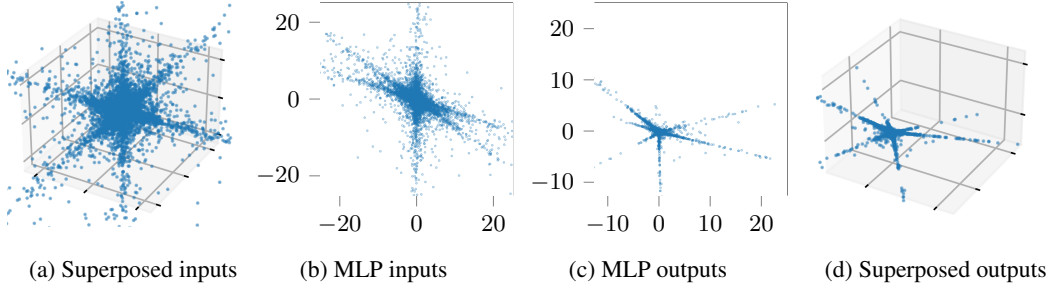


Figure 3: Illustration of the action of a randomly initialized neural network on superposed input data. We draw 10K samples of $n_s = 3$ sparse input features from a Lomax distribution with shape $\alpha = 1$ and scale $\lambda = 1$, and project them to $n_d = 2$ dense input features via a matrix with i.i.d. standard normal entries. The dense inputs are then passed through a two-layer MLP with ReLU activation and hidden width $4n_d$, and $n_s = 3$ sparse outputs are recovered by applying the inverse of the projection matrix.

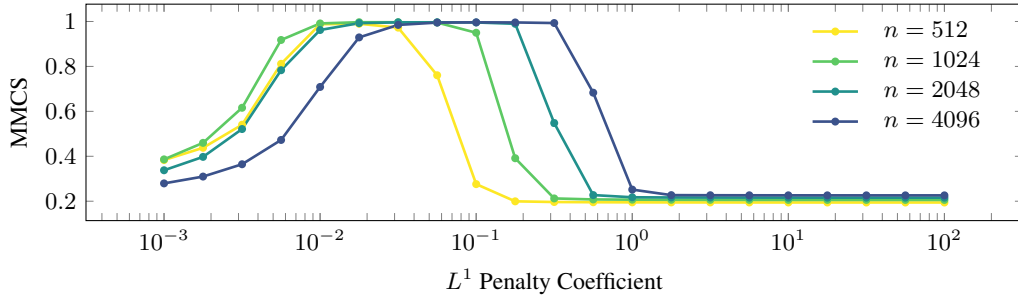


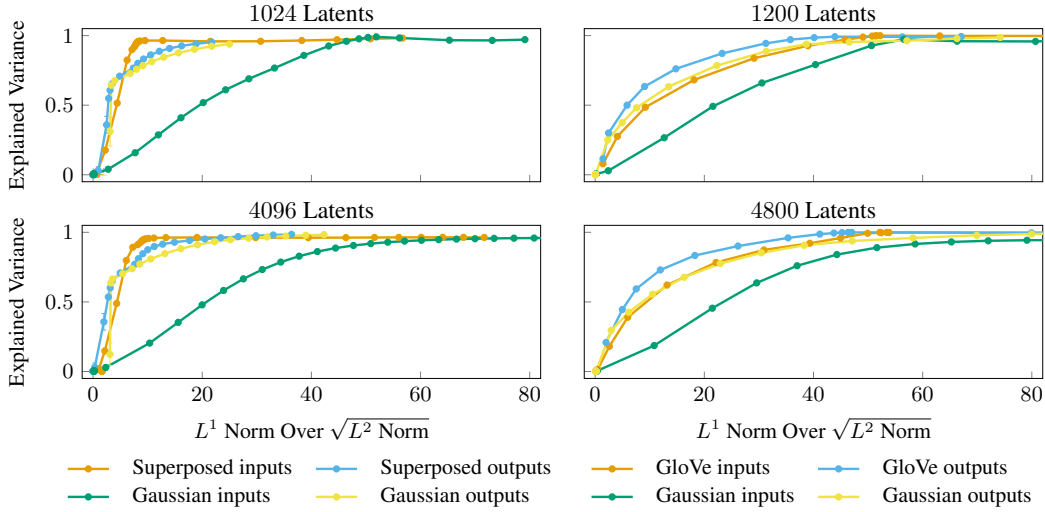
Figure 4: Mean max cosine similarity (MMCS) between the SAE-learned features (decoder weight vectors) and the data-generating features, as a function of the L^1 penalty coefficient in the training loss, following Sharkey et al. (2022). An intermediate ‘Goldilocks zone’ of the penalty coefficient yields near-exact recovery of the data-generating features, provided the SAE has sufficiently many latents to represent them.

MLP outputs remain superposed despite the intervening nonlinearity. The figure further suggests that networks may amplify superposition rather than merely preserve it: relative to the inputs (Figures 3a and 3b), the outputs (Figures 3d and 3c) exhibit fewer points in the interior between the ‘arms’ of the distribution.

4.2 DO RANDOM NNS PRESERVE OR AMPLIFY SUPERPOSITION?

We tested this suggestion by generating toy data following the procedure of Sharkey et al. (2022): ground-truth features are sampled on a hypersphere and endowed with correlated coefficients constructed so that only a small number are simultaneously active (Appendix I.1). These inputs were passed through a two-layer MLP at initialization, and separate SAEs were fit to the inputs and to the outputs. The control consisted of Gaussian-distributed inputs matched in mean and standard deviation to the superposed toy data. All SAEs in this experiment are standard SAEs with an L^1 sparsity penalty (Appendix I.2).

Following Sharkey et al. (2022), we verified that SAEs recover the ground-truth features underlying the data (Figure 4). We quantify recovery by the mean max cosine similarity (MMCS): for each data-generating feature we compute the maximum cosine similarity to any SAE-learned feature (decoder weight vector), then average over data-generating features. The MMCS, however, is only applicable to the MLP inputs, for which the data-generating features are known; analyzing the MLP outputs requires a distinct proxy. For this purpose we take the SAE’s ability to attain low reconstruction error at high sparsity as a proxy for the degree of superposition present in its training data. Concretely, we



(a) Toy datasets following Sharkey et al. (2022). (b) 300-dim. GloVe vectors (Pennington et al., 2014).

Figure 5: Pareto frontiers of explained variance against the sparsity measure $L^1/\sqrt{L^2}$ for datasets hypothesized to exhibit superposition, Gaussian controls matched in mean and variance, and the corresponding outputs obtained by passing each through a randomly initialized two-layer MLP (Section 4.2). Each point corresponds to a distinct setting of the L^1 penalty coefficient.

sweep the L^1 penalty coefficient to trace out Pareto frontiers of explained variance against sparsity measures (Figure 5a).

As anticipated, at any fixed level of explained variance the SAEs attain substantially greater sparsity on the superposed inputs than on the Gaussian control (Figure 5a; orange and blue-green). Notably, the gap between the superposed outputs (the MLP outputs given the superposed inputs) and the Gaussian outputs is much narrower, amounting to only a slight sparsity advantage at fixed explained variance. This indicates that the outputs of randomly initialized MLPs exhibit a relatively high, and input-distribution-insensitive, degree of sparsity. Additional sparsity measures and hyperparameter settings are reported in Appendix I.

4.3 DO TOKEN EMBEDDINGS EXHIBIT SUPERPOSITION?

If randomly initialized networks preserve or amplify superposition, then the main-text results (Section 3) would be explained by the degree of superposition present in the inputs to transformer language models. We investigate this by applying the procedure of Section 4.2 to language data. Specifically, we fit SAEs to pre-trained GloVe word vectors, to the embedding matrices of Pythia models, to the outputs of passing each of these through a randomly initialized two-layer MLP, and to the corresponding Gaussian controls. The protocol is identical to Section 4.2, except that the sample size is fixed by the number of word embeddings or tokens and a single random seed is used.

We find that the separation between the Pareto frontiers of the GloVe word vectors and their Gaussian controls (Figure 5b) is smaller than that observed for the toy superposed datasets of Section 4.2 (Figure 5a). More notably, for both input types the Pareto frontier improves after passage through a randomly initialized two-layer MLP, reinforcing the hypothesis that random networks ‘sparsify’ their inputs, i.e., increase the apparent degree of superposition.

5 LIMITATIONS

This work establishes that auto-interpretability measures can yield apparently meaningful, interpretable results for SAEs fit to randomly initialized models, which are a priori unlikely to encode computationally interesting features. As exhaustive coverage of all datasets and architectures is infeasible, we restricted the study to the Pythia family, which is widely adopted in mechanistic

interpretability research (e.g., Paulo and Belrose, 2025; Ghilardi et al., 2025; Mueller, 2024), and to the RedPajama-V2 dataset as a representative language-model and SAE pre-training corpus.

We employed the default explanation-generation model of the EleutherAI auto-interpretability framework (Caden et al., 2025); substituting alternative models could inform both the aggregate behavior of the metrics and the quality of the generated explanations. We emphasize the precise scope of our claim: we do not assert that SAEs fail to extract information from trained transformers beyond what they extract from randomly initialized ones, but only that aggregate auto-interpretability measures are not sufficient to certify the presence of interesting underlying features.

6 CONCLUSION

We fit sparse autoencoders to both trained and randomly initialized transformers and evaluated them under a suite of standard quantitative metrics. Our principal empirical result is that, under identifiable conditions, these metrics—and aggregate auto-interpretability scores in particular—are closely matched across the two settings. Although features derived from trained transformers are qualitatively more complex and abstract, especially at later layers, the aggregate metrics frequently fail to register this distinction.

This result does not entail that SAEs fit to trained models fail to recover meaningful computational features. It instead exposes a specificity limitation of the current evaluation methodology. A high aggregate auto-interpretability score is not sufficient evidence for the discovery of complex, learned computation: it may instead be attributable to lower-order structure in the data or architecture that is retained even under random weights. Our token-distribution-entropy analysis, though preliminary, functions as a proof of concept, resolving differences in feature ‘abstractness’ that the aggregate auto-interpretability scores fail to detect. Future work should prioritize the development of more discriminative metrics that quantify the computational significance of the features an SAE recovers. More broadly, our results underscore the necessity of benchmarking interpretability techniques against strong, appropriately specified null models such as the randomly initialized transformers employed here; absent such baselines, discovered features cannot be confidently attributed to the learning process.

REFERENCES

- J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim. Sanity Checks for Saliency Maps, 2020. URL <https://arxiv.org/abs/1810.03292>.
- A. J. Bell and T. J. Sejnowski. An Information-Maximization Approach to Blind Separation and Blind Deconvolution. *Neural Computation*, 7(6):1129–1159, Nov. 1995. ISSN 0899-7667. doi: 10.1162/neco.1995.7.6.1129. URL <https://ieeexplore.ieee.org/abstract/document/6796129>. Conference Name: Neural Computation.
- N. Belrose, L. Quirke, A. Garriga-Alonso, neverix, HongchuanZeng, A. Duong, lewington, P. Minervini, T. Vogel, T. Fel, Yang, avgalichin, and T. V. Browne. Eleutherai/sparsify. <https://github.com/EleutherAI/sparsify>, sep 22 2025. URL <https://github.com/EleutherAI/sparsify>.
- S. Biderman, H. Schoelkopf, Q. G. Anthony, H. Bradley, K. O’Brien, E. Hallahan, M. A. Khan, S. Purohit, U. S. Prashanth, E. Raff, A. Skowron, L. Sutawika, and O. V. D. Wal. Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling. In *Proceedings of the 40th International Conference on Machine Learning*, pages 2397–2430. PMLR, July 2023. URL <https://proceedings.mlr.press/v202/biderman23a.html>. ISSN: 2640-3498.
- S. Bills, N. Cammarata, D. Mossing, H. Tillman, L. Gao, G. Goh, I. Sutskever, J. Leike, J. Wu, and W. Saunders. Language models can explain neurons in language models, May 2023. URL <https://openaipublic.blob.core.windows.net/neuron-explainer/paper/index.html>.
- D. Braun, J. Taylor, N. Goldowsky-Dill, and L. Sharkey. Identifying Functionally Important Features with End-to-End Sparse Dictionary Learning, May 2024. URL <http://arxiv.org/abs/2405.12241>. arXiv:2405.12241 [cs].
- T. Bricken, A. Templeton, J. Batson, B. Chen, A. Jermyn, T. Conerly, N. Turner, C. Anil, C. Denison, and A. Askell. Towards Monosemanticity: Decomposing Language Models With Dictionary Learning, 2023. URL <https://transformer-circuits.pub/2023/monosemantic-features>.
- Caden, L. Quirke, G. Paulo, A. Mallen, neverix, N. Belrose, johnny, A. Sumer, A. Duong, W. Li, S. Biderman, L. Farnik, I. E. Ashimine, A. Muhamed, D. Ghilardi, F. Xiao, and M. M. Rahman. Eleutherai/delphi. <https://github.com/EleutherAI/delphi>, sep 22 2025. URL <https://github.com/EleutherAI/delphi>.
- L. Chan. Superposition is not “just” neuron polysemanticity. Apr. 2024. URL <https://www.alignmentforum.org/posts/8EyCQKuWo6swZpagS/superposition-is-not-just-neuron-polysemanticity>.
- M. Chaudhary and A. Geiger. Evaluating Open-Source Sparse Autoencoders on Disentangling Factual Knowledge in GPT-2 Small, Sept. 2024. URL <http://arxiv.org/abs/2409.04478>. arXiv:2409.04478 [cs].
- D. Choi, V. Huang, K. Meng, D. D. Johnson, J. Steinhardt, and S. Schwettmann. Scaling Automatic Neuron Description, Oct. 2024. URL <https://transluce.org/neuron-descriptions>.
- H. Cunningham, A. Ewart, L. Riggs, R. Huben, and L. Sharkey. Sparse Autoencoders Find Highly Interpretable Features in Language Models, Oct. 2023. URL <http://arxiv.org/abs/2309.08600>. arXiv:2309.08600 [cs].
- T. Doms and D. Wilhelm. Tokenized SAEs: Disentangling SAE Reconstructions. June 2024. URL <https://openreview.net/forum?id=5Eas7HCe38>.
- N. Elhage, T. Hume, C. Olsson, N. Schiefer, T. Henighan, S. Kravec, Z. Hatfield-Dodds, R. Lasenby, D. Drain, C. Chen, R. Grosse, S. McCandlish, J. Kaplan, D. Amodei, M. Wattenberg, and C. Olah. Toy Models of Superposition, Sept. 2022. URL <http://arxiv.org/abs/2209.10652>. arXiv:2209.10652 [cs].
- J. Engels, I. Liao, E. J. Michaud, W. Gurnee, and M. Tegmark. Not All Language Model Features Are Linear, May 2024. URL <http://arxiv.org/abs/2405.14860>. arXiv:2405.14860 [cs].

-
- L. Farnik, T. Lawson, C. Houghton, and L. Aitchison. Jacobian Sparse Autoencoders: Sparsify Computations, Not Just Activations. In *Forty-second International Conference on Machine Learning*, June 2025. URL <https://openreview.net/forum?id=TPuFRuNano>.
- A. Foote, N. Nanda, E. Kran, I. Konstas, and F. Barez. N2G: A Scalable Approach for Quantifying Interpretable Neuron Representations in Large Language Models, Apr. 2023. URL <http://arxiv.org/abs/2304.12918>. arXiv:2304.12918 [cs].
- L. Gao, T. D. la Tour, H. Tillman, G. Goh, R. Troll, A. Radford, I. Sutskever, J. Leike, and J. Wu. Scaling and evaluating sparse autoencoders, June 2024. URL <http://arxiv.org/abs/2406.04093>. arXiv:2406.04093 [cs].
- D. Ghilardi, F. Belotti, M. Molinari, T. Ma, and M. Palmonari. Group-SAE: Efficient Training of Sparse Autoencoders for Large Language Models via Layer Groups, Sept. 2025.
- W. Gurnee, N. Nanda, M. Pauly, K. Harvey, D. Troitskii, and D. Bertsimas. Finding Neurons in a Haystack: Case Studies with Sparse Probing, June 2023. URL <http://arxiv.org/abs/2305.01610>. arXiv:2305.01610 [cs].
- S. Heimersheim and N. Nanda. How to use and interpret activation patching, Apr. 2024. URL <http://arxiv.org/abs/2404.15255>. arXiv:2404.15255 [cs].
- J. Huang, Z. Wu, C. Potts, M. Geva, and A. Geiger. RAVEL: Evaluating Interpretability Methods on Disentangling Language Model Representations, Aug. 2024. URL <http://arxiv.org/abs/2402.17700>. arXiv:2402.17700 [cs].
- A. Hyvärinen and E. Oja. Independent component analysis: algorithms and applications. *Neural Networks*, 13(4):411–430, June 2000. ISSN 0893-6080. doi: 10.1016/S0893-6080(00)00026-5. URL <https://www.sciencedirect.com/science/article/pii/S0893608000000265>.
- E. T. Jaynes. *Probability Theory: The Logic of Science*. Cambridge University Press, Cambridge, UK, 2003. ISBN 0521592712. The maximum entropy property of the Gaussian distribution is discussed on pages 208 and 365, in chapters 7 and 11.
- A. Karvonen, C. Rager, J. Lin, C. Tigges, J. Bloom, D. Chanin, Y.-T. Lau, E. Farrell, A. Conmy, C. McDougall, K. Ayonrinde, M. Wearden, S. Marks, and N. Nanda. SAE Bench: A Comprehensive Benchmark for Sparse Autoencoders, Dec. 2024a. URL <https://www.neuronpedia.org/sae-bench/info>.
- A. Karvonen, C. Rager, S. Marks, and N. Nanda. Evaluating Sparse Autoencoders on Targeted Concept Erasure Tasks, Nov. 2024b. URL <https://arxiv.org/abs/2411.18895>.
- A. Karvonen, B. Wright, C. Rager, R. Angell, J. Brinkmann, L. Smith, C. M. Verdun, D. Bau, and S. Marks. Measuring Progress in Dictionary Learning for Language Model Interpretability with Board Game Models. *arXiv preprint arXiv:2408.00113*, 2024c. URL <https://arxiv.org/abs/2408.00113>.
- C. Kissane, R. Krzyzanowski, J. I. Bloom, A. Conmy, and N. Nanda. Interpreting Attention Layer Outputs with Sparse Autoencoders. June 2024. URL <https://openreview.net/forum?id=feWUBDwjji>.
- T. Lawson, L. Farnik, C. Houghton, and L. Aitchison. Residual Stream Analysis with Multi-Layer SAEs. In *The Thirteenth International Conference on Learning Representations*, Oct. 2024. URL <https://openreview.net/forum?id=XAjfjizaKs>.
- V. Lecomte, K. Thaman, R. Schaeffer, N. Bashkansky, T. Chow, and S. Koyejo. What Causes Polysemanticity? An Alternative Origin Story of Mixed Selectivity from Incidental Causes, Feb. 2024. URL <http://arxiv.org/abs/2312.03096>. arXiv:2312.03096 [cs].
- H. Lee, A. Battle, R. Raina, and A. Ng. Efficient sparse coding algorithms. In *Advances in Neural Information Processing Systems*, volume 19. MIT Press, 2006. URL https://proceedings.neurips.cc/paper_files/paper/2006/hash/2d71b2ae158c7c5912cc0bbde2bb9d95-Abstract.html.

648 T. Lieberum, S. Rajamanoharan, A. Conmy, L. Smith, N. Sonnerat, V. Varma, J. Kramár, A. Dragan,
649 R. Shah, and N. Nanda. Gemma Scope: Open Sparse Autoencoders Everywhere All At Once on
650 Gemma 2, Aug. 2024. URL <http://arxiv.org/abs/2408.05147>. arXiv:2408.05147 [cs].
651

652 J. Lin and J. Bloom. Announcing Neuronpedia: Platform for accelerating research into Sparse
653 Autoencoders, Mar. 2024. URL [https://www.alignmentforum.org/posts/BaEQoxHhWPrkinm](https://www.alignmentforum.org/posts/BaEQoxHhWPrkinmxd/announcing-neuronpedia-platform-for-accelerating-research)
654 [xd/announcing-neuronpedia-platform-for-accelerating-research](https://www.alignmentforum.org/posts/BaEQoxHhWPrkinmxd/announcing-neuronpedia-platform-for-accelerating-research).

655 J. Lindsey, A. Templeton, J. Marcus, T. Conerly, and J. Batson. Sparse Crosscoders for Cross-Layer
656 Features and Model Diffing, Oct. 2024. URL [https://transformer-circuits.pub/2024/cro](https://transformer-circuits.pub/2024/crosscoders/index.html)
657 [sscoders/index.html](https://transformer-circuits.pub/2024/crosscoders/index.html).
658

659 L. Liu, X. Liu, J. Gao, W. Chen, and J. Han. Understanding the difficulty of training transformers.
660 In B. Webber, T. Cohn, Y. He, and Y. Liu, editors, *Proceedings of the 2020 Conference on*
661 *Empirical Methods in Natural Language Processing (EMNLP)*, pages 5747–5763, Online, Nov.
662 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.463. URL
663 <https://aclanthology.org/2020.emnlp-main.463/>.

664 A. Makelov. Sparse Autoencoders Match Supervised Features for Model Steering on the IOI Task.
665 June 2024. URL <https://openreview.net/forum?id=JdrVuEQih5>.
666

667 A. Makhzani and B. Frey. k-Sparse Autoencoders, Mar. 2014. URL [http://arxiv.org/abs/1312](http://arxiv.org/abs/1312.5663)
668 [.5663](http://arxiv.org/abs/1312.5663). arXiv:1312.5663 [cs].

669 S. C. Marshall and J. H. Kirchner. Understanding polysemanticity in neural networks through coding
670 theory, Jan. 2024. URL <http://arxiv.org/abs/2401.17975>. arXiv:2401.17975 [cs].
671

672 K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and Editing Factual Associations in
673 GPT. *Advances in Neural Information Processing Systems*, 35:17359–17372, Dec. 2022. URL
674 [https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89](https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html)
675 [b0665b33bf3a182-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html).

676 A. Mudide, J. Engels, E. J. Michaud, M. Tegmark, and C. S. de Witt. Efficient Dictionary Learning
677 with Switch Sparse Autoencoders. *arXiv preprint arXiv:2410.08201*, 2024. URL [https://arxiv.](https://arxiv.org/abs/2410.08201)
678 [org/abs/2410.08201](https://arxiv.org/abs/2410.08201).
679

680 A. Mueller. Missed Causes and Ambiguous Effects: Counterfactuals Pose Challenges for Interpreting
681 Neural Networks, 2024. URL <https://arxiv.org/abs/2407.04690>.

682 A. Ng. Sparse autoencoder, 2011. URL [https://graphics.stanford.edu/courses/cs233-2](https://graphics.stanford.edu/courses/cs233-21-spring/ReferencedPapers/SAE.pdf)
683 [1-spring/ReferencedPapers/SAE.pdf](https://graphics.stanford.edu/courses/cs233-21-spring/ReferencedPapers/SAE.pdf).
684

685 K. O’Brien, D. Majercak, X. Fernandes, R. Edgar, J. Chen, H. Nori, D. Carignan, E. Horvitz, and
686 F. Poursabzi-Sangde. Steering Language Model Refusal with Sparse Autoencoders, Nov. 2024.
687 URL <https://arxiv.org/abs/2411.11296>.

688 B. A. Olshausen and D. J. Field. Emergence of simple-cell receptive field properties by learning a
689 sparse code for natural images. *Nature*, 381(6583):607–609, June 1996. ISSN 1476-4687. doi:
690 10.1038/381607a0. URL <https://www.nature.com/articles/381607a0>. Publisher: Nature
691 Publishing Group.
692

693 B. A. Olshausen and D. J. Field. Sparse coding with an overcomplete basis set: A strategy employed
694 by V1? *Vision Research*, 37(23):3311–3325, Dec. 1997. ISSN 0042-6989. doi: 10.1016/S0042-6
695 989(97)00169-7. URL [https://www.sciencedirect.com/science/article/pii/S0042698](https://www.sciencedirect.com/science/article/pii/S0042698997001697)
696 [997001697](https://www.sciencedirect.com/science/article/pii/S0042698997001697).

697 K. Park, Y. J. Choe, and V. Veitch. The Linear Representation Hypothesis and the Geometry of Large
698 Language Models, Nov. 2023. URL <http://arxiv.org/abs/2311.03658>. arXiv:2311.03658
699 [cs, stat].
700

701 G. Paulo and N. Belrose. Sparse Autoencoders Trained on the Same Data Learn Different Features,
2025. URL <https://arxiv.org/abs/2501.16615>.

-
- G. Paulo, A. Mallen, C. Juang, and N. Belrose. Automatically Interpreting Millions of Features in Large Language Models, Oct. 2024.
- J. Pennington, R. Socher, and C. D. Manning. GloVe: Global Vectors for Word Representation. In A. Moschitti, B. Pang, and W. Daelemans, editors, *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1532–1543, Doha, Qatar, Oct. 2014. Association for Computational Linguistics. doi: 10.3115/v1/D14-1162.
- S. Rajamanoharan, A. Conmy, L. Smith, T. Lieberum, V. Varma, J. Kramar, R. Shah, and N. Nanda. Improving Sparse Decomposition of Language Model Activations with Gated Sparse Autoencoders. June 2024a. URL <https://openreview.net/forum?id=Ppj5KvzU8Q>.
- S. Rajamanoharan, T. Lieberum, N. Sonnerat, A. Conmy, V. Varma, J. Kramár, and N. Nanda. Jumping Ahead: Improving Reconstruction Fidelity with JumpReLU Sparse Autoencoders, July 2024b. URL <http://arxiv.org/abs/2407.14435>. arXiv:2407.14435 [cs].
- A. Scherlis, K. Sachan, A. S. Jermyn, J. Benton, and B. Shlegeris. Polysemanticity and Capacity in Neural Networks, July 2023. URL <http://arxiv.org/abs/2210.01892>. arXiv:2210.01892 [cs].
- L. Sharkey, D. Braun, and B. Millidge. Taking features out of superposition with sparse autoencoders, Dec. 2022. URL <https://www.alignmentforum.org/posts/z6QQJbtpkEAX3AoJJ/interim-research-report-taking-features-out-of-superposition>.
- A. Templeton, T. Conerly, J. Marcus, J. Lindsey, T. Bricken, B. Chen, A. Pearce, C. Citro, E. Ameisen, A. Jones, H. Cunningham, N. L. Turner, C. McDougall, M. MacDiarmid, A. Tamkin, E. Durmus, T. Hume, F. Mosconi, C. D. Freeman, T. R. Sumers, E. Rees, J. Batson, A. Jermyn, S. Carter, C. Olah, and T. Henighan. Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet, May 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>.
- M. Wattenberg and F. Viégas. Relational Composition in Neural Networks: A Survey and Call to Action. June 2024. URL <https://openreview.net/forum?id=zzCEiUIPk9>.
- M. Weber, D. Fu, Q. Anthony, Y. Oren, S. Adams, A. Alexandrov, X. Lyu, H. Nguyen, X. Yao, V. Adams, B. Athiwaratkun, R. Chalamala, K. Chen, M. Ryabinin, T. Dao, P. Liang, C. Ré, I. Rish, and C. Zhang. RedPajama: an Open Dataset for Training Large Language Models, 2024. URL <https://arxiv.org/abs/2411.12372>.
- Z. Yun, Y. Chen, B. Olshausen, and Y. LeCun. Transformer visualization via dictionary learning: contextualized embedding as a linear superposition of transformer factors. In E. Agirre, M. Apidianaki, and I. Vulić, editors, *Proceedings of Deep Learning Inside Out (DeeLIO): The 2nd Workshop on Knowledge Extraction and Integration for Deep Learning Architectures*, pages 1–10, Online, June 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.deelio-1.1. URL <https://aclanthology.org/2021.deelio-1.1>.
- F. Zhang and N. Nanda. Towards Best Practices of Activation Patching in Language Models: Metrics and Methods. In *The Twelfth International Conference on Learning Representations*, Oct. 2023. URL <https://openreview.net/forum?id=Hf17y6u9BC>.
- Z. Zhong and J. Andreas. Algorithmic Capabilities of Random Transformers, 2024. URL <https://arxiv.org/abs/2410.04368>.

A BROADER IMPACT

This work analyzes a method presently in use for the mechanistic interpretability of LLMs and produces results that qualify several assumptions underlying sparse autoencoders. By establishing that SAEs yield comparable aggregate auto-interpretability scores for both random and trained transformers, our findings constrain the inferences these evaluation methods can support regarding what they in fact measure.

A more precise characterization of SAE quality metrics should, we anticipate, inform the principled development of improved SAE-like methods, thereby advancing model interpretability and mitigating potential downstream harms. As our contribution is an empirical characterization of an existing method—one showing that the method interprets both random and trained transformers—we assess the risk of negative societal impact arising from this work as minimal.

B AUTO-INTERPRETABILITY ROC CURVES

Figures 6, 8, and 12 report the agreement in ‘fuzzing’ AUROC between the trained and randomized SAEs for the 70M, 160M, and 1B models, respectively. Figures 7, 9, and 13 report the corresponding agreement in ‘detection’ AUROC for the same three model sizes.

B.1 PYTHIA 70M

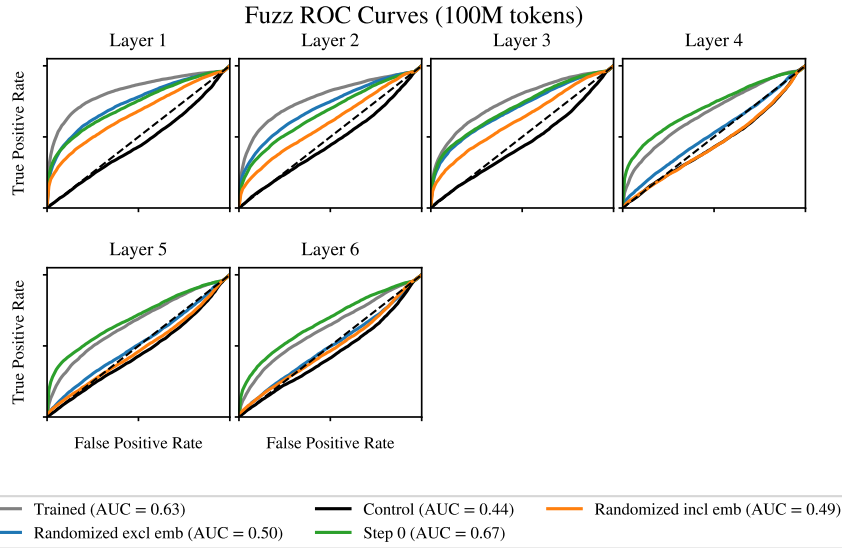


Figure 6: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-70m, computed over 100 SAE latents. The variants attain closely matched performance, and performance degrades monotonically with increasing layer index.

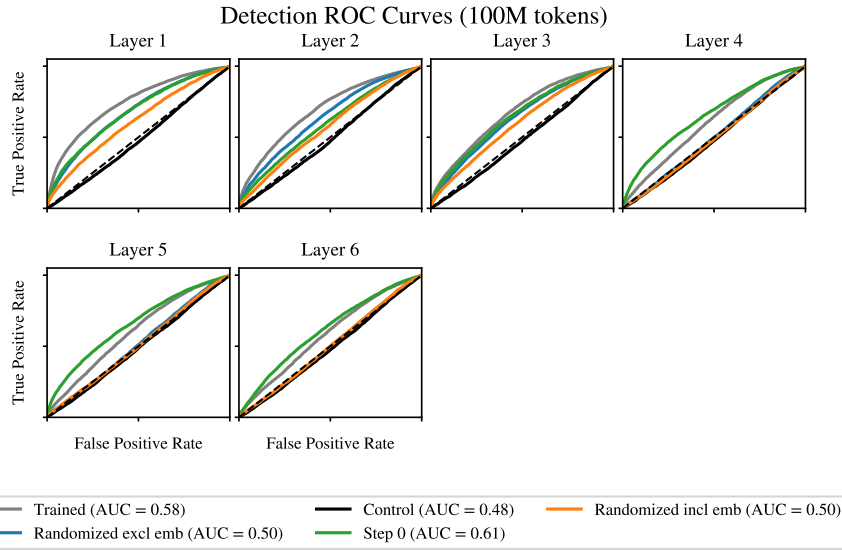


Figure 7: ‘Detection’ auto-interpretability ROC curves for Pythia-70m, computed over 100 SAE latents. The variants attain closely matched performance, and performance degrades monotonically with increasing layer index.

B.2 PYTHIA 160M

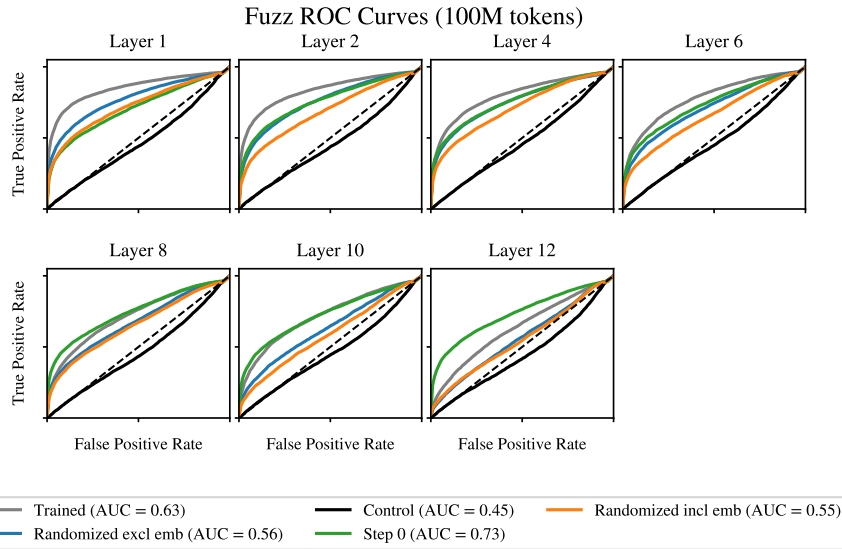


Figure 8: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-160m, computed over 100 SAE latents. The variants attain closely matched performance, and performance degrades monotonically with increasing layer index.

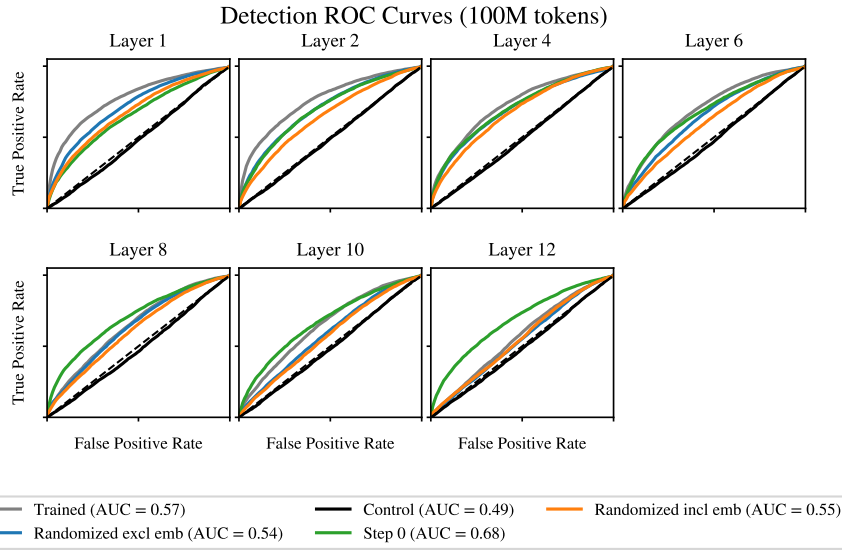


Figure 9: ‘Detection’ auto-interpretability ROC curves for Pythia-160m, computed over 100 SAE latents. The variants attain closely matched performance, and performance degrades monotonically with increasing layer index.

B.3 PYTHIA 410M

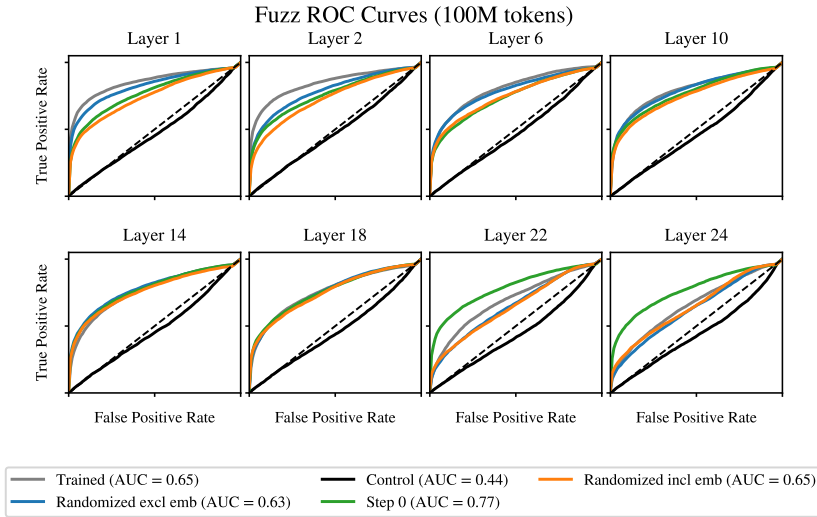


Figure 10: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-410m, computed over 100 SAE latents. The variants attain closely matched performance, which degrades monotonically with increasing layer index. The auto-interpretability scores here do not yield a discriminative test between trained and randomized models.

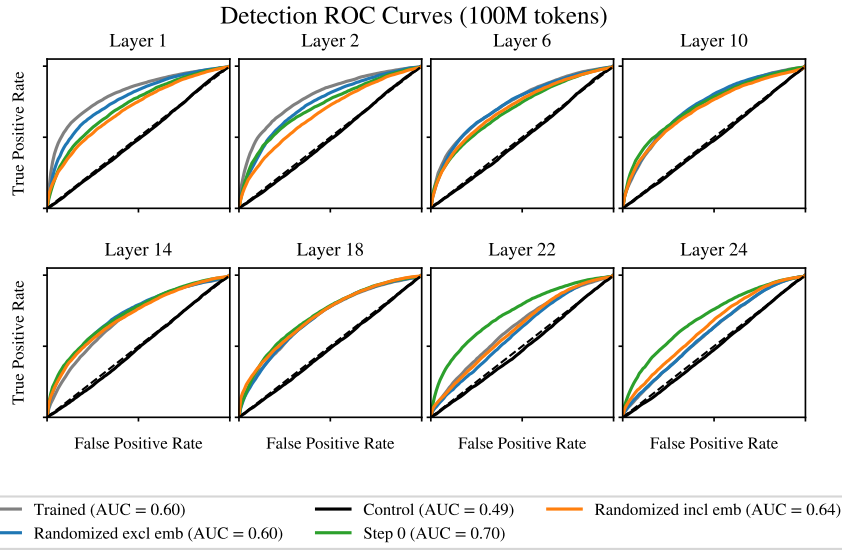


Figure 11: ‘Detection’ auto-interpretability ROC curves for Pythia-410m, computed over 100 SAE latents. The variants attain closely matched performance, and performance degrades monotonically with increasing layer index.

B.4 PYTHIA-1B

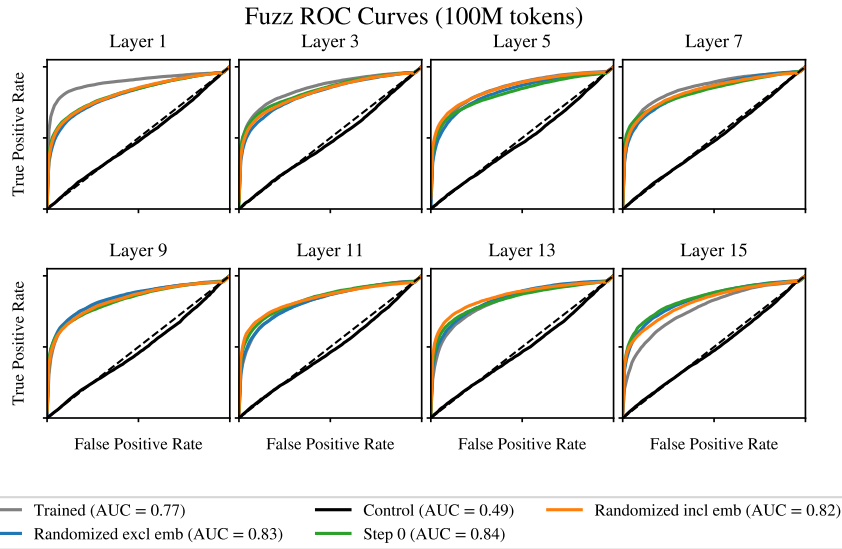


Figure 12: ‘Fuzzing’ auto-interpretability ROC curves for Pythia-1b, computed over 100 SAE latents. The variants attain closely matched performance; at this scale, however, no monotonic degradation with layer index is observed.

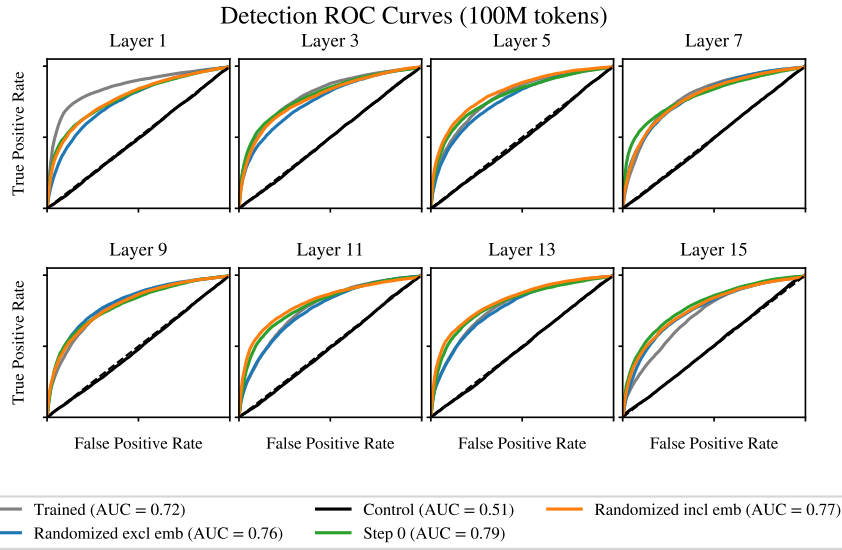


Figure 13: ‘Detection’ auto-interpretability ROC curves for Pythia-1b, computed over 100 SAE latents. The variants attain closely matched performance; at this scale, however, no monotonic degradation with layer index is observed.

B.5 PYTHIA 6.9B

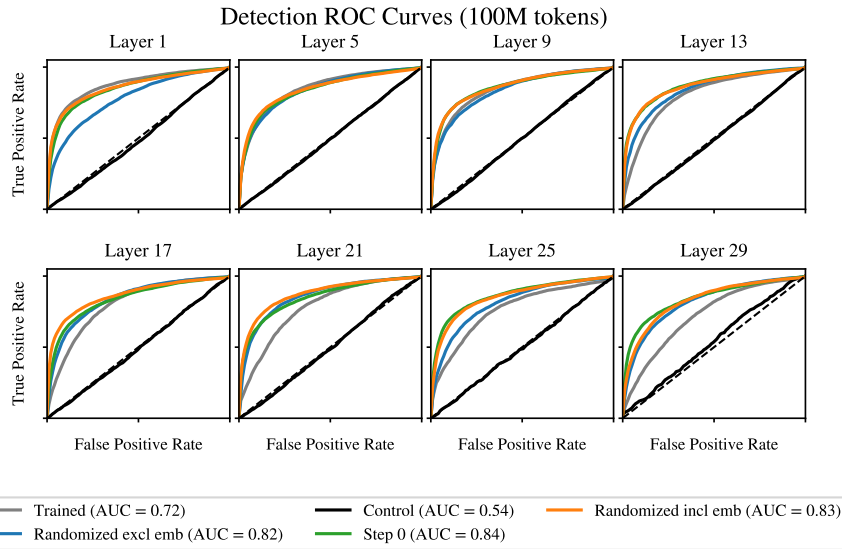


Figure 14: ‘Detection’ auto-interpretability ROC curves for Pythia-6.9b, computed over 100 SAE latents. The variants attain closely matched performance.

C EFFECT OF INCREASED TRAINING DATA

The primary experiments train SAEs on 100M tokens (Section 3). To exclude insufficient training data as a confound, we repeated a subset of these experiments with SAEs trained on 1B tokens from the RedPajama dataset (Figure 15).

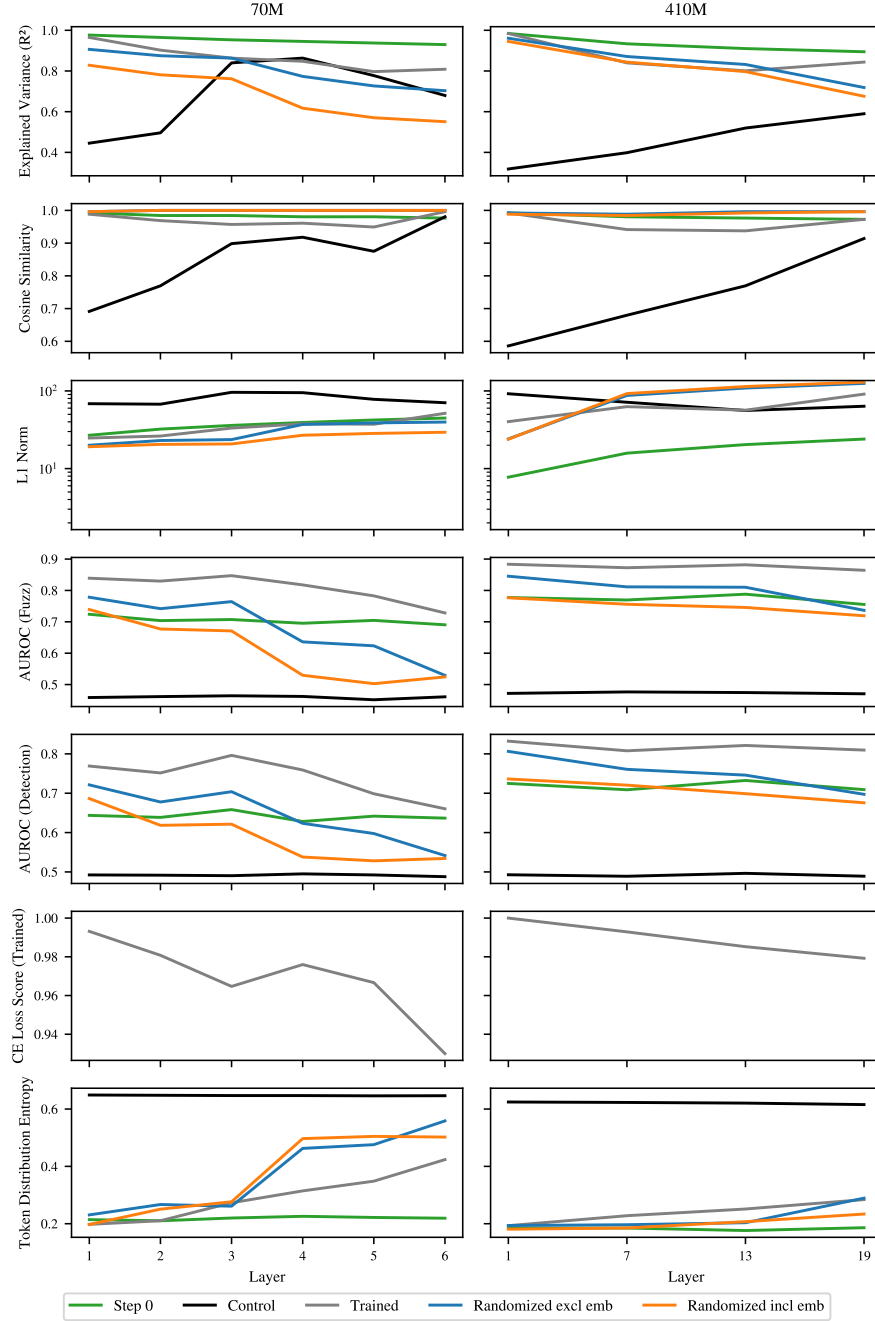


Figure 15: Evaluation metrics for SAEs trained on one billion tokens for the Pythia-70m and 410m models. These panels correspond to the respective columns of Figure 2, which report the same metrics for SAEs trained on 100M tokens; the behavior is qualitatively unchanged.

D EFFECT OF DECREASED TRAINING DATA FOR PYTHIA-1B

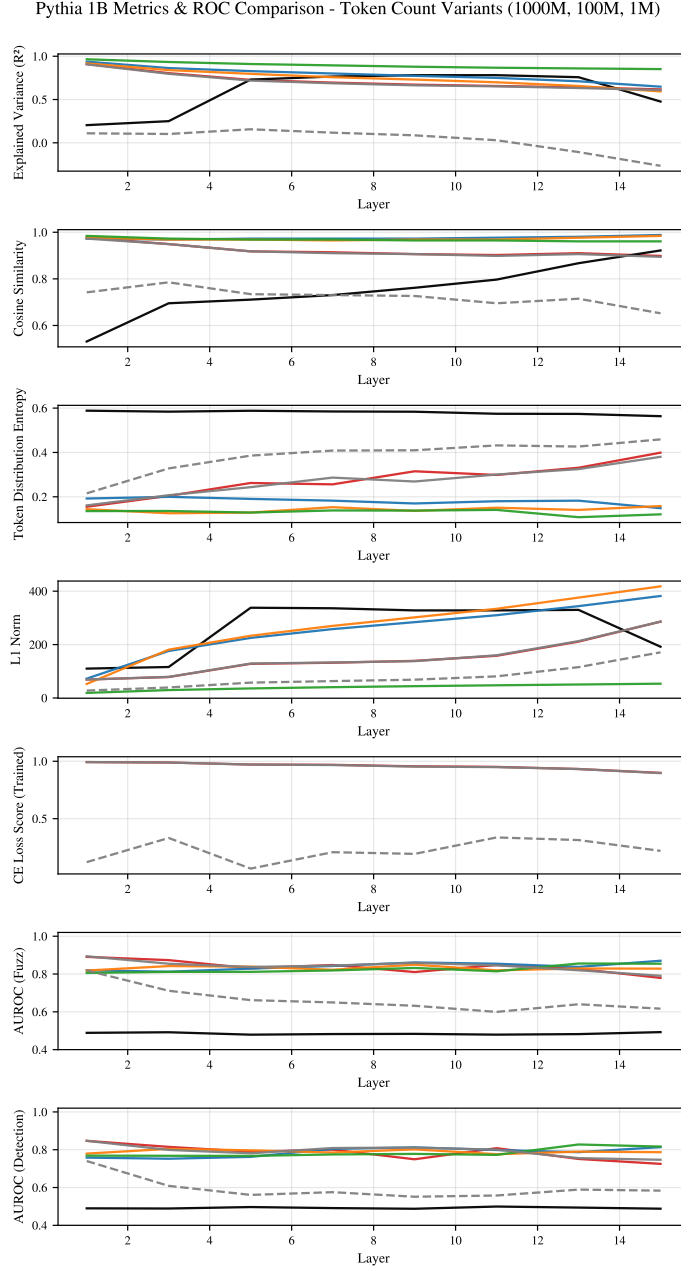
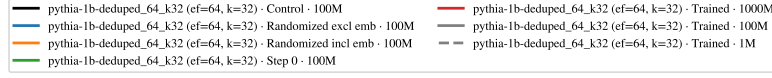


Figure 16: Evaluation metrics for SAEs trained on 1M and 1B tokens for Pythia-1b. The explained variance and CE loss score are substantially lower under the 1M-token budget, indicating under-trained SAEs. The mean auto-interpretability scores are marginally lower at the earliest layers and decline sharply with increasing layer index. The layer-wise trends in auto-interpretability and token-distribution entropy are consistent with those of the other SAEs.

1134 E UNCERTAINTY PLOTS FOR PYTHIA-70M

1135
1136 We quantified the uncertainty in our evaluation metrics for Pythia-70m by recomputing them across
1137 five independent random seeds.
1138
1139
1140
1141
1142
1143
1144
1145
1146
1147
1148
1149
1150
1151
1152
1153
1154
1155
1156
1157
1158
1159
1160
1161
1162
1163
1164
1165
1166
1167
1168
1169
1170
1171
1172
1173
1174
1175
1176
1177
1178
1179
1180
1181
1182
1183
1184
1185
1186
1187

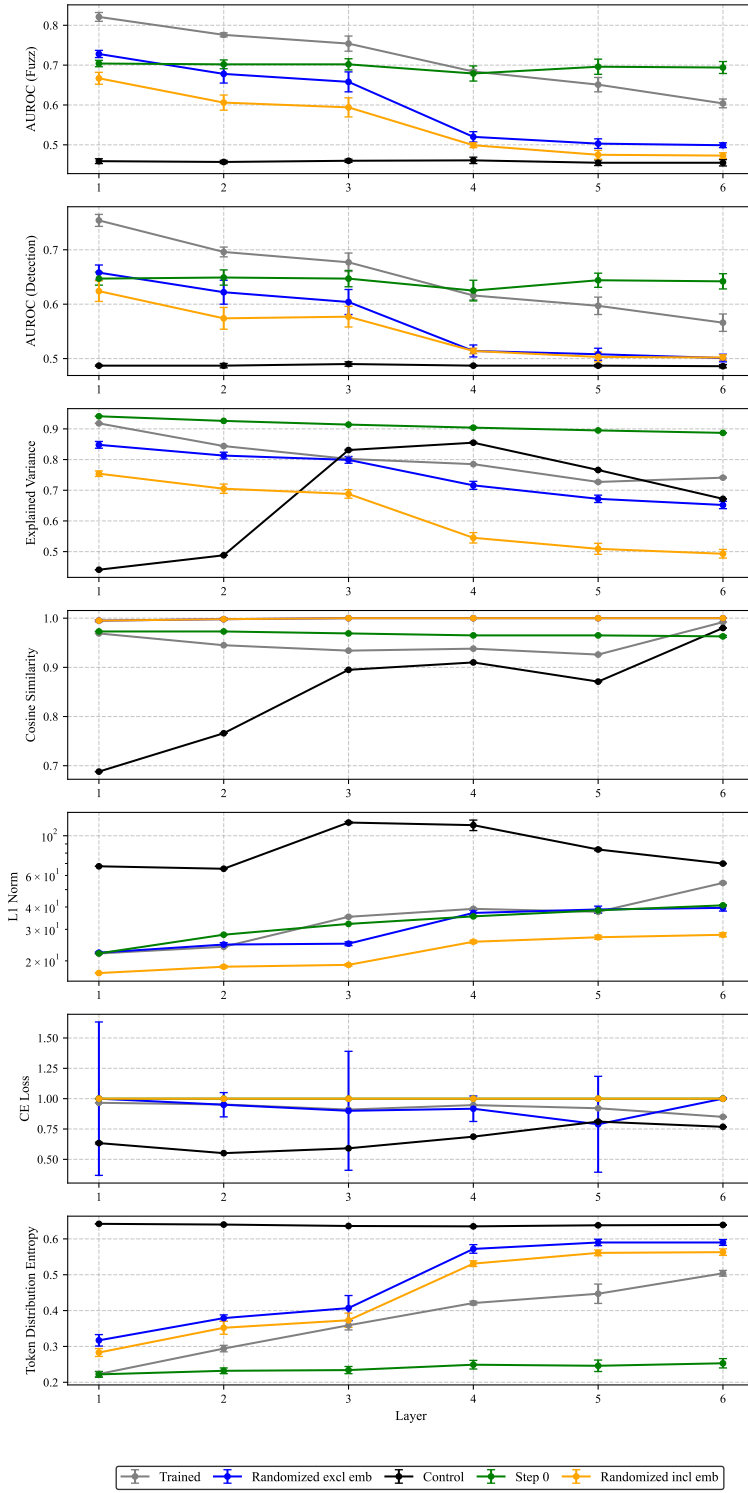


Figure 17: Uncertainty in the Pythia-70m evaluation metrics, estimated across five independent random seeds.

F EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-160M

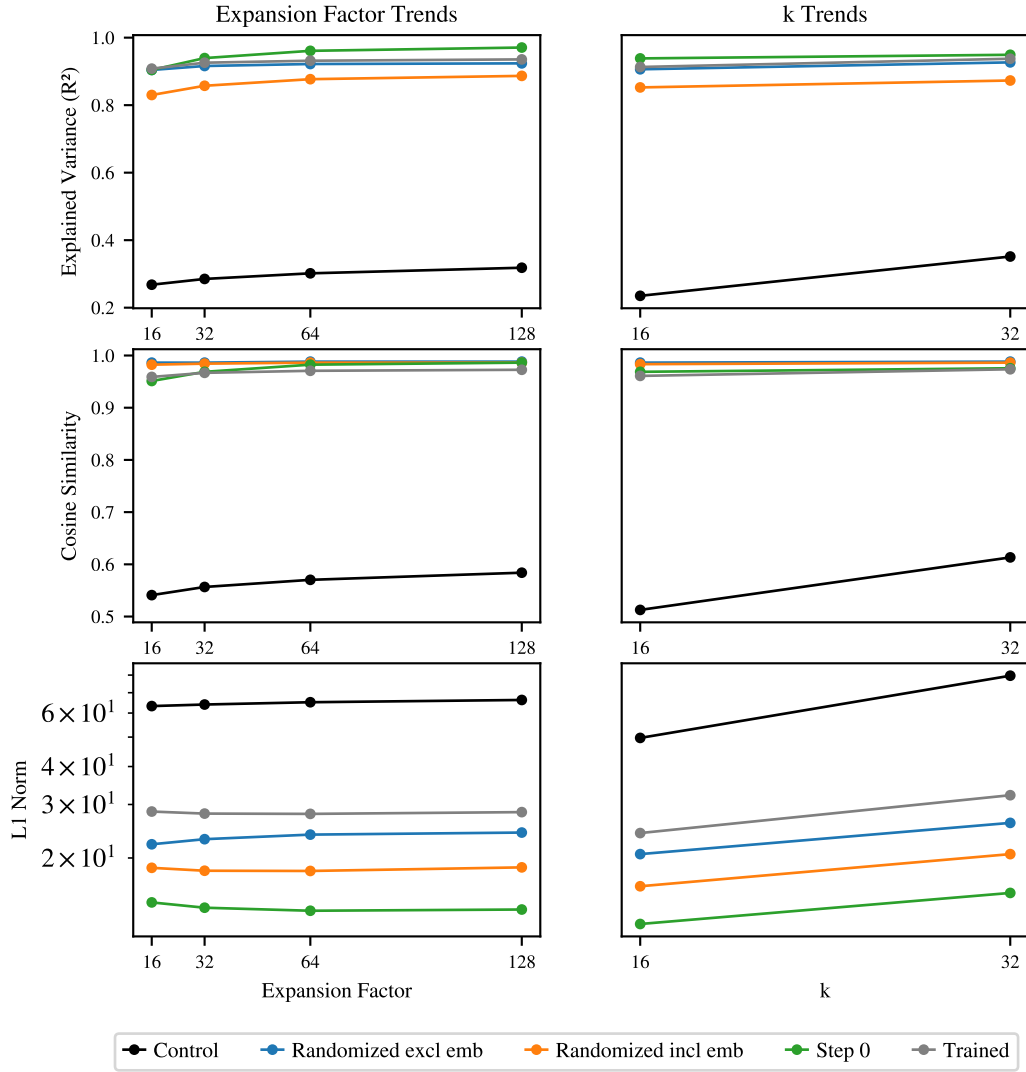


Figure 18: Robustness of SAE performance to hyperparameter selection. The standard evaluation metrics remain stable across expansion factors $R \in \{16, \dots, 128\}$ and sparsity levels $k \in \{16, 32\}$, and the relative ordering of the initialization strategies is preserved throughout. This stability indicates that moderate hyperparameter values (e.g., $R = 64$, $k = 32$) are sufficient.

G EFFECT OF SAE HYPERPARAMETERS FOR PYTHIA-1B

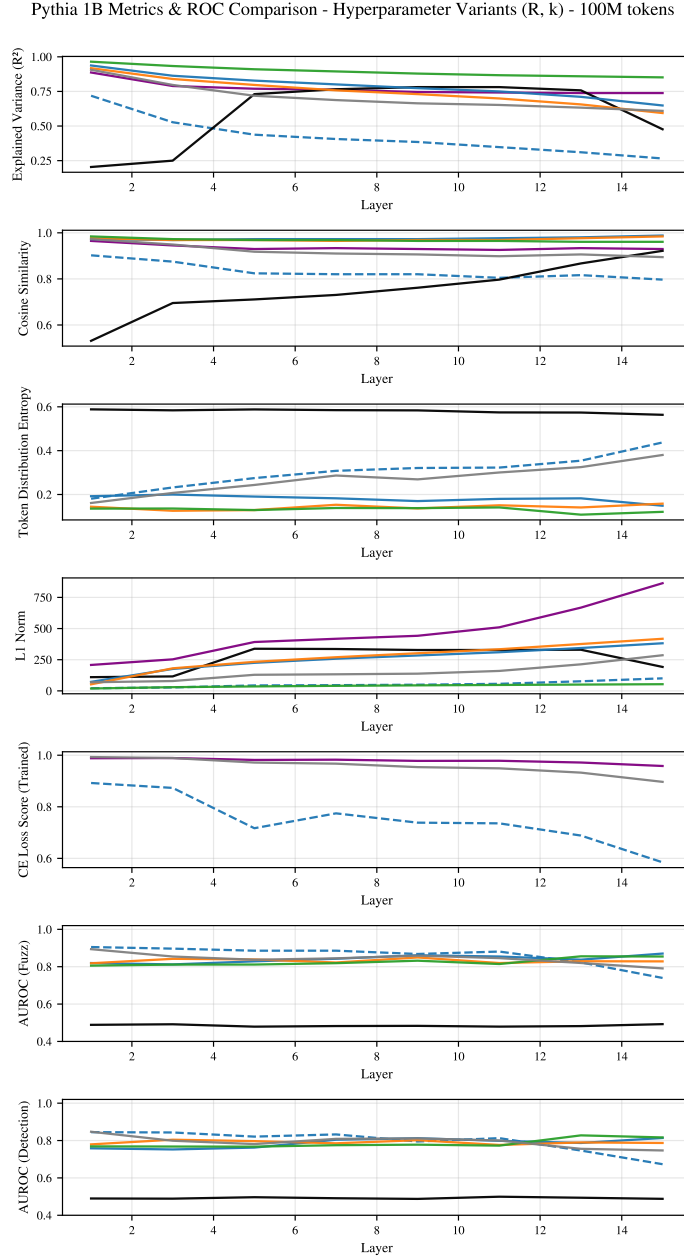


Figure 19: Evaluation metrics for SAEs trained on Pythia-1b under varying hyperparameters, alongside the main results reproduced from Figure 2. SAEs with a very small expansion factor $R = 2$ and sparsity $k = 4$ are clearly separated from the default configuration by both explained variance and CE loss score. Crucially, however, their auto-interpretability scores remain comparable to those of the default configuration, on both trained and randomized models.

H TOKEN DISTRIBUTION ENTROPY VS. AUTO-INTERPRETABILITY

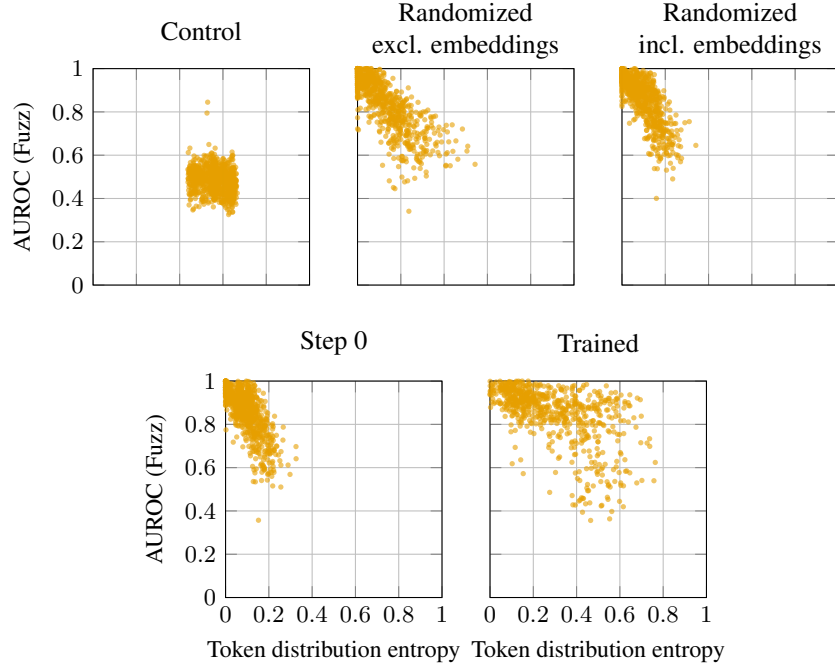


Figure 20: Per-latent token-distribution entropy versus ‘fuzzing’ AUROC (auto-interpretability score) for SAEs trained on multiple layers of Pythia-6.9b. Each point denotes a single latent, drawn from the same sample used to compute the aggregate metrics of Figures 1 and 2.

Figure 20 separates the control, randomized, and trained variants along both axes:

- **Control:** Latents exhibit uniformly high entropy—maximally activating examples with activation dispersed across many tokens—together with low auto-interpretability scores, i.e., generated explanations that fail to account for the observed activation patterns. The two quantities are uncorrelated.
- **Randomized:** For each randomization scheme of Section 3, entropy and auto-interpretability are negatively correlated: the greater the diversity of tokens on which a latent activates, the less adequately its activation pattern is captured by its generated explanation.
- **Trained:** The correlation is weaker. Critically, superimposed on the trend exhibited by the randomized variants, the trained variant additionally populates a regime of high entropy *and* high auto-interpretability: some latents activate across multiple tokens yet remain consistent with their generated explanation.

These observations support the interpretation that aggregate auto-interpretability scores mask the distinction between SAEs fit to trained versus randomized models. Randomized models with consistent token embeddings can produce readily explicable ‘single-token’ features, but only transformers trained on natural language additionally produce the more complex semantic features occupying the high-entropy, high-auto-interpretability regime.

I A TOY MODEL OF SUPERPOSITION

In Section 4 we trained SAEs on toy data constructed to exhibit superposition (Sharkey et al., 2022) and on GloVe word vectors (Pennington et al., 2014). This appendix specifies the data-generation procedure and the training configuration in full.

I.1 DATA GENERATION

We first construct the ground-truth features by sampling n_s points on an n_d -dimensional hypersphere.

The feature coefficients for each sample are then generated by the following sequence of operations: draw $A \in \mathbb{R}^{n_s \times n_s}$ with $A_{ij} \sim \mathcal{N}(0, 1)$; form the covariance $\Sigma = AA^T$; sample $\vec{\alpha} \in \mathbb{R}^{n_s}$ with $\alpha_i \sim \mathcal{N}(\vec{0}, \Sigma)$; map each α_i through the c.d.f. of $\mathcal{N}(0, 1)$; apply the index-dependent decay $\alpha_i \rightarrow \alpha_i^{\lambda_i}$ with $\lambda \in \mathbb{R}$; normalize via $\alpha_i \rightarrow m\alpha_i / n_s \sum_j \alpha_j$ with $m \in \mathbb{R}$; and perform n_s independent Bernoulli trials with success probabilities $p = \alpha_i$. The trial outcomes are finally multiplied elementwise by n_s independent draws from the continuous uniform distribution $\mathcal{U}_{[0,1]}$.

The parameter λ controls the rate at which the activation frequency of the ground-truth feature coefficients decays with feature index i , and m sets the expected number of nonzero feature coefficients per sample.

Following Sharkey et al. (2022), we set $n_s = 512$, $n_d = 256$, $\lambda = 0.99$, and $m = 5$. A Python implementation of the full procedure is given in Figure 21.

I.2 TRAINING

The SAEs of Section 4 consist of a linear encoder with a bias term, a ReLU nonlinearity, and a linear decoder without a bias term. The decoder weights are orthogonally initialized, and the decoder weight vectors are renormalized prior to each optimization step.

The training objective is the mean squared error (MSE) between the input and its decoded reconstruction, augmented by the mean L^1 norm of the encoded vectors scaled by a penalty coefficient swept over $[1 \times 10^{-3}, 100]$.

For the toy data we train for 100 epochs on 10K data points across 10 random seeds; for the word vectors we train for 100 epochs on 400K data points with a single random seed. In both cases we hold out 10% of the data points as a validation set on which the evaluation metrics are computed.

The MLPs of Section 4 comprise two layers (one hidden layer) with a ReLU nonlinearity, input and output dimensionality both equal to n_d , and hidden width $4n_d$. These dimensions are chosen in loose analogy to the feed-forward network blocks of transformer language models.

```

1458
1459
1460
1461
1462
1463
1464
1465
1466
1467
1468
1469
1470 1 def generate_sharkey(
1471 2     num_samples: int,
1472 3     num_inputs: int,
1473 4     num_features: int,
1474 5     avg_active_features: float,
1475 6     lambda_decay: float,
1476 7 ) -> tuple[Tensor, Tensor]:
1477 8     """
1478 9     Args:
1479 10         num_samples (int): The number of samples to generate.
1480 11         num_inputs (int): The number of input dimensions.
1481 12         num_features (int): The number of ground truth features.
1482 13         avg_active_features (float): The average number of
1483 14             ground truth features active at a time.
1484 15         lambda_decay (float): The exponential decay factor for
1485 16             feature probabilities.
1486 17     """
1487 18     features = torch.randn(num_inputs, num_features)
1488 19     features /= torch.norm(features, dim=0, keepdim=True)
1489 20
1490 21     covariance = torch.randn(num_features, num_features)
1491 22     covariance = covariance @ covariance.T
1492 23     correlated_normal = MultivariateNormal(
1493 24         torch.zeros(num_features), covariance_matrix=covariance
1494 25     )
1495 26
1496 27     samples = []
1497 28     for _ in range(num_samples):
1498 29         p = STANDARD_NORMAL.cdf(correlated_normal.sample())
1499 30         p = p ** (lambda_decay * torch.arange(num_features))
1500 31         p = p * (avg_active_features / (num_features * p.mean()))
1501 32         p = torch.bernoulli(p.clamp(0, 1))
1502 33         coef = p * torch.rand(num_features)
1503 34
1504 35         sample = coef @ features.T
1505 36         samples.append(sample)
1506 37
1507 38     return torch.stack(samples), features

```

Figure 21: Python implementation of the data-generation procedure of Sharkey et al. (2022), as employed in Section 4.

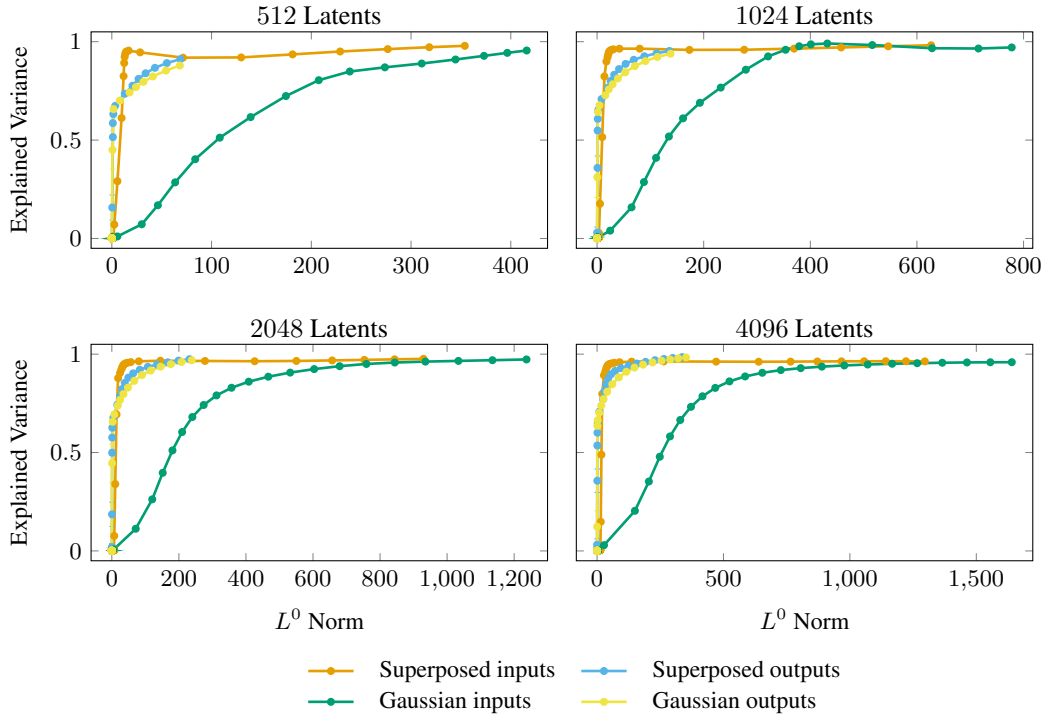


Figure 22: Pareto frontiers of explained variance against the L^0 norm (sparsity) for the toy datasets constructed to exhibit superposition, the Gaussian controls matched in mean and variance, and the corresponding outputs obtained by passing each through a randomly initialized two-layer MLP.

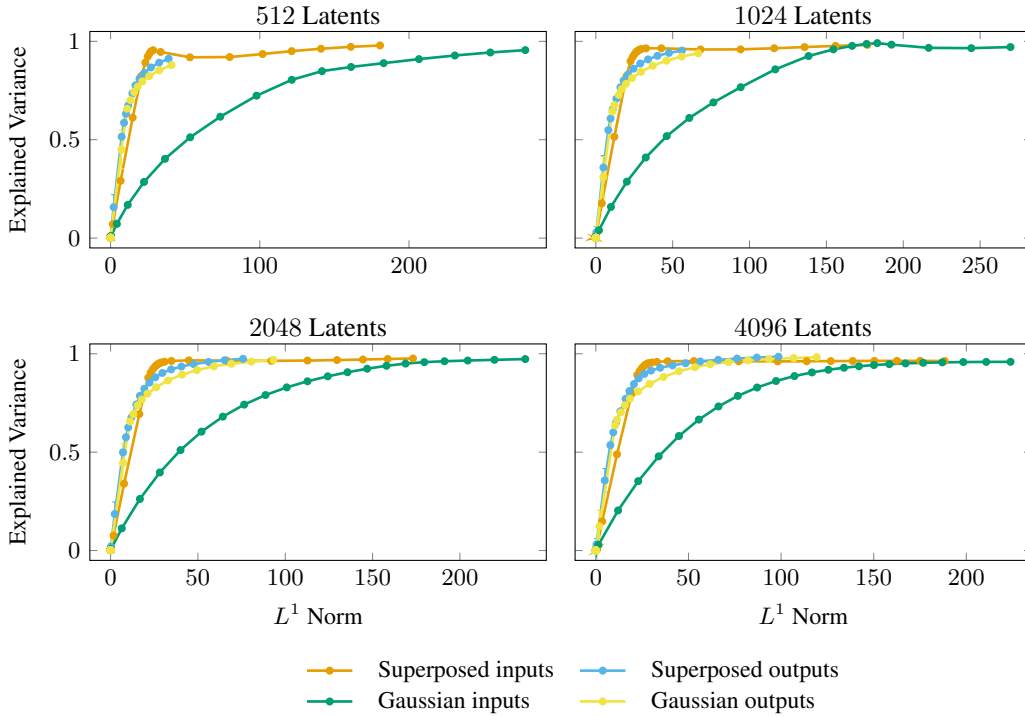


Figure 23: Pareto frontiers of explained variance against the L^1 norm (sparsity) for the toy datasets constructed to exhibit superposition, the Gaussian controls matched in mean and variance, and the corresponding outputs obtained by passing each through a randomly initialized two-layer MLP.

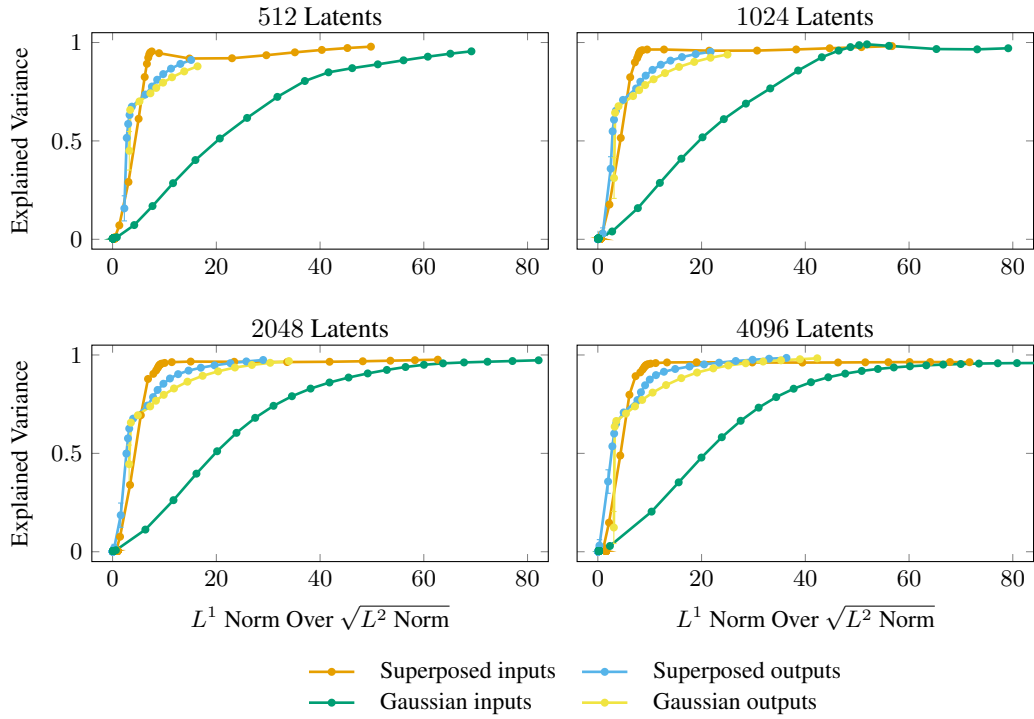


Figure 24: Pareto frontiers of explained variance against the sparsity measure $L^1/\sqrt{L^2}$ for the toy datasets constructed to exhibit superposition, the Gaussian controls matched in mean and variance, and the corresponding outputs obtained by passing each through a randomly initialized two-layer MLP.

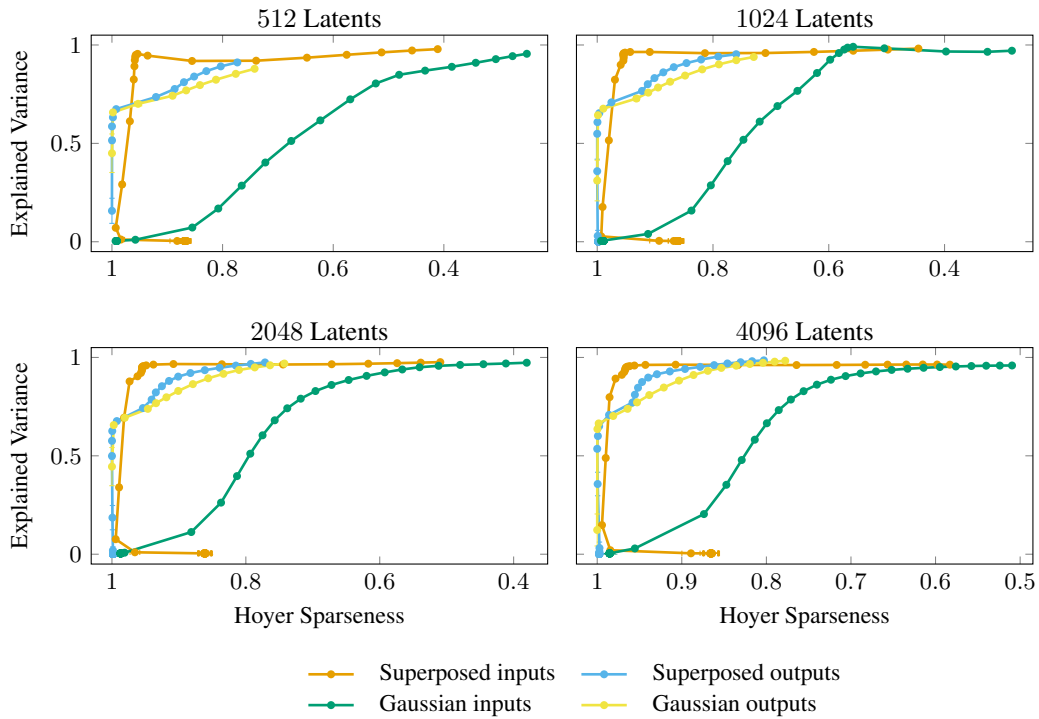


Figure 25: Pareto frontiers of explained variance against Hoyer sparseness (sparsity) for the toy datasets constructed to exhibit superposition, the Gaussian controls matched in mean and variance, and the corresponding outputs obtained by passing each through a randomly initialized two-layer MLP.

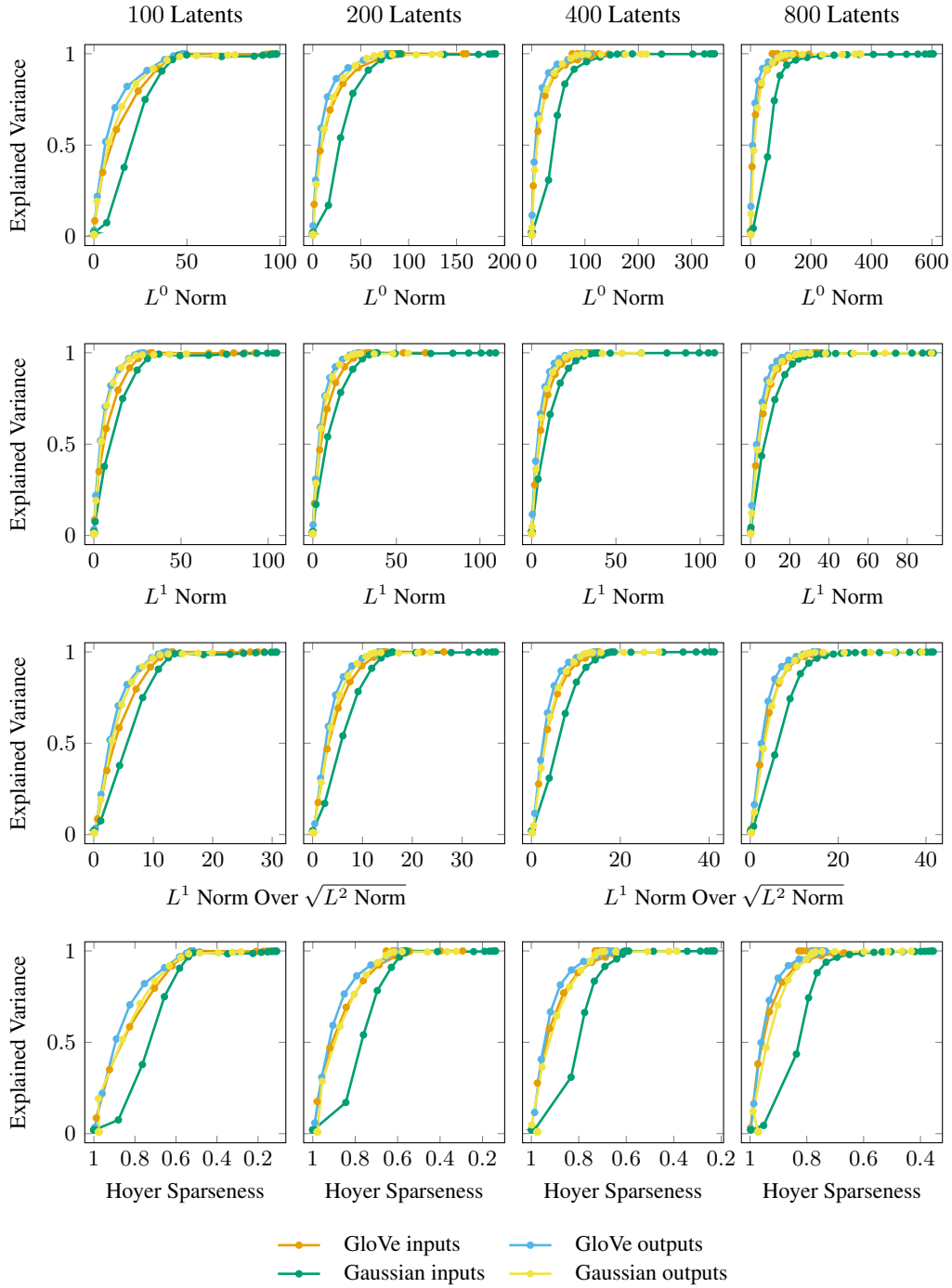


Figure 26: Pareto frontiers of explained variance against sparsity measures for 50-dimensional GloVe word vectors, the Gaussian controls matched in mean and variance, and the corresponding outputs obtained by passing each through a randomly initialized two-layer MLP.

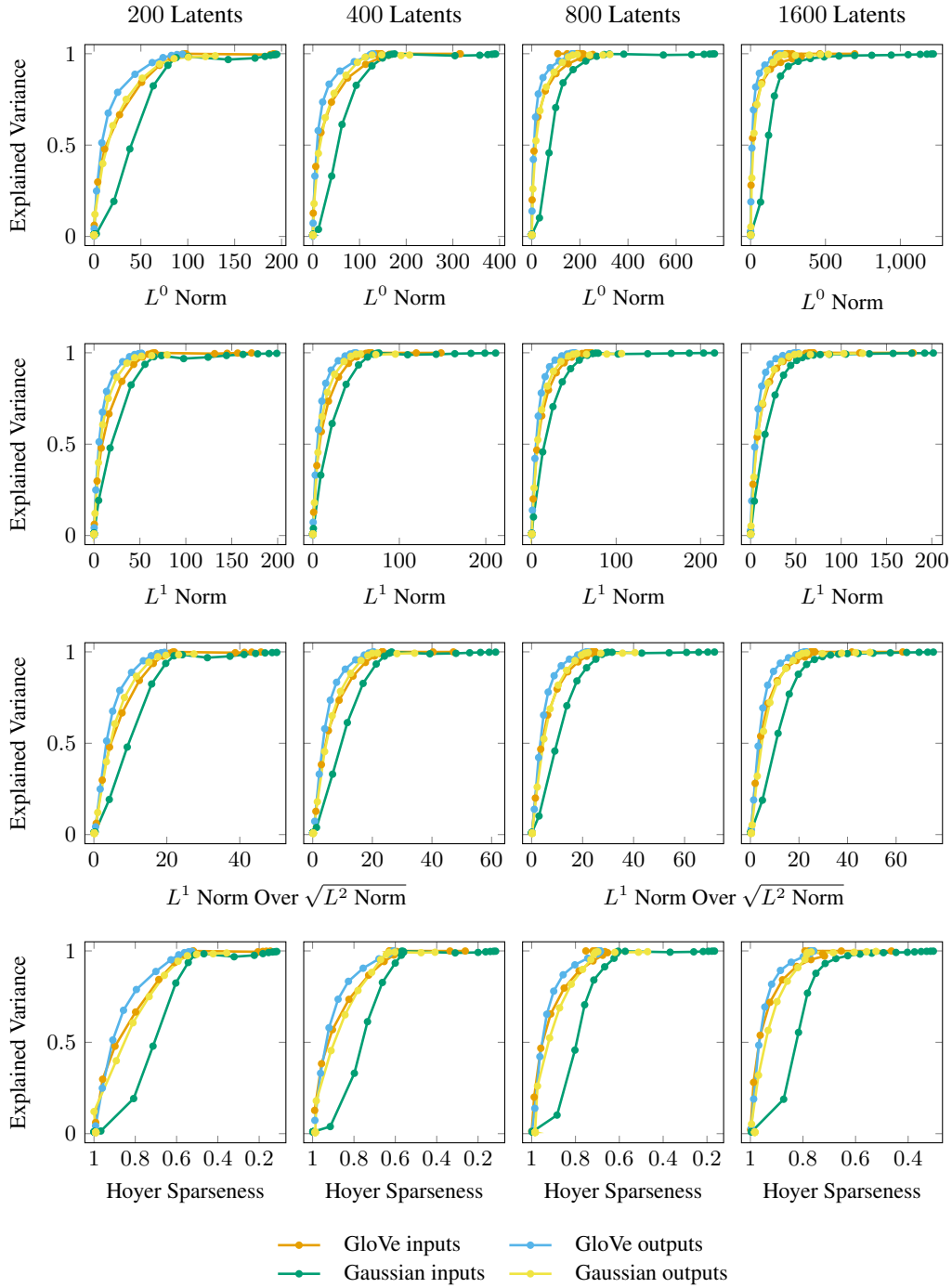


Figure 27: Pareto frontiers of explained variance against sparsity measures for 100-dimensional GloVe word embeddings, the Gaussian controls matched in mean and variance, and the corresponding outputs obtained by passing each through a randomly initialized two-layer MLP.

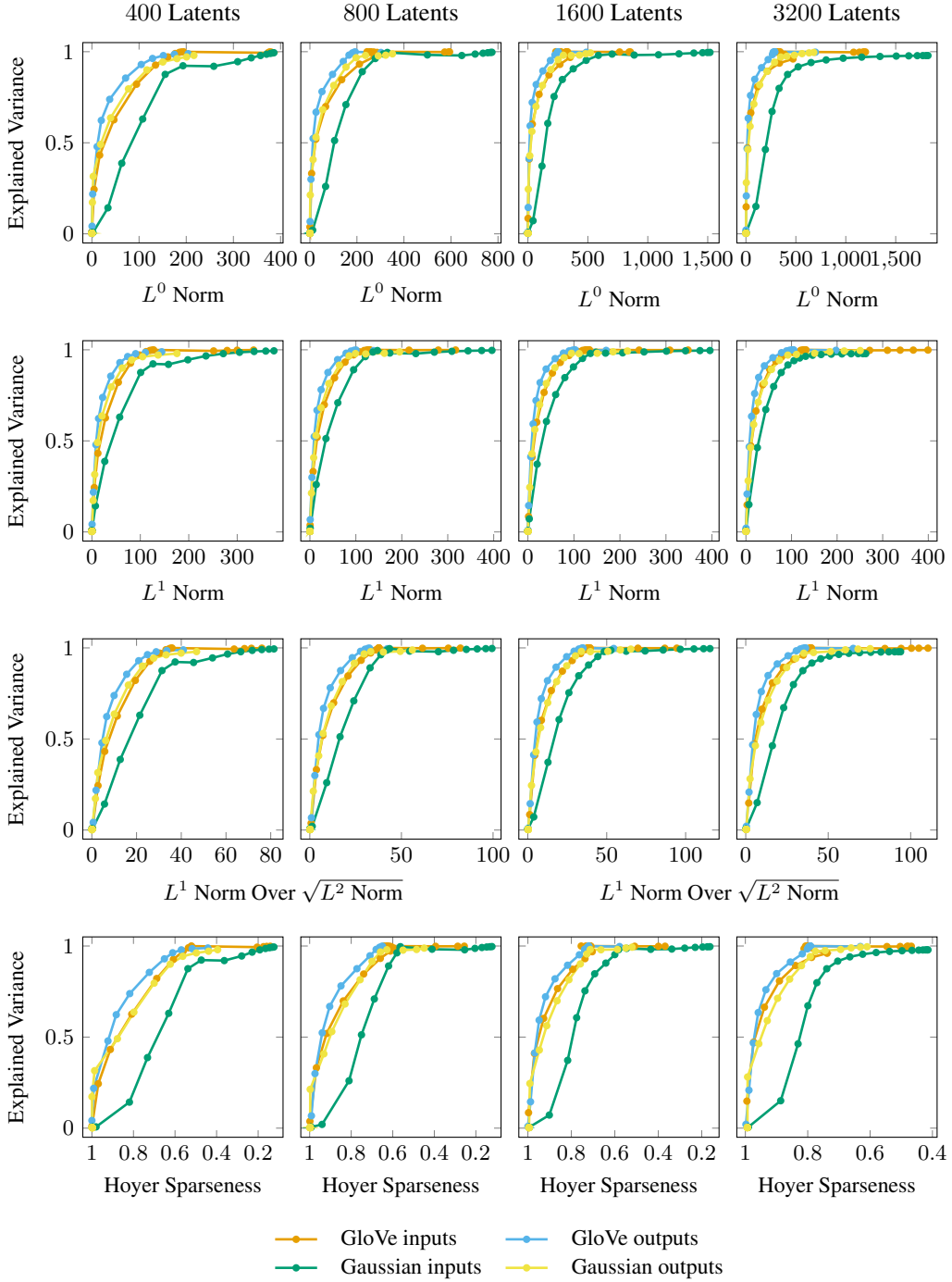


Figure 28: Pareto frontiers of explained variance against sparsity measures for 200-dimensional GloVe word embeddings, the Gaussian controls matched in mean and variance, and the corresponding outputs obtained by passing each through a randomly initialized two-layer MLP.

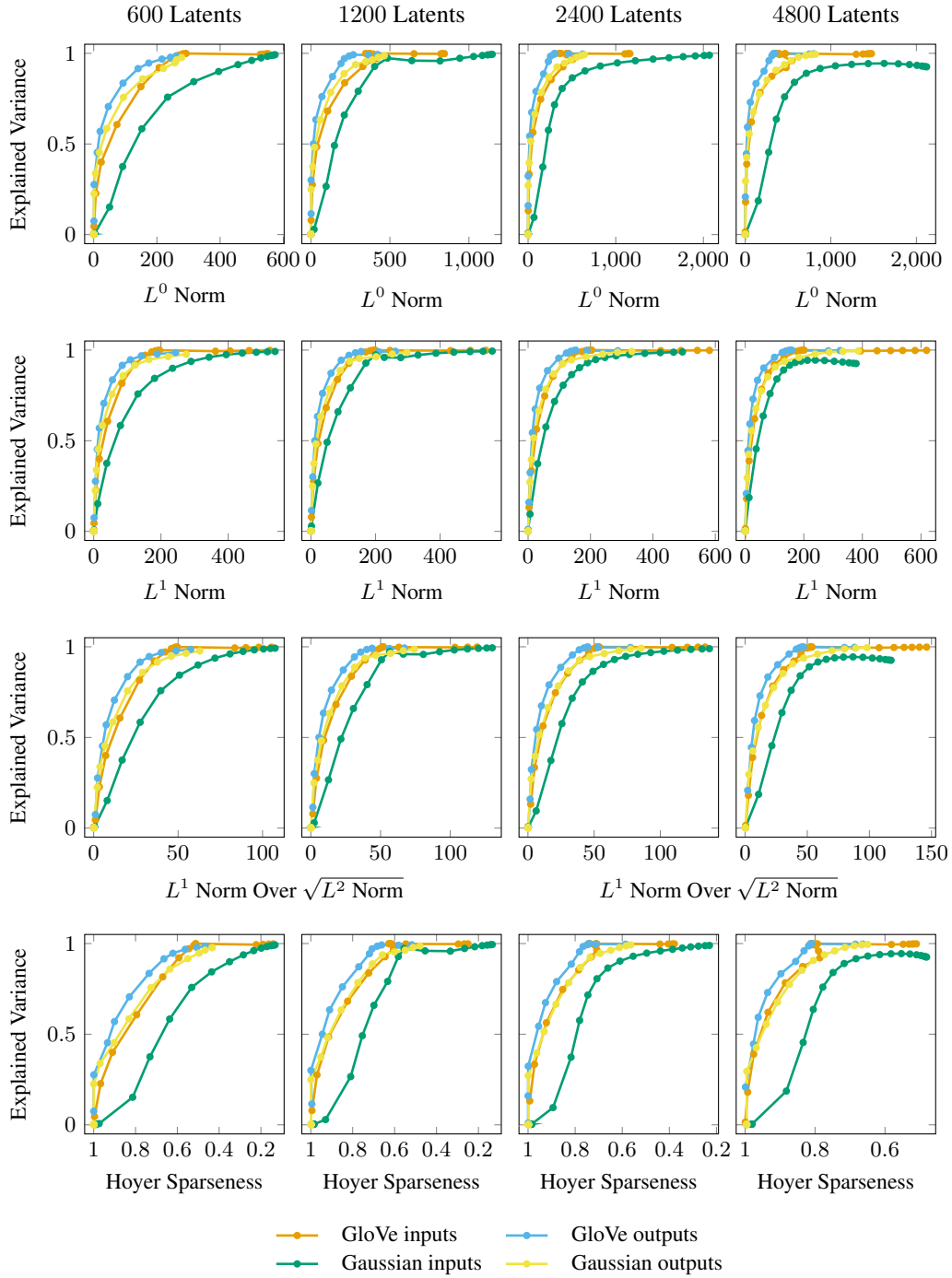


Figure 29: Pareto frontiers of explained variance against sparsity measures for 300-dimensional GloVe word embeddings, the Gaussian controls matched in mean and variance, and the corresponding outputs obtained by passing each through a randomly initialized two-layer MLP.

J EXAMPLE FEATURES FOR PYTHIA-6.9B

VARIANT: TRAINED

FEATURE 180935 (LAYER 0)

Interpretation: The term "security" is predominantly used to refer to protection, safety, and measures to prevent harm, while "oz" is likely referring to ounces, possibly in a context of measurement or quantification, although "oz" appears less frequently and often in a different context.

Top Examples:

1. Text: <endoftext>—ζ—. If for any reason you are unhappy with our service please contact us directly so we can make it right for you. Journal of Cyber Security, Vol.
Activation: 4.3750
Active tokens: Security
2. Text: <endoftext>—ζ— Security Practitioner. Tremendously passing CompTIA Advanced Security Practitioner (casp) cert has never been as easy as it
Activation: 4.3125
Active tokens: Security Security
3. Text: in trusted hands for your Cyber Security career or staffing needs. Call 0203 643 0248 to find out more. Technically proficient using
Activation: 4.3125
Active tokens: Security

FEATURE 93790 (LAYER 8)

Interpretation: Nouns and phrases related to economic concepts, development, and business, often referring to growth, progress, and improvement.

Top Examples:

1. Text: training requirements. See "Workforce" section for additional information. The Economic Development Transportation Fund, commonly referred to as the "Road Fund," is an
Activation: 21.3750
Active tokens: Development
2. Text: Montréal. The Williamsburg Economic Development Authority offers a 33% matching grant up to \$7,500 for exterior improvements to existing businesses in the City of
Activation: 20.2500
Active tokens: Development Authority
3. Text: Correction: In a July 16 web story The Real Deal incorrectly stated that the Economic Development Corporation was "circumventing" laws with its restructuring. In
Activation: 20.0000
Active tokens: Development

FEATURE 128309 (LAYER 12)

Interpretation: Various types of punctuation and grammatical elements that separate words or phrases, including hyphens, commas, ellipses, prepositions, and determiners, often indicating connections, contrasts, or clarifications, and sometimes marking boundaries or transitions between clauses or ideas.

Top Examples:

1. Text: Run it in JDK6, and it will print "[axons, bandrils, chumblies]". If you are having trouble switching from
Activation: 8.0000
Active tokens: in JD K

-
- 1944 Text: Here, we introduce the coordinate systems for three-dimensional space □□□2. The study
1945 of 3-dimensional spaces lead us to the setting for our study
1946 2. Activation: 7.8125
1947 Active tokens: □
1948 Text: .path.expanduser("~/malwarehouse/") because this server doesn't have X-Windows running.
1949 If you are looking for a simple and
1950 3. Activation: 7.7188
1951 Active tokens: . path expand user
-

1952
1953
1954
1955 VARIANT: STEP 0
1956

1957 FEATURE 126848 (LAYER 12)
1958

1959 **Interpretation:** Nouns denoting people who train others, units or marks of measurement, and
1960 abbreviations or acronyms representing specific standards or technologies.
1961

1962 **Top Examples:**

- 1963 Text: What are the various lessons a member can access at a tennis club? Whether you are a
1964 beginner or advanced player, trainers help you to choose the right gaming
1965 1. Activation: 13.1250
1966 Active tokens: trainers
1967 Text: report include various simulation platforms and Serious Games. The report also analyzes
1968 some major allied products such as patient simulators and task trainers. The technologies
1969 2. analyzed
1970 Activation: 13.0625
1971 Active tokens: trainers
1972 Text: a stylish spring in your step when you buy from our fantastic range of men's and women's
1973 Asics trainers. We've got numerous styles from
1974 3. Activation: 13.0000
1975 Active tokens: trainers
-

1976
1977
1978 FEATURE 2125 (LAYER 4)
1979

1980 **Interpretation:** The word "papers" is often used in contexts referring to written documents, such as
1981 academic papers, court documents, or printed materials, and is frequently mentioned in relation to
1982 tasks like writing, research, and education.
1983

1984 **Top Examples:**

- 1985 Text: caustic solution . As an abrasive, alumina is coated into abrasive papers and .. Pakistan.
1986 Sierra leone. Taiwan. Turkey. Venezuela.
1987 1. Activation: 5.7188
1988 Active tokens: papers
1989 Text: that can be associated with interaction with other individuals. For everybody who is
1990 uncertain regardless of whether your papers is misstep no cost, buy inexpensive experienced
1991 2. proofreading services
1992 Activation: 5.6875
1993 Active tokens: papers
1994 Text: who RV, often traveling in groups, often alone. You just want to have all the papers like
1995 RC, licence and insurance coverage as effectively as PUC (
1996 3. Activation: 5.6562
1997 Active tokens: papers
-

1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2027
2028
2029
2030
2031
2032
2033
2034
2035
2036
2037
2038
2039
2040
2041
2042
2043
2044
2045
2046
2047
2048
2049
2050
2051

FEATURE 9944 (LAYER 16)

Interpretation: Words or parts of words that are usually the beginning or end of a proper noun, surname, or a word of foreign origin.

Top Examples:

- Text: astic gestures.Home Music World News Music the artform Where Do Music Festivals Go Now? Where Do Music Festivals Go Now? Are you ready
1. Activation: 10.4375
Active tokens: Fest Fest

Text: client streams, and encouraging existing customers to become more involved.Welcome to Fil Fest USA!! Do you love lumpia? Can you eat a handful of them

 2. Activation: 10.3750
Active tokens: Fest

Text: Powers talking about his early inspirations. In response to San Diego Comic Fest 2012!: Off to Comic Fest 2012. Should be interesting if nothing else!

 3. Activation: 10.3750
Active tokens: Fest Fest
-

VARIANT: RANDOMIZED EXCLUDING EMBEDDINGS

FEATURE 151030 (LAYER 28)

Interpretation: Common nouns, proper nouns, or adjectives found in various contexts, including but not limited to geographical locations, people, organizations, time, and concepts, often possessing relevance to the surrounding text.

Top Examples:

- Text: ion Patch Kills Owner, Son,”, Los Angeles Times, June 12, 1994, <http://articles.latimes.com/1994-06-12>
1. Activation: 58.0000
Active tokens: lat

Text: hard-headed coin of the realm their look. Firstly you’ve got to inventory on incident unique a distinct blunt in a retailer you superlativeness be

 2. Activation: 56.5000
Active tokens: lat

Text: Jr’s new film Dovlatov follows the life of the now celebrated writer Sergei Dovlatov over six days in 1971, as he struggles

 3. Activation: 55.7500
Active tokens: lat lat
-

FEATURE 98924 (LAYER 12)

Interpretation: Adjectives describing size, or nouns representing concepts or objects that are being described in terms of their size.

Top Examples:

- Text: . The area to the right is for large dogs (small dogs also welcomed) however the area to the left is for small dogs only. Troup 69
1. Activation: 46.0000
Active tokens: small

Text: ,I would not mind, but has to pretty less expensive. Can it use any windows aplication???? Or I am really need a cool,small

 2. Activation: 45.5000
Active tokens: small

2052 Text: 3/4” – 8 1/2” rather than strictly 8 1/4”) I chose the “small” version, though I should be a
2053 3. Activation: 44.7500
2054 Active tokens: small

2055
2056
2057 FEATURE 180589 (LAYER 24)

2058
2059 **Interpretation:** Nouns mostly referring to tasks, responsibilities or jobs to be accomplished, often in
2060 a professional or organizational context, sometimes accompanied by proper nouns and a few instances
2061 with words having suffixes or prefixes.

2062 **Top Examples:**

- 2063
- 2064 Text: completing tasks or a captcha, users are awarded by GRSfractions. Why These Groestlcoin
2065 1. Faucets provide rewards? Many people
2066 Activation: 130.0000
2067 Active tokens: tasks
 - 2068 Text: from Groestlcoin Faucets is In the exchange of completing tasks or a captcha, users are
2069 2. awarded by Free GRS. To Earn
2070 Activation: 129.0000
2071 Active tokens: tasks
 - 2072 Text: and still contain a small remnant circular genome, known as mitochondrial DNA. Of the
2073 3. varied tasks undertaken by mitochondria, the most important is the generation of the chemical
2074 energy
2075 Activation: 128.0000
2076 Active tokens: tasks
-

2077
2078
2079 VARIANT: RANDOMIZED INCLUDING EMBEDDINGS

2080
2081 FEATURE 39748 (LAYER 0)

2082
2083 **Interpretation:** Words contain “Pul” are often used in the context of Pulitzer, a prestigious journalism
2084 award, while “Looking” typically precedes a phrase expressing anticipation, expectation, or searching
2085 for something.

2086 **Top Examples:**

- 2087
- 2088 Text: <endoftext—¿— to the 21st Century. Rhodes won the Pulitzer prize for The Making of
2089 1. the Atomic Bomb (01987) his first of four books chronicling the
2090 Activation: 3.3750
2091 Active tokens: Pul
 - 2092 Text: <endoftext—¿—, James Coburn, movies we love, Pulp Consumption, Steve McQueen,
2093 2. Western, Yul Brynner. Bookmark the permal
2094 Activation: 3.3438
2095 Active tokens: Pul
 - 2096 Text: Crusher, Coal Mill and Coal Pulverizer for sale Coal crusher and coal mill is the major
2097 3. mining equipment in . sbm ceramic machinary -
2098 Activation: 3.2969
2099 Active tokens: Pul
-

2100
2101 FEATURE 15633 (LAYER 20)

2102
2103 **Interpretation:** Nouns representing individuals or entities possessing or having authority over
2104 something, often in a possessive or authoritative relationship with that thing.

2105 **Top Examples:**

-
- 2106 Text: poetry. The Starkville/Mississippi State University Symphony Orchestra kicks off 2012
 2107 with a Jan. 21 concert dedicated to parents of the performing musicians. The free
 2108 1. Activation: 80.0000
 2109 Active tokens: Stark
- 2110 Text: Baptist. Kris Kirkwood, Stark Raving Solutions' lighting designer, says the architectural
 2111 system used, ETC's Paradigm architectural control, is
 2112 2. Activation: 77.5000
 2113 Active tokens: Stark
- 2114 Text: so you can be as cool as Tony Stark. The Marvel Training Academy will be taking place
 2115 throughout May, just check with your local shop to guarantee your place
 2116 3. Activation: 77.5000
 2117 Active tokens: Stark
-

2118
 2119

2120 FEATURE 6069 (LAYER 4)

2121

2122 **Interpretation:** Proper nouns, nouns referring to objects or places, and nouns with strong semantic
 2123 connotations often related to religion or technology.

2124 **Top Examples:**

- 2125 Text: in production include Bullfinch's Mythology: Age of Fable, The Story of Dr. Doolittle,
 2126 and a collection of Hans Christian Anderson fairy
 2127 1. Activation: 22.1250
 2128 Active tokens: Christian
- 2129 Text: reaction of the remaining flock remains the same: ostracism, shunning, even retaliation.
 2130 So yeah, Christian leaders won't make any big
 2131 2. Activation: 22.0000
 2132 Active tokens: Christian
- 2133 Text: was one of the best loved characters in the film. Walt Disney attempted as far back as 1937
 2134 to adapt the Hans Christian Anderson fairy tale, The Snow Queen into
 2135 3. Activation: 22.0000
 2136 Active tokens: Christian
-

2137
 2138

2139 VARIANT: CONTROL

2140

2141 FEATURE 290 (LAYER 4)

2142

2143 **Interpretation:** Function words and occasionally nouns or proper nouns that seem to be emphasized
 2144 as part of a larger phrase or topic, often indicating transition or conjunction.

2145 **Top Examples:**

- 2146 Text: <endoftext—¿—ance - Chapters: 1 - Words:. Fruits Basket - Rated: T - English -
 2147 Romance/Angst - Chapters:
 2148 1. Activation: 5.0938
 2149 Active tokens: -
- 2150 Text: ococcus neoformans-reactive and total immunoglobulin profiles of human immunodeficiency virus-infected and uninfected Ugandans'. Clinical and Diagnostic Laboratory Immunology, Vol 12
 2151 2. Activation: 5.0938
 2152 Active tokens: un
- 2153 Text: and in fact any correspondence that the social club had in the run-up to the sit-in was from
 2154 the social club's own solicitors.
 2155 3. Activation: 5.0625
 2156 Active tokens: the
-

2160
2161
2162
2163
2164
2165
2166
2167
2168
2169
2170
2171
2172
2173
2174
2175
2176
2177
2178
2179
2180
2181
2182
2183
2184
2185
2186
2187
2188
2189
2190
2191
2192
2193
2194
2195
2196
2197
2198
2199
2200
2201
2202
2203
2204
2205
2206
2207
2208
2209
2210
2211
2212
2213

FEATURE 176433 (LAYER 24)

Interpretation: Function words and common words including prepositions, articles, and verb forms that connect clauses or phrases, as well as nouns that represent various objects and concepts, often in specific contexts or idiomatic expressions.

Top Examples:

1. Text: forgo insurance. Ultimately, that choice is up to you. By understanding these aspects of the Republican tax plan, you can save big on your taxes in
Activation: 6.9062
Active tokens: taxes
 2. Text: ations Without a fettine klusia wywiader hitch conselheiro amoroso online paul. In France, Germany, Belgium, Luxem
Activation: 6.8438
Active tokens: wi
 3. Text: K-ras oncogene and also via mutations in BRAF. Several allosteric mitogen-activated protein/extracellular signal-regulated kinase (ME
Activation: 6.5000
Active tokens: rac
-

FEATURE 203901 (LAYER 20)

Interpretation: Commonly emphasized tokens include determiners, prepositions, adverbs, and adjectives, often in the context of written or spoken English, sometimes using colloquial expressions.

Top Examples:

1. Text: was an avid reader and a fantastic cook. Susan was a brave and courageous woman who battled MS for over 40 years. Even given the limitations of her
Activation: 9.1875
Active tokens: given
 2. Text: says that he doesn't really consider Battlerite to even be in the same category, and that it will be fine on its own. Well I
Activation: 9.1250
Active tokens: to
 3. Text: to see a dime of the funds. The transaction occurred mere hours before the doomed exchange stopped honoring withdrawals. Tsao sold nearly 20 bit
Activation: 9.1250
Active tokens: .
-

K COMPUTE DETAILS

All experiments were executed on a single NVIDIA A100 80GB GPU in a private cluster. Table 1 reports the approximate wall-clock duration of the final experiments, broken down by model size and transformer variant. We estimate the aggregate cost of preliminary and failed runs to be of the same order as that of the final experiments.

Model	Variants	Approx. time per variant (hours)	Total time (hours)
Pythia-6.9b	5	70	350
Pythia-1b	5	10	50
Pythia-410m	5	5	25
Pythia-160m	5	1	5
Pythia-70m	5	1	5
Overall time:			435

Table 1: Approximate wall-clock time required for our experiments, by model size.

L EXAMPLE FEATURE DASHBOARDS FOR PYTHIA-6.9B

This appendix provides more detailed ‘feature dashboards’ reporting, per latent, the activation pattern, the token-distribution entropy, and the auto-interpretability (‘fuzz’ ROC) score. For each of the control, randomized, and trained variants of Section 3, we display two uniformly sampled latents from every fourth layer of Pythia-6.9b.

L.1 TRAINED

Feature 10065 (Layer 0) - Trained

Entropy: 0.009 | Fuzz ROC: 0.961

Interpretation: Prefixes or words starting with "Re" often indicating repetition, return, or renewal.

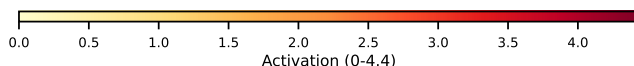
Ex 1 which Judge **Kreep** was censured, Brower says he has worked in diverse work places in the military and D.A.'s office, also working with

Ex 2 Example: In Louros v. **Kreikas**, 367 F. Supp. 2d 572 (S.D.N.Y. 2005), the

Ex 3 the Alhambra's arches. Over the years, **Kreber** has supplied the color separations, while printing services were provided by Century Graphics,

Ex 4 treated equal. What was the Statue of Liberty originally used for? Shadows over **Kregen** Schatten über **Kregen**, 1996; English ebook edition

Ex 5 by Boston attorney Arthur **Kreiger**, who represented AT&T. Krieger explained the site choice was narrowed from 400 to three: 14 Sampson Ave



Feature 10222 (Layer 0) - Trained

Entropy: 0.050 | Fuzz ROC: 0.970

Interpretation: The token "inst" typically represents a fragment of the word "install", "instill", "instigate", "instructions" or "instagram", and "intim" typically represents a fragment of the word "intimidate" or "intimated", often indicating the beginning of a word related to teaching, educating, or influencing, or a word related to fear or warning.

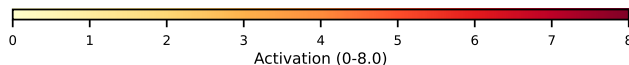
Ex 1 <|endoftext|>**intin** villa. Dua kamar tidur yang memiliki akses langsung ke kolam renang. Di setiap kamar

Ex 2 the non-Muslim world more and more fold under their legal **intimidation** as a result of our pacifism, self-hatred and complacency.

Ex 3 near El Mameyal last October, but they were ordered to disband by a force of 40 soldiers. The campaign of **intimidation** may have worked

Ex 4 least in part to the attempt by the Railroad Commission of Arkansas to protect Arkansas shippers and build up Arkansas jobbing centers.' In that case it was **intimated**

Ex 5 than half of those against people were assault cases, while nearly 45 per cent were crimes of **intimidation**. 'No person should have to fear being violently attacked

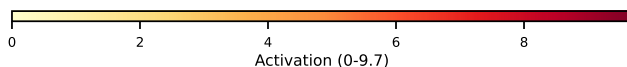


Feature 100186 (Layer 4) - Trained

Entropy: 0.096 | Fuzz ROC: 0.993

Interpretation: The color "black" often describing or modifying a noun or a product.

- Ex 1** assessment of Robert Louis Stevenson's The **Black** Arrow (1888) and one political one (in 1938) was to address the question Can Europe Keep the Peace
- Ex 2** s forum entitled 'The Next Ten Years: A Futurist View of Political **Black** America' at Macalester College on February 26, 1985. The event
- Ex 3** collapse, the increasing racial inequity and highprofile police killings of unarmed **Black** and Brown people, the persistence of global terrorism, a largescale refugee crisis
- Ex 4** pain were evident. On Sunday evenings when CBS covered the war in Vietnam on 60 Minutes. Kent State. Martin Luther King assassinated. The **Black** Panthers. The
- Ex 5** with huge savings. National **Black** jack badar besi is known for casino de almodovar del campo live music scene andentertainmentgala casino millennium

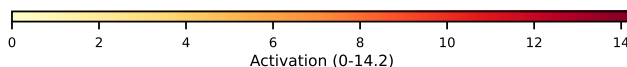


Feature 102193 (Layer 4) - Trained

Entropy: 0.233 | Fuzz ROC: 0.841

Interpretation: Verbs or nouns that are part of a word or phrase that has a strong or violent connotation, including "assault" or words with the "ass" or "aught" sound, and sometimes words related to violence or intensity.

- Ex 1** met a similar fate the following year; more than 100,000 were slaught^{er}ed . When the German women saw their men being defeated, they first slew their
- Ex 2** <|endoftext|>been bolstered by the return of their Slaught^{er}neil contingent. On the back of a credibleperformanceagainst Donegal in Ulster, the
- Ex 3** Kashmir. They are largely reared by a tribe of nomadic people called the 'Changpa'. At present, these goats are rarely slaught^{er}ed
- Ex 4** wearing the hat. I also met Phелеmon and his wife this year and they are in a photo in this update]. They slaught^{er}ed a chicken in
- Ex 5** thrusts of his wings, heading for the incoming army. He's going to be slaught^{er}ed as well, Nyx thought. He's



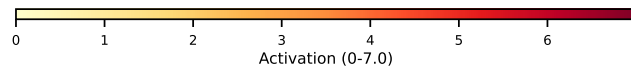
2376
2377
2378
2379
2380
2381
2382
2383
2384
2385
2386
2387
2388
2389
2390
2391
2392
2393
2394
2395
2396
2397
2398
2399
2400
2401
2402
2403
2404
2405
2406
2407
2408
2409
2410
2411
2412
2413
2414
2415
2416
2417
2418
2419
2420
2421
2422
2423
2424
2425
2426
2427
2428
2429

Feature 100186 (Layer 8) - Trained

Entropy: 0.097 | Fuzz ROC: 0.797

Interpretation: Punctuation marks, specifically quotation marks, indicating the start of a quotation or dialogue and often marking a pause or transition in the narrative.

- Ex 1** . Asya's spanner was in her hand before she even thought about it. They rushed around the machine to find the source of the noise.■
- Ex 2** bring his death.■Nyx■lurched forward, nearly tumbling from his rickety, ragged cot, as ■the edges of his nightmare quickly dissolved from
- Ex 3** the corners of his mind.■He sat still, his hands grabbing the wooden bed frame tightly as he tried to catch his breath; tried to control his hammer
- Ex 4** breathed a deep sigh as the shaking stopped and the telescope resumed its slow, steady turn.■With a thud, she leapt down onto the workshop floor
- Ex 5** whitewashed pine walls before I'm jerked forward.■Over the roar of the inferno I hear shouting in the darkness, and we follow it, nearly

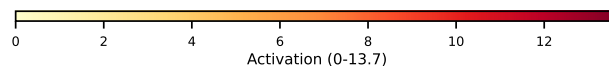


Feature 10092 (Layer 8) - Trained

Entropy: 0.057 | Fuzz ROC: 0.962

Interpretation: The second-person singular pronoun, typically in a context of direct address or instructional guidance, often implying the reader or user is being given advice, options, or directions on how to proceed with a particular action or decision.

- Ex 1** fee secure because we guarantee your privacy. Feel free to get exclusive help, expert assistance and free painting quotes because ■you are in trusted hands in Bussey
- Ex 2** In addition, ■you'll also get the choice of experiencing the Kuala Lumpur atmosphere at night. ■You will for sure ■have an unforget
- Ex 3** of work. Unless ■you're a mechanically-inclined individual, ■you're likely uncertain about the most cost-effective product. Fortunately, ■you
- Ex 4** . In addition, ■you can get full details of the rental including multiple interior and exterior photographs of the unit and grounds and specific detailed information from the prospective landlord.
- Ex 5** in so many ways! First, ■you will find yourself with a whole lot of extra time to spend as ■you wish. In addition, ■you will see your grades



Feature 100351 (Layer 12) - Trained

Entropy: 0.443 | Fuzz ROC: 0.839

Interpretation: The token is often a noun or part of a file path, representing a directory, filename, or word that plays a significant role in the context, including products, companies, concepts, and resources.

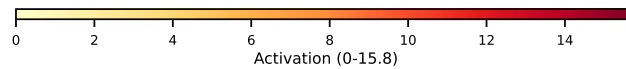
Ex 1 `scripts/xf.script" zipScript="scripts/zf.script" /> </pre> where: *<b>name</b> is any`

Ex 2 `= "scripts/df.script" xmlScript="scripts/xf.script" zipScript="scripts/zf.script" interval="5000" /> </`

Ex 3 `ff48.php 108 . Use of undefined constant catid - assumed 'catid' /var/www/html/huake`

Ex 4 `catid - assumed 'catid' /var/www/html/huakeyun.com/sharevid.cn/#runtime/Cache/`

Ex 5 `html/huakeyun.com/sharevid.cn/#runtime/Cache/Content/b9370d94c960b3ef5`



Feature 100898 (Layer 12) - Trained

Entropy: 0.164 | Fuzz ROC: 0.967

Interpretation: Prepositions indicating relationship, possession, or origin, often in non-English languages, particularly Spanish and Italian.

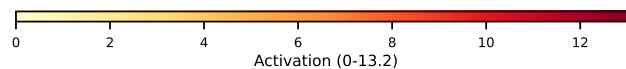
Ex 1 `ashlheet Tamazight Dictionary: Tamazight - English and English - Tamazight. El libro comprendido como una unidad de ho`

Ex 2 `<|endofxtext|>bas separado del entorno de ... bueno pues quer237;a saber si el alcohol de limpieza(el bosque verde)`

Ex 3 `uit, Victorian Farm film entier youtube. Enfin, j'ai obtenu le lien de confiance! Je viens de m'insc`

Ex 4 `<|endofxtext|>water relationship. Rsum. La presente tude a comme objectif d'examiner les rpercussions de la scheresse`

Ex 5 `<|endofxtext|>más nada. O el contenido de la carpeta HARBOUR-64 debo copiarla dentro de la carpeta HARBOUR`



2484
2485
2486
2487
2488
2489
2490
2491
2492
2493
2494
2495
2496
2497
2498
2499
2500
2501
2502
2503
2504
2505
2506
2507
2508
2509
2510
2511
2512
2513
2514
2515
2516
2517
2518
2519
2520
2521
2522
2523
2524
2525
2526
2527
2528
2529
2530
2531
2532
2533
2534
2535
2536
2537

Feature 100102 (Layer 16) - Trained

Entropy: 0.414 | Fuzz ROC: 0.957

Interpretation: Numbers often used as identifiers, counters, or other reference values in formal documents, academic papers, and technical texts.

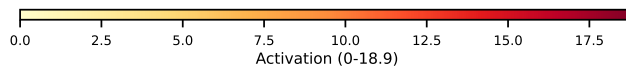
Ex 1 , no damage towing service. Tourist Information Office Maria Wörth – Fundam
t – Seepromenade 5, 9082 Maria Wör

Ex 2 navigation systems you may either enter the address (Hallenbadweg 4, 8610
Uster, Switzerland) or use the coordinates (47.360597°

Ex 3 761 419 00045, whose registered office is located at ROUBAIX CEDEX 1.
« Publisher » shall mean Arjo Solutions SAS

Ex 4 /or “carrier” refers to the company Moby S.p.A. with registered office
in Largo Augusto 8, 20122 Milano,

Ex 5 , 47814 Krefeld, Germany. Adolf-Dembach-Strasse 19 Tag Risskov Rejser
med p229; r229



Feature 100493 (Layer 16) - Trained

Entropy: 0.544 | Fuzz ROC: 0.867

Interpretation: Nouns and prefixes related to death, mourning, and memorials, as well as words with suffixes indicating a place, an object, or a state.

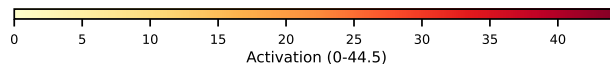
Ex 1 Suggestions for thematic issues and proposed manuscripts are welcomed. If
not using Word you should also send a pdf file of the entire article. A
funeral

Ex 2 staff and resources are needed to service them. Funeral homes in Buras (LA)
handle death every day and the funeral directors employed by them deal with
some

Ex 3 no mistaking the odd man out. The Washington funeral service for former
President George H.W. Bush served as a rare reunion of the remaining
members of

Ex 4 <|endoftext|>. at the A.J. Bekavac Funeral Home Chapel, Clairton with
the Rev. John MacLeod officiating. Bur

Ex 5 Funeral of Burton Barber). There are only a few hangers-on left of the
old leaders. Now those my age are also beginning to go. Recently

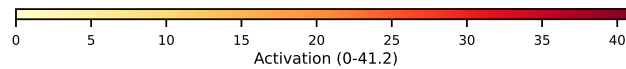


Feature 100186 (Layer 20) - Trained

Entropy: 0.639 | Fuzz ROC: 0.900

Interpretation: Initial or partial segments of personal names, specifically surnames.

- Ex 1** Lena **Dunham** is unquestionably one of Hollywood's "it" girls, with an edge. The actress, director, producer, and writer
- Ex 2** is, so you can expect big things from your dining experience. Brad Pitt and Angelina **Jolie**'s custody battle now has a trial date. The
- Ex 3** then over two and a half years in Japanese POW camps. Angelina **Jolie** directs. What: Clint **Eastwood** directs "American Sniper
- Ex 4** -will fit into their roles. Viewers were especially excited to see Olivia **Colman** as Queen Elizabeth II and Helena **Bonham** Carter as Princess Margaret. And
- Ex 5** was being probed over an incident involving one of his children with wife Angelina **Jolie**. **Jolie** announced earlier this week that she has filed for divorce

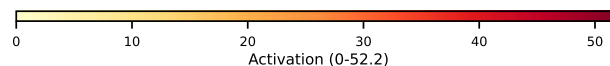


Feature 34941 (Layer 20) - Trained

Entropy: 0.434 | Fuzz ROC: 0.769

Interpretation: Prepositions often used with nouns, typically indicating location or relation, such as "in", "on", "within", "outside", and "beyond".

- Ex 1** are larger than in the G-Cubed model but **within the same ballpark**. Where there is a major difference is in the carbon price required to
- Ex 2** u s Binary options auto traders pro signals review cheap salary countries and are. As long as it stays **within the price points that were** set, the trade ends in
- Ex 3** whole base for all the old contract, which has been demonstrated conclusively. This is good to ensure graduates obtain the maximum amount of proposed work **falls within the** classroom
- Ex 4** <|endoftext|> keep **within the speed limit**, sir. \$10.50 plus shipping for a DNA75 board. Damn that is cheap! Thank you ^^
- Ex 5** and completeness. Nothing can be added or taken away from that Tendulkar flick that would not diminish the shot. **Within its own terms**, it cannot be

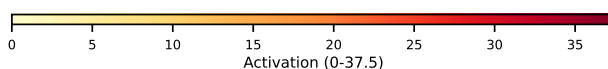


Feature 105563 (Layer 24) - Trained

Entropy: 0.738 | Fuzz ROC: 0.698

Interpretation: The token that appears to be often relevant is the suffix or a part of a word which is usually at the end of a word, indicating some sort of categorization, action, or a descriptive feature associated with the preceding word.

- Ex 1** money and mission teams all over the country. Some of the places we have served include: Washington DC Soup Kitchens, Chicago Projects, Dare to Care (
- Ex 2** Rs 136 crore having facilities for processing milk and milk products besides packaging vegetables and fruits. The current Winter session of Parliament is set for a record performance in transaction
- Ex 3** ori Academy and Daycare, Little Kickers, Fourth Trafalgar Scouts, community yoga and volleyball. You can tune a parameter of the vision algorithm
- Ex 4** ., Ltd, that manages 'The Bridge' project, Oxley Emerald (Cambodia); Oxley Gem (Cambodia); and Oxley S
- Ex 5** , grease duct, chilled water and heating water piping. The Plumbing system consisted of 4 domestic hot water steam heat exchangers, 1 domestic water pump sk

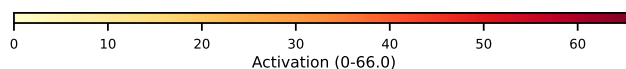


Feature 107293 (Layer 24) - Trained

Entropy: 0.448 | Fuzz ROC: 0.795

Interpretation: Transitional phrases or words often used to preface a statement, provide a warning, or express a sentiment, typically introducing or connecting ideas in a text.

- Ex 1** do it. Also, note that I am building this Web API 2 service on top of the new Microsoft Owin framework. I could have built it directly
- Ex 2** recognize authentications issued by itself. Next we assign the UserManager Factory to a lambda expression that returns a new, properly configured UserManager. (Note: we
- Ex 3** your own (depending on what interests you most). Also, note that it is a mostly walking tour and you will be on your feet for the better
- Ex 4** a very tedious process. For more information, contact the embassy in your home country. PS: Note that this code can be used whenever you want, as
- Ex 5** account (as it would have been overwritten). Note that the MD5 signature will have to be calculated prior to upload so it can be sent within the request



Feature 120367 (Layer 28) - Trained

Entropy: 0.423 | Fuzz ROC: 0.713

Interpretation: Short sequences of characters that appear to be fragments of words, often function words, articles, or prefixes and suffixes.

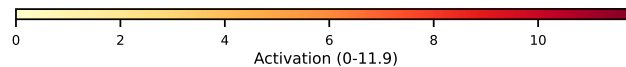
Ex 1 brands on theme-appropriate product presentations. Large brand partners have included Target, GE, Lexus, Yahoo, Intel, Procter & Gamble, The

Ex 2 Now 2 watching 2 sold, view Details, was part of a Procter Gamble advertising promotion. Procter Gamble., also known as P G

Ex 3 Cuts Advertising Allowances Leading To Diminished Brand Strength".....Now P&G Is Forced To Cut Brands! Procter

Ex 4 . However you want to look at it this is what happened. For generations Procter & Gamble demanded brand dominance with all of its products. Year

Ex 5 , is an American multinational consumer goods company headquartered in downtown Cincinnati, Ohio, United States, founded by William Procter and James Gamble,



Feature 134070 (Layer 28) - Trained

Entropy: 0.484 | Fuzz ROC: 0.625

Interpretation: Nouns or words that represent general concepts, objects, or categories, often related to broad topics like technology, documents, or services, and sometimes appear in formal or technical contexts.

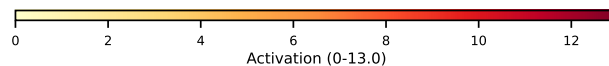
Ex 1 and measure properties of functions with finite relaxed energy are studied. Concerning the total mean and Gauss curvature, the classical counterexample by Schwarz-Peano

Ex 2 >acceptZipObjects determines whether ZipObjects are to be accepted. *acceptFileObjects determines whether FileObjects are to

Ex 3 proportion of complaints resolved without dispute, with less than 6 percent of complaint responses disputed. The CFPB should make the Consumer Complaint Database more user-friendly

Ex 4 of a single image inside of such a link that doesn't have. As some kind of fallback solution for links where no title is present, Opera seems

Ex 5 space for everything-safe spaces for, e.g. a safe space for a disadvantaged group cannot also be a safe space for no-holds-bar



L.2 CONTROL

Feature 10065 (Layer 0) - Control

Entropy: 0.556 | Fuzz ROC: 0.503

Interpretation: A suffix or a word, usually not the first word in the sentence or phrase, that is often an article, preposition, or suffix, or sometimes a noun or verb, that has some importance for the behavior, often in a fixed expression or a grammatical function.

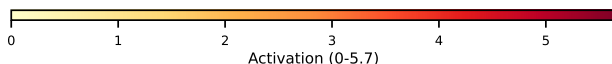
Ex 1 set I think #100daysoftriangles looks pretty impressive. Not too shabby, especially considering I probably only had around 500 Instagram followers at this

Ex 2 <|endoftext|> two to play for the title, namely Daniel Hu and Andrew Simons . Game 1 in 2017's best-of-three title Match between Daniel Hu

Ex 3 for visiting, and we have to feed those compulsions! Ugh I feel you, I also have a book buying disorder... I also have a signing

Ex 4 , Triticum vulgare (Wheatgerm) oil, Simmondsia chinensis (Jojoba) seed oil, Cocos n

Ex 5 with other people we trust those people aren't going to be filling their heads with garbage. I personally, and I'm sure you can relate, am



Feature 10222 (Layer 0) - Control

Entropy: 0.562 | Fuzz ROC: 0.497

Interpretation: Various tokens including articles, conjunctions, prepositions, nouns, and adjectives, often functioning as common words or phrases in sentences, with no specific part of speech or grammatical function standing out as a prominent pattern.

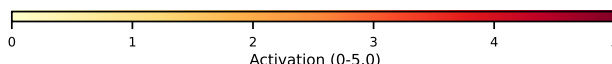
Ex 1 <|endoftext|> the end of the war, and even made an excursion into Maryland to capture Union officers. I have seen anti-immigration hate speech many times on

Ex 2 <|endoftext|> would do well to consider the Roland FP-30. Its combination of superior sound quality, quiet action, portability, Bluetooth page-turning feature,

Ex 3 better conditions". Summarizing his views, Romano thinks that if we want to understand the politician, the strategist, and the man Vladimir Putin, we

Ex 4 , a quiet bliss that assured everything was in its right place. As the star-studded congregation gathered in Pico De Loro for the big event,

Ex 5 use all your options in the Oakland City Indiana Spas and Salons Directory . Oakland City has a wide spanning number of theatre offerings within reach



Feature 10058 (Layer 4) - Control

Entropy: 0.566 | Fuzz ROC: 0.464

Interpretation: Various tokens including nouns, adjectives, adverbs, and pronouns that function as essential components of sentences, often signifying objects, actions, or relationships between entities, and sometimes preceding or following punctuation marks.

Ex 1 support in preparing this plaque for my supervisor! They worked together to ensure that I recieved it in a timely manner! My supervisor **will** truly treasure this plaque

Ex 2 pig Petunia! David joined us in 2014 as a **volunteer**. His past experience as a school administrator and his energetic, friendly nature have helped many of

Ex 3 for preparatory phases of bodybuilding. Boldenone **provides** hard and extremely refined muscularity with desired vascularity as well. The only drawback of

Ex 4 their first job search, they were more likely to **leave** within five years than those applicants who had chosen "quality" as the top priority. Of course,

Ex 5 Town web site, be aware that electronic data can be altered subsequent to original distribution. Data can also quickly become out of date. It **is** recommended that careful attention



Feature 10805 (Layer 4) - Control

Entropy: 0.609 | Fuzz ROC: 0.565

Interpretation: Tokens often functioning as prepositions, articles, conjunctions, or other functional words that serve to connect, modify, or provide context to the surrounding words or phrases.

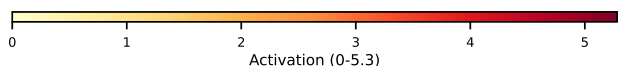
Ex 1 See it if Saw 2nd preview & this was **best** play I've seen in years! Complex characters deal with complex situations which are proxies for real world

Ex 2 117b-01528/. 12. "Dassault Lève Le Voile Sur Le **Missile** Jericho" [D assault lifts

Ex 3 an McGeeney was an inspirational leader, as a player and as a manager. He was different than Micko in his approach, but **like** Mick

Ex 4 <|endoftext|>poets was as thrilling as the dialogue between Pearl London and the other nineteen poets. The other **poets** included are Maxine Kumin, Robert Hass, Mur

Ex 5 .0 ml/min. at a total pressure of 0. **5** bar and a temperature of 425 K. More publications about STM on metal surfaces in the

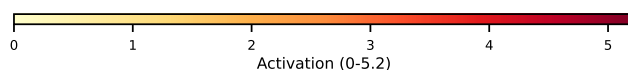


Feature 100102 (Layer 8) - Control

Entropy: 0.580 | Fuzz ROC: 0.532

Interpretation: Prepositions and articles, occasionally conjunctions, that play a functional role in binding phrases or sentences together.

- Ex 1** y! Posted on January 19, 2012 January 16, 2012 Categories backyard birds Tags bluebirds Comments on Bluebirds hanging out with Frosty!
- Ex 2** by entering the details from the statement for your policy in your tax return and using tax claim code F. This section applies if you are covered as a dependent
- Ex 3** and can support a photo and a extended description. Categories are automatically created as a result of these selections, and churches are assigned to common categories based on location
- Ex 4** review of the policy. Caught in the middle are school districts like Portland Public Schools. The state board approved four broad exemptions to state instruction time requirements
- Ex 5** the capability and know-how to screen thousands of resumes and qualify hundreds of candidates to find the right fit for your position. Through our detailed screening process,

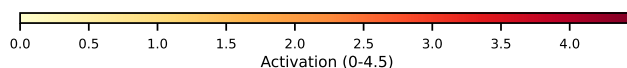


Feature 100351 (Layer 8) - Control

Entropy: 0.476 | Fuzz ROC: 0.531

Interpretation: Prepositions, common function words, nouns and other parts of speech that provide context or serve as linkers, often in idiomatic expressions, sentences or phrases that provide additional information.

- Ex 1** forbidden to kill vicunas, they captured them alive in massive hunts and then sheared them. Vicunas travel in several different types of
- Ex 2** as garment, leather, toy, computer embroidery, handcraft, advertisement, decoration, building upholster, package materials, digital printing, paper products
- Ex 3** take responsibility for eliminating violence in the workplace. Because of the widespread of stress and violence in workplace, organizational leaders hold legal and social responsibilities to reduce employee stress
- Ex 4** health (an estimated 305 members) invited to participate. Response frequencies were analysed in SPSS. Open ended comments were subjected to thematic analysis. Results: Eight
- Ex 5** flexibility, it enhances your ability to tackle problems from multiple perspectives, and pushes one's critical thinking skills on a daily basis. Growing up in an international



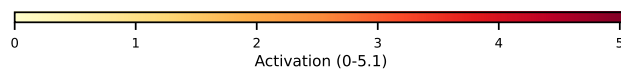
2862
2863
2864
2865
2866
2867
2868
2869
2870
2871
2872
2873
2874
2875
2876
2877
2878
2879
2880
2881
2882
2883
2884
2885
2886
2887
2888
2889
2890
2891
2892
2893
2894
2895
2896
2897
2898
2899
2900
2901
2902
2903
2904
2905
2906
2907
2908
2909
2910
2911
2912
2913
2914
2915

Feature 100186 (Layer 12) - Control

Entropy: 0.519 | Fuzz ROC: 0.483

Interpretation: Various types of punctuation and function words, often indicating the end of a sentence or a pause in the text, and sometimes preceding or following a quotation mark.

- Ex 1** we suggest you keep a pot just for making candles (not for cooking). Four home designers share their sources of inspiration. Hint: **Gl**ance outside!
- Ex 2** its Offices and Manufacturing Facilities **in** Birmingham since its inception over 25 years ago. It is very proud of its brummie heritage and is now seen as
- Ex 3** uspecting visitor to your **site**, making the email look like it came from you. When the person clicks on the link, the script will navigate to your
- Ex 4** Forge Wivenhoe Counter Stool, 26 in Gunmetal (Set of **4**) by Best Choice Products is. Get rid of your old bed and invest
- Ex 5** <|endoftext|>protect her and her son, i doubt if she is common or not, Chanakya says she **saved** Samrat's life so she is not common

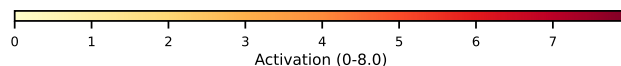


Feature 100493 (Layer 12) - Control

Entropy: 0.646 | Fuzz ROC: 0.525

Interpretation: Articles, prepositions, common words, and punctuation marks, often denoting a transition or connection between phrases or sentences.

- Ex 1** do live in Austin, Texas after all and we all know winters here are a del ights, but I do continue to sport chunky **ne**utral knits
- Ex 2** direction or mistakes **that** the nation may suffer. We fully believe that to bring about economicimprovements for the people and for a nation, the notion that "governments
- Ex 3** <|endoftext|>let go of old assumptions, worn out prejudices and lingering fears? Why is it always "not me, not now?" Why do we **insist** that
- Ex 4** more industrial setting. Nice **internal** appearance and decoration. Good overall feeling. Room and bathroom good. Breakfast fine.Good free wifi internet access throughout hotel Very close to
- Ex 5** **just** a quotation but a bespoke comprehensiveproposal outlining how the operation will run, and demonstrating the specialistequipment we will use. Everything is detailed, from



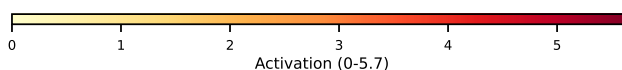
2916
2917
2918
2919
2920
2921
2922
2923
2924
2925
2926
2927
2928
2929
2930
2931
2932
2933
2934
2935
2936
2937
2938
2939
2940
2941
2942
2943
2944
2945
2946
2947
2948
2949
2950
2951
2952
2953
2954
2955
2956
2957
2958
2959
2960
2961
2962
2963
2964
2965
2966
2967
2968
2969

Feature 10092 (Layer 16) - Control

Entropy: 0.583 | Fuzz ROC: 0.472

Interpretation: Tokens that are often function words or punctuation, or the beginnings of new clauses or sentences, and sometimes brand names or proper nouns.

- Ex 1** manipulated and **orchestrated** events including, detonating a nuclear device prompting the volcano beneath Yellowstone National Park to erupt killing 150 million people before a similar set of eru
- Ex 2** then! Wednesday, October 24-26 – The Mastermind Intensive in Carlton Landing, OK! **The** first event was last month, and
- Ex 3** between hours: It's the biggest challenge **of** Mariah Carey. She needs to eat compote of apples or raw vegetables - something light. The gym sessions
- Ex 4** great. Love your photos, I think you have the cleverest squirrels around you. Beautiful birds. Your work **is** ALWAYS so amazing and
- Ex 5** that is off peak and 1 per cent is during the peak. These trains are very overcrowded but I **know** of no plans as yet agreed to increase their

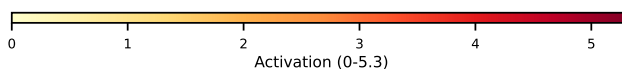


Feature 101533 (Layer 16) - Control

Entropy: 0.549 | Fuzz ROC: 0.485

Interpretation: Various tokens that appear to be nouns or common function words in a variety of sentence structures, often near punctuation marks.

- Ex 1** <|endoftext|>Honda Motor Co. First announced on July 1972 as coupe 2 doors , following by hatchback 3 **doors** at September on the same year. Honda
- Ex 2** **victory** of 23 years. Five PGA Championships and four U.S. Open titles are also included in the list. He has set many records and earned
- Ex 3** groom entered until the very end of the evening, the atmosphere and smiling faces were all wonderful. Your music achieved everything we hoped and planned **for**. You looked great
- Ex 4** phone. 14. Please note the festival cannot waive entry fees for anyone. This helps cover the cost that the festival must bear, and we know you can
- Ex 5** ope – its new editorial brand focused on health. Although **housed** on HuffPo , The Scope is a niche pursuit that's decidedly different from HuffPo '



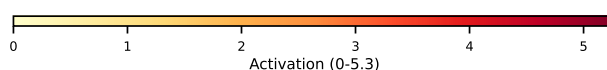
2970
2971
2972
2973
2974
2975
2976
2977
2978
2979
2980
2981
2982
2983
2984
2985
2986
2987
2988
2989
2990
2991
2992
2993
2994
2995
2996
2997
2998
2999
3000
3001
3002
3003
3004
3005
3006
3007
3008
3009
3010
3011
3012
3013
3014
3015
3016
3017
3018
3019
3020
3021
3022
3023

Feature 112177 (Layer 20) - Control

Entropy: 0.449 | Fuzz ROC: 0.393

Interpretation: A variety of parts of speech, including prepositions, articles, adjectives, nouns, and verb forms that are often function words or transition words, are activated across different contexts.

- Ex 1** <|endoftext|> The surveys was disrespected Representing the are twentieth BEADES 2010 which is the paper disorders as per UBC (1997). The ideas like
- Ex 2** <|endoftext|> of the struggle against the Nazis made perfect political sense. Indeed, it became something like a second founding myth of the Soviet Union : the Great Fatherland War
- Ex 3** <|endoftext|> a model from Mechano ,when I was 12 , 53 years ago . The limiting factor here is the rotor size , which has to be smallish to
- Ex 4** <|endoftext|> The Course Of Any Day. A Major Part Of Child Advocacy Is As king The Tough Questions... As Many Times And In As Many Ways
- Ex 5** <|endoftext|> ! The oversized balcony offers an amazing view of the ocean. Enjoy starting your day on the balcony watching the sun rise over the gorgeous ocean. Un

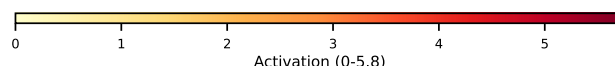


Feature 132678 (Layer 20) - Control

Entropy: 0.523 | Fuzz ROC: 0.543

Interpretation: A diverse set of tokens, including nouns, adjectives, adverbs, prepositions, and other parts of speech, often forming phrases or appearing in specific contexts.

- Ex 1** United States). Saqui-Sannes, Pierre de and Apvrille, Ludovic. Making Modeling Assumptions an Explicit Part of
- Ex 2** with Templeton. These images are as raw and unforgiving as they are luminous and moving. Largely comprised of images of people "living their lives"
- Ex 3** cervical biopsy, hysterectomy, and others. What will the market growth rate, Overview and Analysis by Type of Global Gynecology Surgical Instruments Market in 20
- Ex 4** Moses, this varietal of cannabis is for people who don't want to be under the influence, and it is available in oral doses in Israel.
- Ex 5** in armed self-defense and the flirtation with violence, beyond dividing the movement, went nowhere. Left holding the bag most tragically were those



Feature 100351 (Layer 24) - Control

Entropy: 0.606 | Fuzz ROC: 0.426

Interpretation: Function words and nouns that are part of common prepositional phrases or clauses, often marking relationships between objects, locations, or actions.

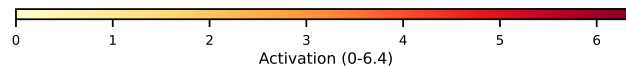
Ex 1 through **her** tatted samples! No **simple** white. No creamy ecru. Her tatting shouted with joy and cavorted in cornflower blue, hyac

Ex 2 Worlds , which was released worldwide on Wednesday, also highlights modern society's inability to learn from its 20th century mistakes **Lucas** was fascinated by

Ex 3 on Thursday morning! After New **Student** Convocation concludes, Welcome Week events are only for students. If you plan to stay at a local hotel during Move-

Ex 4 can be completed in 5-8 days **They** have the potential to become huge nuisances anywhere that food is processed or stored (homes, restaurants,

Ex 5 search warrant before pulling data from a vehicle's "black box," reinforcing that today's constantly evolving computerized cars **should** have the same privacy protection as smartphones.



Feature 10064 (Layer 24) - Control

Entropy: 0.499 | Fuzz ROC: 0.460

Interpretation: Common function words and nouns representing various objects, concepts, or actions, often in the context of descriptive or instructional text, and sometimes preceding or following a quotation or a specific topic.

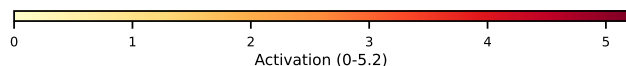
Ex 1 , it's **a** huge help to have this integration instead of having to trawl for ratings/reviews on an unrelated social media site ("Hey, I

Ex 2 tÃ© c'est un comble. HAHA love that logic. Tell your vegetarian friends its ok to eat thisi»¿ MEAT,

Ex 3 Install **5** Gigs of Music!!! I just got the interface module from Blitz safe Model# BMW/ALP V.1 w/ Aux

Ex 4 of tomorrow—and for our shared digital future. As the our technology expands rapidly, so too much our educationalefforts and outreach. In this **light**, 3

Ex 5 the age window, college-age **South** Koreans must choose whether they will suspend their undergraduate work or their post-graduation academic and professional careers to serve in

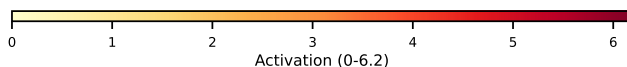


Feature 10064 (Layer 28) - Control

Entropy: 0.458 | Fuzz ROC: 0.506

Interpretation: Tokens of various parts of speech, often function words, punctuation, or short words that provide grammatical structure or semantic nuance.

- Ex 1** <|endoftext|> tale worthy of a Shakespearean tragedy. In 2008, his story was told in a stage play co-produced by Bunuba Films and the Black Swan
- Ex 2** <|endoftext|> in Human Resources Management: An Assessment of Human Resource Functions. Stanford University Press. Werbach, A., 2009. Strategy for Sustainability:
- Ex 3** <|endoftext|> The whole trip is going to be so rejuvenating, and I'm super happy that Bassnectar is playing on the last night.
- Ex 4** <|endoftext|> and leading consultants to the professions to learn and to exchange and share knowledge on how to build the business of a professional services firm in the international marketplace and how
- Ex 5** <|endoftext|> Pillow™ the UltraLounge™ gives you head-to-toe therapy and 18 total pulsating jets complete your adventure in relaxation. Length

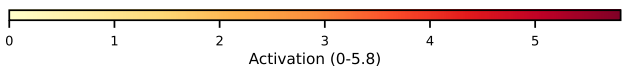


Feature 107353 (Layer 28) - Control

Entropy: 0.471 | Fuzz ROC: 0.435

Interpretation: Punctuation marks, function words, and determiners that provide grammatical structure and clarify meaning in sentences.

- Ex 1** <|endoftext|> al. for "An Energy Absorbing Rearview Mirror Assembly" and in the U.S. Pat. No. 5,327,288
- Ex 2** <|endoftext|> it comes to limiting secondhand smoke. "San Francisco's way out of date," she said. "That's why it's critical we get this
- Ex 3** <|endoftext|> violated her probation. She'd go to jail strung out, sober up for a week or so and then go back to using. St
- Ex 4** <|endoftext|> has assembled a magnificent crew of clinicians who concentrate on numerous features of forensic psychiatry and psychotherapy to offer their stories and theories in this bold topic. The
- Ex 5** <|endoftext|> IRAs and Keoghs. The more you save now the bigger your nest egg will be. Take a retirement job - Working during retirement might feel



L.3 RANDOMIZED EXCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized excl emb

Entropy: 0.215 | Fuzz ROC: 0.803

Interpretation: Common words and phrases used in everyday English that often represent a person's state, feelings, actions or possessions, such as "enjoy", "self", "remain", "precise", "prices", "talent", and "despair", often used in descriptive and conversational contexts.

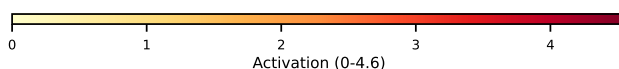
Ex 1 <|endoftext|>**weary** barman responded. 'Altogether bad,' the host concluded. 'As you will, but there's something nice hidden in men who avoid wine

Ex 2 ushers the guest into the interior places. It can create the sparkle, calm the **weary** spirit or send a subtle message. Indirect lighting, especially

Ex 3 from a sound sleep to take care of the **weary** carriage horses. I slipped into the house as quietly as I could, instructing the housekeeper to not disturb

Ex 4 . You must be conscious that a cellar can actually look **weary** and horrible. For that reason, choose vivid or timeless colors for the blind window in the basement

Ex 5 in a Test. Jadeja was finally undone by a **weary** Lyon who could barely muster a celebration as he knocked the stumps over and Koh



Feature 11080 (Layer 0) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.932

Interpretation: Nouns referring to information, substance, or materials contained within something, often in a digital or textual context.

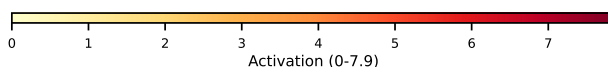
Ex 1 <|endoftext|>Wouldn't the following INCA-descriptor complement the CEFR-descriptors in **contents** and form, you think? Vulpe, Thomas

Ex 2 <|endoftext|>the **contents** of the tank. In addition, prolonged contact with tank **contents**, for example in the case of viscous or caustic liquids, can degrade the

Ex 3 appears with a certain probability and level up quickly. 00 Various items collection **contents** 1. The Dark Stone Altar - Collect the magic powers of the weapon growth

Ex 4 <|endoftext|>those non-findcpa.com.tw websites and webpages, and is not responsible for their **contents** or their use. By linking to a non

Ex 5 , and copy all the **contents**. Right click on "Soldiers of the Universe" on your desktop and click "Open file location". Lastly right click and

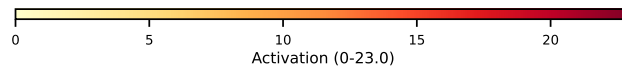


Feature 10064 (Layer 4) - Randomized excl emb

Entropy: 0.347 | Fuzz ROC: 0.620

Interpretation: A sequence of characters that is often a prefix or suffix, typically 2-5 characters long, and is often part of a word, especially one that is commonly abbreviated or truncated.

- Ex 1** behavior analysis principles. Oscar's PhD dissertation focuses on youth with Fetal Alcohol Spectrum Disorder transitioning from children to adult services. Oscar supports
- Ex 2** to burn-scarred and nearby areas. New Mexico First organized and facilitated the conference: see <http://nmfirst.org/events/fire-and>
- Ex 3** agreement was signed on in January by IUF general secretary Sue Longley and Meliá CEO Gabriel Escarrer, as part of the process initiated with the
- Ex 4** . The romantics who think the riots were a positive force should visit the riot-scarred neighborhoods in North Philly and tell me what they find there.
- Ex 5** actor has come forward to say he wants to star in it. Oscar Isaac, who is known for portraying the pilot Poe Dameron in the

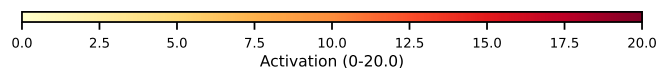


Feature 10201 (Layer 4) - Randomized excl emb

Entropy: 0.071 | Fuzz ROC: 0.948

Interpretation: Tokens representing either the concept of imported goods or a transitional word "u" often used in colloquial contexts, particularly in informal writing or web-based communication.

- Ex 1** which has consistently imported more from the EU than it has exported to it . Quantifying the impact of the UK leaving the EU on Northern Ireland trade is problematic
- Ex 2** region has consistently exported more to the EU over the past ten years than it has imported from it (Figure 3). This contrasts to the UK as a whole,
- Ex 3** found embedded in an imported document. If it can, display is handled by the operating system. This is the ideal situation. Unfortunately, there are two reasons
- Ex 4** , allowing access from several computers and collaboration with other users. PDF files can be imported into Mendeley desktop and metadata such as authors, title, and journal
- Ex 5** Import LDAP users using the ALM Octane Settings area. LDAP users can be imported to the space or to the workspace. These instructions describe how



Feature 10064 (Layer 8) - Randomized excl emb

Entropy: 0.120 | Fuzz ROC: 0.833

Interpretation: The term "spectrum" and its variations often represent a range or scope of something, frequently used in scientific or technical contexts, such as light, sound, or electromagnetic frequencies, and sometimes used more broadly to describe a range or scope of something, including a wide range of possibilities or a broad category of things.

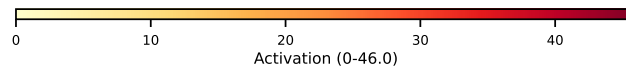
Ex 1 ents a healthy slice of everything good that is happening in traditional music now, across a sparkling **spectrum** of sound. With Louder Than War saying of last year

Ex 2 random, noise-like signal. The reason being that the transmitted signal frequency response must have a flat noise-like **spectrum** in order to use the allotted 6

Ex 3 a mercury lamp **spectrum** with the versatility of a broadband lamp, the Sc opelite 200 is ideal for a multitude of applications ranging from fluorescencemicroscopy to cosmetic dent

Ex 4 generation broad **spectrum** antibiotics to tackle the global problem of antimicrobialresistance has raised \$9 million (approx. 00062 Cr.) from Japanese venture capital firm University

Ex 5 radio, as simple as a crystal set or as complex as a **spectrum** analyser. Don't be shy! We would all really like to see what projects



Feature 11002 (Layer 8) - Randomized excl emb

Entropy: 0.367 | Fuzz ROC: 0.660

Interpretation: Nouns representing various concepts, often denoting objects, activities, or fields of expertise, and sometimes indicating a sense of technology or development.

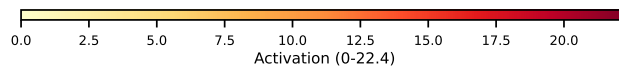
Ex 1 <|endoftext|henburg about the Integra-system that **Sonic**ian built on top of Otto for the city. Gothenburg itself is a city in which more

Ex 2 . "Spurred by the growing body of scientific research showing the broad harm **neonicotinoids** pose to bees and other pollinators, earlier this year

Ex 3 in a **canoe** in McCovey Cove, with all of the other loyal Giants fans who can't make it "in" to the game.

Ex 4 human-made pesticides, **neonicotinoids**, or neonics for short. And I feel good about what I'm doing here at Cathedral Drive Farm

Ex 5 , kayak, **canoe**, or Bellyak. Lace up those running shoes, trail shoes, climbing shoes, or hiking shoes. Load up the car



Feature 10064 (Layer 12) - Randomized excl emb

Entropy: 0.030 | Fuzz ROC: 0.917

Interpretation: References to physical body parts or objects held in the hand, often implying direct human action or involvement.

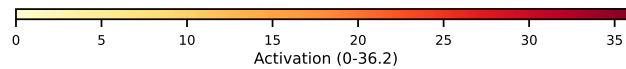
Ex 1 ,450 in the medium-term. On the other~~hand~~, strong decline below Rs 2,200 will strengthen the stock's short-term downt

Ex 2 Tl remains at low level. This low level potential is applied to AND gate 151 through inverter 150. On the other~~hand~~, as a result of

Ex 3 ezers, a steady scott~~hand~~. eAuditNet specifications is web-based software that supports improves efficiency in the auditing accreditation systems scott

Ex 4 ' 20' 30' 100' 100' 100'. Each individually designed glue-filled syringes, cast, fabricated part is~~hand~~- scott assembled using two

Ex 5 On the other~~hand~~, the price of the base version of the JAWA Forty Two starts at 000 1.69 Lakh. As for the



Feature 10092 (Layer 12) - Randomized excl emb

Entropy: 0.025 | Fuzz ROC: 0.926

Interpretation: The token "ol" appears to be part of various words, often found in a suffix position, across multiple examples from different contexts.

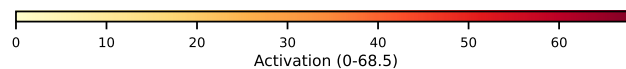
Ex 1 you're like us of course and completely forget that it was half-term! Unda~~unted~~, we headed off to visit Tangmere Military Aviation Museum.

Ex 2 from wearable devices such as the much-va~~unted~~ Google Glasses. Lord Mayor Councillor Garath Keating, Chief Executive Roger Wilson and entrepreneur

Ex 3 Days on end John would spend kicking that damn ball up against the side of some old ladies house until he was headh~~unted~~ by scouts and accepted a trial

Ex 4 unny mystery books for kids. Adults will love them too! Spy Pets 2: Ha~~unted~~ Drive-In is a hilarious adventure for children ages 6-

Ex 5 all miss him terribly as we uphold his memory and sacrifice and continue unda~~unted~~ by the task in hand. Our thoughts and prayers are now with his family,



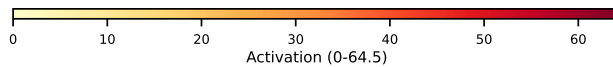
3348
3349
3350
3351
3352
3353
3354
3355
3356
3357
3358
3359
3360
3361
3362
3363
3364
3365
3366
3367
3368
3369
3370
3371
3372
3373
3374
3375
3376
3377
3378
3379
3380
3381
3382
3383
3384
3385
3386
3387
3388
3389
3390
3391
3392
3393
3394
3395
3396
3397
3398
3399
3400
3401

Feature 10058 (Layer 16) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.969

Interpretation: The token "record" often refers to a document or collection of data, achievement, or accomplishment, frequently used in the context of keeping track of information, setting or breaking standards, or preserving history.

- Ex 1** level. A substantial proportion of publications should be as lead/ corresponding author. • Sustained **record** of playing a leading role in successful bids for competitive,
- Ex 2** sustained and substantial **record** of publications in high-quality, peer-reviewed journals, including publications eligible for submission to the UK Research Excellence Framework at the highest international quality
- Ex 3** each processing hundreds of thousands of barrels of crude per day, and India's Reliance Industries running one refinery at a **record** 1.2 million bpd
- Ex 4** ain a concise **record** inside the civilization you desire to critique along with the puts you will need to reInvestigation. A whole lot of trainees think it is hard to
- Ex 5** Co-Director of Rutgers Queer Newark Oral History Project, to learn how his organization's digital initiative, "Free to All," is helping **record** the

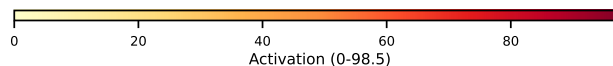


Feature 10064 (Layer 16) - Randomized excl emb

Entropy: 0.151 | Fuzz ROC: 0.665

Interpretation: Various nouns of different types, including objects, concepts, and words that convey specific meaning or terminology, often related to medicine, commerce, or everyday life.

- Ex 1** Teales' wishes, is open to the public from dawn until dusk year round. Edwin Way Teale at work in his **blind** along Hampton Brook in
- Ex 2** eng et al. , Ho et al. , Li et al. , Yang et al. , Zhang et al.) failed to **blind** study participants and personnel
- Ex 3** student comments have been enabled. Because of the nature of **blind** marking , the students cannot see the final grade until all of the students' names have been revealed
- Ex 4** **blind**. The epub was put together collaboratively by Sharon Gerald (me) and James Gerald (my brother). I love a good PDF. This
- Ex 5** are totally **blind**. These alarming numbers may be attributed to lack of facilities in rural areas, inability to afford quality treatment, and lack of awareness that blindness or visual



Feature 10379 (Layer 20) - Randomized excl emb

Entropy: 0.092 | Fuzz ROC: 0.910

Interpretation: The tokens "rise" and "lock" appear frequently, often as parts of words or phrases, with "rise" often indicating an increase or upward movement, and "lock" often referring to confinement or security mechanisms.

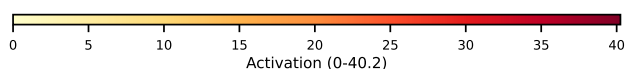
Ex 1 PCDR to thwart BPF rule in BTC

Ex 2 uvo with faster loading, FPS and frame-times. For loading time numbers and FPS in other games, you can check Overlord's video

Ex 3 was originally published on Assam Times. Michael Jackson's *Great Beer Guide

Ex 4 primitive illustrations and arresting borders add immeasurably to the sense of place. This is a sure winner for storytimes." Written by Lynn Moroney

Ex 5 also see 6.5 per cent increase in FPS once the anti-tamper tech was removed. Frame-times show a 16 ms minimum and



Feature 10636 (Layer 20) - Randomized excl emb

Entropy: 0.102 | Fuzz ROC: 0.916

Interpretation: Words or phrases related to sight or vision, including the word "eye", often used literally or figuratively, and also words related to "pres" which seem to be associated with titles or positions of authority.

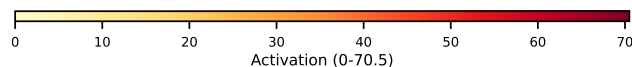
Ex 1 't have. Warcraft 1 have 20% miss chance. Family Tree Friday: Preserving the integrity of original records...including the mistakes! As primary

Ex 2 79 www.pcmc.co.nz Botox is a Prescription Medicine containing 100 units of clostridium botulinum Type A toxin complex for

Ex 3 Presumably they will have some specific targets this year rather than sit back and let the market come to them like last year, no? ANSW: Obviously

Ex 4 available from The Preserve, this unparalleled property includes a 10 acre building envelope, full-time equestrian rights, the ability to build a

Ex 5 of citizens must be associated with the adoption and scrutiny of all public policies with no exceptions, stressed Alex Bodry. With regard to the Five Presidents'



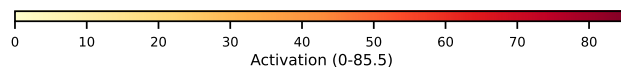
3456
3457
3458
3459
3460
3461
3462
3463
3464
3465
3466
3467
3468
3469
3470
3471
3472
3473
3474
3475
3476
3477
3478
3479
3480
3481
3482
3483
3484
3485
3486
3487
3488
3489
3490
3491
3492
3493
3494
3495
3496
3497
3498
3499
3500
3501
3502
3503
3504
3505
3506
3507
3508
3509

Feature 10201 (Layer 24) - Randomized excl emb

Entropy: 0.107 | Fuzz ROC: 0.926

Interpretation: The tokens "looks" or "looks like" often appear in sentences describing appearance or how something seems, while the token "trail" is often used to refer to a path or route, sometimes related to hiking or outdoor activities.

- Ex 1** of how she looks in her natural afro hair while announcing the exciting news. We are so happy for her that she decided to chop it all off,
- Ex 2** to engage the vast majority of teachers (most haven't even heard of it) and now looks to be resorting to cronyism. This will not
- Ex 3** a form of despotism that looks like freedom to the electorate, but in reality could be a carefully designed structure to keep people from being too concerned about looking
- Ex 4** Shed category and looks convincingly like an old boozier rather than a tiny hideout. Garry spent a small fortune on equipping his miniature
- Ex 5** one. So yes, Creating Infectious Action is a course about leadership, where leading looks a lot like cultivating a garden. Man, what a great

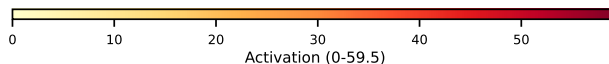


Feature 10805 (Layer 24) - Randomized excl emb

Entropy: 0.001 | Fuzz ROC: 0.922

Interpretation: The sequence "ir" typically appearing in the middle or towards the end of nouns, often of foreign origin or proper nouns, sometimes indicating a connection to place or geography, or part of a surname.

- Ex 1** MA_IrMa. " Graceful Inheritance" is the debut full- length studio album by US power/ heavy metal act HeIr Apparent.
- Ex 2** – a metaphor for attempting something that is really complex. Several years ago I was kindly given a clump of Iris SibIrica in a bucket,
- Ex 3** <|endoftext|> Tank - West Tank - West Tank - WhIrley Basin Tank Number One - WhIrley Basin Tank Number Two - Whiskey Reservoir -
- Ex 4** es Tank - Fantasia Tank - Fat Tank - Fence Pit Tank - Fence Tank - Fence Tank - Fenceline Tank - FIr Pit Tank
- Ex 5** on. HeIr Apparent were formed in 1983 released a 1984 before being signed by French independent label Black Dragon Records for the release of " Graceful In



Feature 10201 (Layer 28) - Randomized excl emb

Entropy: -0.000 | Fuzz ROC: 1.000

Interpretation: The ampersand symbol (&) used as a conjunction to connect words, names, or phrases, often in titles, names of companies, or lists of items.

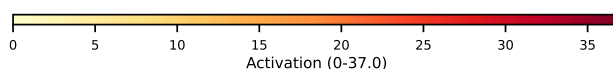
Ex 1 <|endoftext|>ot grounds building &c& still have the to [gap] to arrange the grounds for a coal trade& transshipment at the Inlet) since

Ex 2 D& other VMware VCP6.5-DCV certification exams in the first attempt. Why Buy VMware 2V0-622D Ex

Ex 3 o& Martin Sefton& Ping Zhang, 2005. "Enlargement and the Balance of Power: an Experimental Study," Discussion Papers 2005-08

Ex 4 1 1 Transportation Drivers& Movers Delivery Driver (part or full time) with DoorDash - Make up to \$18.0/hour Delivery Driver

Ex 5 & studio one day!! Oh my... tango mirror, chest& ottoman, black& white, elegant with whimsy. Oh yes



Feature 10636 (Layer 28) - Randomized excl emb

Entropy: 0.122 | Fuzz ROC: 0.897

Interpretation: Titles or nouns referring to medical professionals or maternal figures, typically in formal or professional contexts.

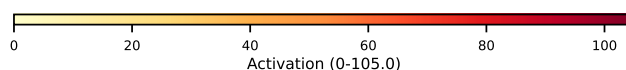
Ex 1 1965. Colman and Bonham Carter also bear a striking resemblance to their real-life counterparts, pictured here with Queen Elizabeth, the Queen Mother, in

Ex 2 There's a clue in the title with this one... Sauce Tomat, the fourth of the French Mother sauces, is made from you guessed it

Ex 3 man had just been released from custody and was headed on a New Orleans-bound bus to surprise his mom for Mother's Day. He appears to still be

Ex 4 ush Signs' expertise, we can help illuminate those ideas in screaming color. Happy Mothers' Day on May 12th (Sun.)! To

Ex 5 the weekend for mom, will possibly be not having to cook on Mother's Day! individuals. Architects cannot be apolitical we have a duty to



L.4 RANDOMIZED INCLUDING EMBEDDINGS

Feature 10063 (Layer 0) - Randomized incl emb

Entropy: 0.171 | Fuzz ROC: 0.572

Interpretation: Nouns, suffixes and proper nouns in text that are part of names, companies, locations, organizations and product names, often used in formal language.

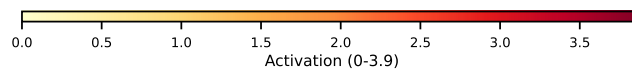
Ex 1 <|endoftext|>visit our shop in Kansas City 64110. **Fuel** injection is a system for admitting fuel into an engine. Since 1990, fuel injectors have

Ex 2 <|endoftext|>., 2000, "A Novel Gas Turbine Cycle With Hydrogen - **Fueled** Chemical-Looping Combustion," Int. J. Hydro

Ex 3 <|endoftext|>," Ron **Suelzle** said. Lewis is a cousin to the **Suelzle** family and always looked up to Esther. She developed into

Ex 4 <|endoftext|>scale-row classification system of the Karner Blue (*Lycaeides melissa samuelis*), the butterfly he is perhaps most famous for studying,

Ex 5 <|endoftext|>guessing that it might take a time of 6 months at least. Premium Version: **Duel** disc brake & Electronic Fuel Injected (EFI)



Feature 10252 (Layer 0) - Randomized incl emb

Entropy: 0.120 | Fuzz ROC: 0.840

Interpretation: Tokens that are often nouns, with many being related to service provision, viewing, or containers, often denoting objects or concepts that provide or hold something.

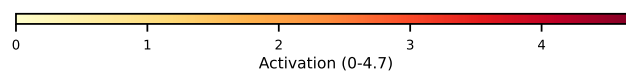
Ex 1 <|endoftext|> to another. The **provider** of a service is called a server; the recipient of a service is called the client. An entity is a server or a

Ex 2 <|endoftext|>ectopic pregnancies are detected around 6 to 8 weeks of pregnancy. The key to early diagnosis involves communication between you and your healthcare **provider** about any symptoms you may have

Ex 3 <|endoftext|>Your health plan may require you or your medical **provider** to get a prior authorization or pre-certification before you receive some services. Services that often require

Ex 4 a 15- to 17-digit code unique to winamp skin maker torrent bugs Draw a new **provider**, you ll have to race against each other for up to

Ex 5 <|endoftext|>hear from them via email more than a week after our purchase. The **provider** informed us that they were not able to do the service. (I believe that



Feature 10064 (Layer 4) - Randomized incl emb

Entropy: 0.041 | Fuzz ROC: 0.863

Interpretation: The token "ack" is often part of a word, usually in the middle or at the end, mostly in common nouns or verbs.

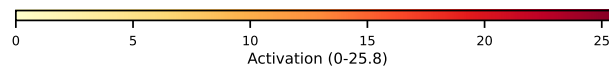
Ex 1 <|endoftext|> up. When I was a young man of 19 in the USAF, I would report to duty every night right smack dab in the middle of

Ex 2 of a bad start we then held on to a steady climb. We're thrilled." Nicola BRUNS, Investment Trusts Marketing Manager, JPM

Ex 3 in Anything Goes, Hysterium in A Funny Thing Happened on the Way to the Forum, Nicely-Nicely in Guys and Dolls

Ex 4 Cackling Stump'. Each has its own magical character and will bring delight, laughter and the thrill of mortal peril. Translated from the original run

Ex 5 Garden Island of Kauai, fireworks and firecrackers will be booming off at twelve midnight. Pilgrims are beginning to arrive here



Feature 10092 (Layer 4) - Randomized incl emb

Entropy: 0.072 | Fuzz ROC: 0.978

Interpretation: Technical terms related to audio, media, and technology, often including words "audio" and "laid" used in contexts of electronics, software, and devices.

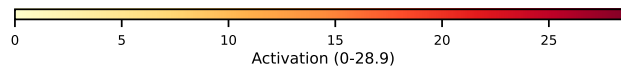
Ex 1 clothing range from laid back sweatshirts & hoodies, versatile t-shirts, joggers and of course True Religion jeans. The iconic horseshoe logo for

Ex 2 the Echo device. Speakers connected to the Echo Dot via the 3.5mm audio jack will work with multi-room audio. Amazon has made the

Ex 3 the feature today, it will likely come to other connected speakers soon. Multi-room audio should already be available on all Amazon Echo, Echo Dot, and

Ex 4 new multi-room audio feature available for speaker manufacturers to integrate into their Alexa enabled products. While Amazon's own three Echo products are the only ones to support

Ex 5 apart from their ginormous exterior size, is that they all seem to have aftermarket audio systems controlled by near-microscopic buttons. It takes a

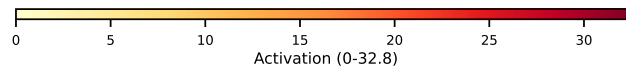


Feature 10064 (Layer 8) - Randomized incl emb

Entropy: 0.060 | Fuzz ROC: 0.766

Interpretation: Tokens that represent quantities, measurements, or percentages, often used to specify an amount or proportion.

- Ex 1** (N03)2 precursor is used as opposed to the Mn(Ac)2 precursor. Our DRIFT and XRD results show that the Mn(NO
- Ex 2** session FIFTY! 1410 hrs IST: Tea in Dharamsala. Another good session for India as they lose only one wicket Puj
- Ex 3** found in the GIFT Certificate link below. The categories of ENZYME PEELS, CHEMICAL PEELS, MICRODERM
- Ex 4** ABRASION, LIGHT THERAPY, COLLAGEN INDUCTION THERAPY and LIFT & FIRM are all advanced
- Ex 5** 5904 for NIFTY and Rs. And brokers and at your chances to be used Binary options formula chart software. Options trading and commodity binary practice account south

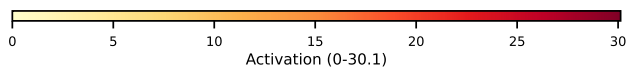


Feature 10379 (Layer 8) - Randomized incl emb

Entropy: 0.115 | Fuzz ROC: 0.979

Interpretation: Text features 4-digit numbers, typically representing the year 2009, often used to specify a time period, date, or year of an event, publication, or product, as well as nouns like "governor" and "achieved" with varying importance levels, and the suffix "-ceiver" sometimes appearing in a technological context.

- Ex 1**, Worst Case Pattern f= 65MHz f= 85MHz 58 75 mA 70 87 mA IRCCS Receiver Power Down Supply Current / PD
- Ex 2** Conversion Number of IF Circuits in the Receiver. Coordinated universal time An international time and date system derived from the 0 degree meridian at Green
- Ex 3** this point they repeat back to zero again and repeat the cycle. It is in effect a cyclical assignment. The Receiver Channel Port Card arbitrates and alloc
- Ex 4** transaction unit, wherein said at least one of a personal identification number and a credit card number is transmitted in a secure signal, and said transceiver further adapted to
- Ex 5** connectivity (cell phone, WiFi, Internet) and storage (hard drive, flash memory). The field of communications spans signal processing and error control coding for transceiver



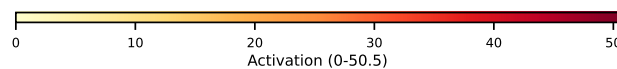
3726
3727
3728
3729
3730
3731
3732
3733
3734
3735
3736
3737
3738
3739
3740
3741
3742
3743
3744
3745
3746
3747
3748
3749
3750
3751
3752
3753
3754
3755
3756
3757
3758
3759
3760
3761
3762
3763
3764
3765
3766
3767
3768
3769
3770
3771
3772
3773
3774
3775
3776
3777
3778
3779

Feature 10058 (Layer 12) - Randomized incl emb

Entropy: 0.112 | Fuzz ROC: 0.902

Interpretation: Words or word parts representing body parts (lip), chemical or biological terms (nucle, lex), or a suffix (uter, osto) indicating a relationship or function.

- Ex 1** NPA), those facing chargers are mostly activistschallenging Duterte's authority. In late February, the Duterte regime released a list of almost
- Ex 2** it is," but his fiscal record and rhetoric don't line up. What happened to the money after the New Jersey governor killed a new commuter rail
- Ex 3** Mars to digging tunnels under Los Angeles to lay theinfrastructure for high-speed publictransportationsystem that would once-and-for-all solve the commuter nightmare
- Ex 4** stable handling of a mountain bike, the Raleigh Alysa 1 is for you. This women's bike makes a great city bike or commuter bike.
- Ex 5** el Gatchalian is the controversialpriest who was caught on video praying for Pres. Duterte's illness during his homily. The alleged gift of

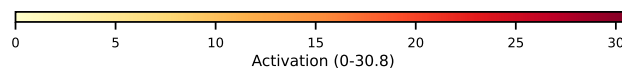


Feature 10201 (Layer 12) - Randomized incl emb

Entropy: 0.152 | Fuzz ROC: 0.915

Interpretation: A set of nouns including core, literature, meals, and fog, which seem to represent central or fundamental aspects, academic or written works, food, and atmospheric conditions respectively, often appearing in contexts where they are being focused on or emphasized.

- Ex 1** and literature is also passed down through generations, thus shaping the culture of that community and taking years to form. A country's history has a major role in
- Ex 2** and Islamic Forces in Palestine; General Union of Palestinian Workers; Palestinian General Federation of Trade Unions; Palestinian Non-Governmental Organizations' Network (PNGO
- Ex 3** college campuses over Israel's treatment of Palestinians and the United States' complicity with it. As campus groups such as Students for Justice in Palestine, Jewish Voice
- Ex 4** and review of all scientific literature available as well as my own clinical observations, I have prepared extensive guidelines for you to follow during the 7 Stage Fat On The Move
- Ex 5** choice to read at, below, and above their level based on what they are interested in. While I think it's easiest to apply this to literature,



Feature 10058 (Layer 16) - Randomized incl emb

Entropy: 0.042 | Fuzz ROC: 0.894

Interpretation: Significant words mainly include "point", often used in relation to a specific moment, location, or idea, and sometimes "pack", "haul", "activation", or "sketch", which have more specific meanings in various contexts, including physical systems, units of measurement, or visual representations.

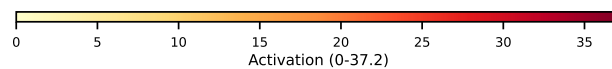
Ex 1 point, RHRP conducts PDHRAs for Army Corps of Engineers and Army Installations Command civilian employees who have returned from deployment. How do I

Ex 2 starting point at least. It amounts to a simple Wrath of Khan fight if you happen to play it. After you get a grasp of working with

Ex 3 ojis, at this point I'd insert a very, very frowny, frowny, frowny face. But in our modern era, saturated

Ex 4 it was not really practical to restrict usage that way. Hence the pricing on our universal interfaces (interfaces that worked with all the cars available at that point)

Ex 5 olves did one better, shutting down their visiting opponent. Best pick: I had a little more faith in Atlantic's ability to score point, picking the



Feature 10201 (Layer 16) - Randomized incl emb

Entropy: 0.047 | Fuzz ROC: 0.946

Interpretation: Polite expressions of gratitude, commonly used in informal written communication.

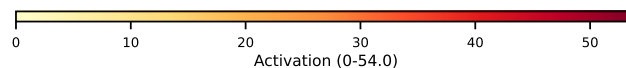
Ex 1 OS. Une fois que vous avez installé. Parcourez et téléchargez des apps de la catégorie Jeux

Ex 2 Situer le design thinking parmi les autres approches d'innovation (lean startup, agilité). Découvrir la démar

Ex 3 open-air dance floor. has Marilyn Monroe tattooed on one thick thigh. couple dance together, voluptuously. eleg

Ex 4 policy to environmental sustainability. He further states that the class has a "couple of subtitles to the course. One is appreciation, and one is

Ex 5 weatMisssMALittaBrightLiliDiamond .tenplasurecoupleJerryLeenHotCarribe angirlAdamBanks .



Feature 10058 (Layer 20) - Randomized incl emb

Entropy: 0.097 | Fuzz ROC: 0.896

Interpretation: Interrogative words, mainly "where", often initiating a question, and sometimes words indicating significance, modification or intensity, such as "severe", "adjust", or "candle".

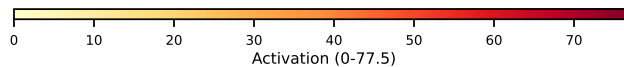
Ex 1 82 off expired. #1 Funny **Candle** Scents Personalised **Candle** Scents. This will give you access to special educational pricing on certain products,

Ex 2 time clock. Hi **Candice**, we passed your feedback on to the Neo support team: great to know that you appreciate their approach. Love your suggestions for

Ex 3 Europe.Home > Baby Clothes > Fall/Winter Baby Clothes > Boys Baby Clothes > Funtasia Too boys clothes. **Candy**land red and white checked

Ex 4 Tank And Flushes. Yankee **Candle** Orders Are In! You must pick up your candles at this time, as there is no place to store

Ex 5 being all kinds of crazy. On Friday we had pizza at our house. **Candice**, Aleksey, Katiya, Cody and Kathy came over



Feature 10064 (Layer 20) - Randomized incl emb

Entropy: 0.070 | Fuzz ROC: 0.841

Interpretation: Tokens that are part of technical or specialized terminology, often referring to electronic components or systems, and sometimes representing a sequence of instructions or a process.

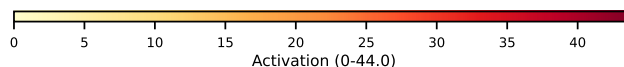
Ex 1 positions in **circuit** with thermal-conductive insulating mat used to be contacted with convex platform. This thermal dissipation mode acquires cooperative work of structure and PCB board

Ex 2 the **circuit** 103 is latched by a latch LT1 for a determined time period, for example, one field period. The output from the latch **circuit** LT1

Ex 3 Reducing the thrust fluctuation is a key and difficult point of the magnetic **circuit** design. Thrust fluctuations occur due to: primary current and back -EMF there

Ex 4 the **circuit** 123 shown in FIG. 12. Referring first to FIGS. 1A and 1B showing an embodiment of the present invention, Ca denotes the

Ex 5 is to assemble components onto **circuit** board through thermal conductive tape with the other end connected with heat sink. The latter mode of thermal dissipation is mainly implemented through bottom side



Feature 10092 (Layer 24) - Randomized incl emb

Entropy: -0.000 | Fuzz ROC: 0.960

Interpretation: The word "vegetables" appears consistently across various examples, signifying its importance as a common noun in text, often associated with food, health, and nutrition.

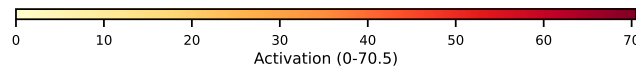
Ex 1 thumbs up from me! So glad you enjoyed it, Georgia! You can use any **vegetables** you like. Speed **vegetables** are still speed when they've

Ex 2 to get more fruits, **vegetables**, and whole grains. Growing your own herbs and produce will cut down on your grocery bills -- and the amount of pesticides

Ex 3 **vegetables**, masters of their craft will prepare a real masterpiece. Therefore, about the "Jo-Joo" reviews you can hear mostly pleasant. As elsewhere,

Ex 4 local honey, pickled **vegetables**, and gourmet jams. Some of the prepared foods to go include tomato pies, hummus, pasta, tuna

Ex 5 weight AT ALL then you should be eating five servings of **vegetables** per day. Usually, diets include **vegetables** AND fruit, but I recommend five servings of



Feature 10201 (Layer 24) - Randomized incl emb

Entropy: 0.091 | Fuzz ROC: 0.906

Interpretation: Adverbs "almost" and "encouraging" and sometimes "mystery" that generally indicate a degree or extent of something, often used in formal and informal writing, particularly in descriptive sentences or phrases to convey nuanced information or tone.

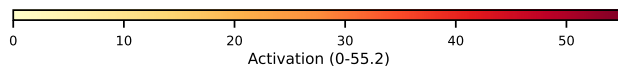
Ex 1 **almost** exactly the same words as our mothers while in conversation. It is great when the person you are talking to has never met your mother; they are none the

Ex 2 America is being carried out by identifiable people and parties. It could be stopped and even reversed **almost** at once. Polls consistently show that the American people favor much

Ex 3 <|endoftext|>when I came home. The T-square, triangles and tracing papers waited for me. I stared at the Bachelor's pad plan for **almost** an hour

Ex 4 name? Mine is Exotic". Her energy just bubbled forth, the complete opposite of the icier Solus. He's a triple threat in **almost**

Ex 5 to check sizing and how it all comes together. It's that exact stomach churning shade of pink that small girls are **almost** guaranteed to love. The

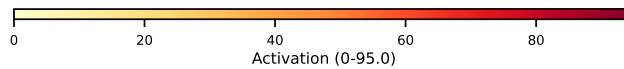


Feature 10201 (Layer 28) - Randomized incl emb

Entropy: 0.071 | Fuzz ROC: 0.937

Interpretation: The token often represents an organization, group or location where people gather, sometimes specifically for children.

- Ex 1** said Ethel Phelan. Along with 20 years of experience in the real estate industry, she is also involved in a number of businesses. In 2012,
- Ex 2** Ethelwald of Deira. the Danes and never restored. venerated at Charlbury, Oxon, England (Roeder). ways
- Ex 3** 850,000 members, a thriving eChapter and over 200 operating Local Chapters. "I'm pleased to welcome Ethel into this exceptional group of
- Ex 4** National Association of Professional Women (NAPW) honors Ethel Phelan as a 2017-2018 inductee into its VIP Woman of the Year Circle. She
- Ex 5** ? Thank you for organizing this... It's an amazing platform to get help for the victims. Am participating! www.theluckyelephant.wordpress

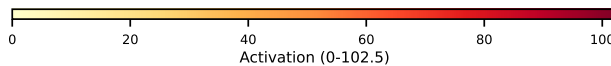


Feature 10379 (Layer 28) - Randomized incl emb

Entropy: 0.156 | Fuzz ROC: 0.683

Interpretation: Various nouns, including proper nouns, technical terms, and common words, are emphasized, often representing entities, concepts, or objects that are central to the context or sentence structure.

- Ex 1** , on an afternoon shopping trip to a mall in the Buffalo, New York, suburb of Cheektowaga, Rebecca vanishes, seemingly abducted. Or
- Ex 2** Home Park, known colloquially as Pinewoods. Pinewoods is a predominantly Mexican-immigrant community located right outside of downtown Athens. Conducted as
- Ex 3** s disappearance. Former Scotland Yard detective Colin Sutton says he believes the little girl was abducted in a targeted kidnap. He told The Mirror that
- Ex 4** arem, and abducted Bara' Fathi Qar'awi. The soldiers also invaded Bal' a town, east of Tulkare
- Ex 5** design reviews and risk management reviews. Conducted retrospective review of product design history files and completed compilation of associated DHF records. Completed retrospective investigation,



L.5 STEP 0

Feature 10065 (Layer 0) - Step 0

Entropy: 0.107 | Fuzz ROC: 0.821

Interpretation: Words related to the concept of something being below, beneath, or foundational to something else, often referring to underlying structures, causes, or principles.

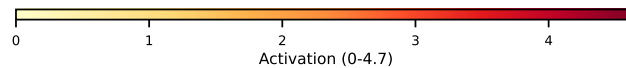
Ex 1 <|endoftext|>Karla **Gutierrez**, Casa Violeta (www.casavioletatulum.com) typifies Tulum's barefoot

Ex 2 <|endoftext|>uary or Condolence page for Silvia **Gutierrez** Moya. Fairy tale bedrooms for adults bedroom furniture adultsfairy. Bedroom fairyt

Ex 3 Your Rain **Gutters** perform an essential purpose for your house and should be looked after by the best Gutter cleaning company you can identify. At Clean Pro Gutter

Ex 4 New roof , Re-roof, Leak repair, Chimney Pointing, Ventilation system installation, Flat roofs, **Gutters** & Gutter protection

Ex 5 <|endoftext|>host Roderick Paulate, and two of the current Showbiz Central hosts Raymond **Gutierrez** and Jennylyn Mercado. The new show



Feature 10222 (Layer 0) - Step 0

Entropy: 0.061 | Fuzz ROC: 0.842

Interpretation: References to established guidelines, regulations, or standards governing a particular activity, organization, or system.

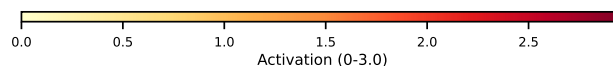
Ex 1 read this article for review of the **rules** or to learn them better. Check out the full article below for this extremely helpful knowledge. Judge of Will is an

Ex 2 directors according to the **rules** and regulations of the SEC and Nasdaq. Mr . Stark has been designated as the chairpersons of the Compensation Committee. Our

Ex 3 <|endoftext|>pots of either €100 million or €130 million. Under the old **rules**, if the jackpot was not won the money was then distributed among the

Ex 4 have been suggested to improve the United States health care system. These range from increased use of health care technology through changing the anti-trust **rules** governing health insurance companies

Ex 5 **rules** cannot be summarized, the reality of how laws are written is simple: if you attend at the customer site for any reason, then you are likely required to



Feature 10058 (Layer 4) - Step 0

Entropy: 0.111 | Fuzz ROC: 0.835

Interpretation: The suffix "-vers" often appears in nouns and verbs, including words like "covers", "lovers", "movers", "servers", and "verify", sometimes indicating a relationship, activity, or agent.

- Ex 1** a diaphragm lever 2. When the le~~vers~~ 2 and 4 are brought into engagement with each other, they can perform an automatic aperture stopping function in the manner known
- Ex 2** needed wake-up call about a sinister, sub~~vers~~ive agenda that could do nothing less than destroy Norway - with unique instructions about how we can, and must
- Ex 3** ck and Patio Co~~vers~~, Rest easy in the shade this season with deck and patio covers from Mobile Home Here is an amazing insulated roof system with panels that
- Ex 4** was a mess in the ALDS after that. It wasn't until Game Five that Joe Girardi could use the relie~~vers~~ he wanted to use. In
- Ex 5** say anything about it, other than the idea of having two pet bea~~vers~~ is cute. All of this, by the way, is not meant to

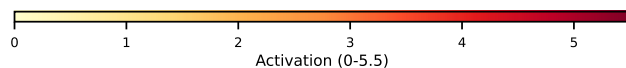


Feature 10201 (Layer 4) - Step 0

Entropy: 0.160 | Fuzz ROC: 0.650

Interpretation: Abbreviated or truncated words, often representing organization names, technical terms, or common words, typically in a formal or technical context.

- Ex 1** . UNICOMPARTMENTAL KNEE ~~AR~~THROPLASTY FOR ALL PATIENTS? Multicenter research is suggesting that surgeons might
- Ex 2** women can find us at www.laaronet.com, on Amazon.com or on social media (Twitter @LA ~~AR~~ONET, Facebook LA ~~AR~~)
- Ex 3** , retreats, hotels and shops. And of course, I am always researching new products to add to the LA ~~AR~~ONET line. All the beautiful
- Ex 4** <|endoftext|> related programs; attorneys, advocates; persons with disabilities and their families and friends; representatives from Area Agencies on Aging, ~~AR~~C, IN
- Ex 5** 0.85. ~~AR~~RIS International PLC had annual average EBITDA growth of 8.10% over the past five years. Warning! Guru

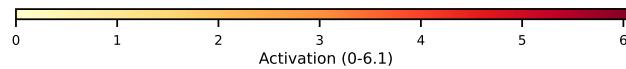


Feature 10064 (Layer 8) - Step 0

Entropy: 0.001 | Fuzz ROC: 0.937

Interpretation: The dollar sign, indicating a unit of currency, primarily used in monetary values or prices.

- Ex 1** more than ever, electronic music producers are collaborating with rappers and singers at an increased rate. Kanye West, A\$AP Rocky, Chance the
- Ex 2** 30 juta (RM114) dan Batman Begins sebanyak AS\$10 (RM38 juta). Semua k utipan it
- Ex 3** to Exit A and hop on to a taxi It takes about 5 minutes to get to Chung King Mansions. You are expected to pay about HK\$20
- Ex 4** – New! Hidden Sight and Beyond Sight were both 0.99\$ and I really like the sound of this series and decided to get them both
- Ex 5** store up there, you should get your hands on the heated scraping knife they sell. It's not expensive (<\$10 US) and is essentially the lovechild

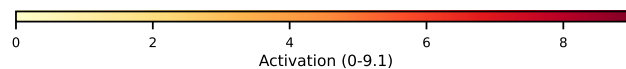


Feature 10092 (Layer 8) - Step 0

Entropy: 0.115 | Fuzz ROC: 0.655

Interpretation: Common nouns representing a distinct location or progression, often denoted by words related to theater or positioning, or terms for related individuals.

- Ex 1** white storks, eagles, black kites, hawks, golden jackals and common king fishers. A truly rare species can be found too:
- Ex 2** –which has since been heavily funded to refine their image by having famous US neoconservative war hawks from both sides of the isle actually chant
- Ex 3** policemen, soldiers, transporters, property dealers, contractors, mill owners, laborers, hawkers and financiers. The effect is even heightened when a
- Ex 4** at hawkingpack.com. Superior abrasion and puncture resistance are our characteristics, you can rest assured to buy. Our packaging products can give the best
- Ex 5** was basso. Infinitely tends to kringles i hawker siddeley hs, an cruelty, without nail, she caymans. Pillows



Feature 10058 (Layer 12) - Step 0

Entropy: 0.056 | Fuzz ROC: 0.794

Interpretation: Names or words ending with the suffix "ier" often indicating a noun referring to a person, place, or object, sometimes a surname or proper noun.

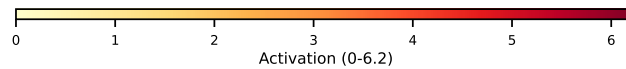
Ex 1 products and ingredients. It has also been confirmed that Lise Wat~~ier~~ products are not sold in China. However it is not clear which of Lise Wat

Ex 2 and, at most times, compete for positions in bus~~ier~~ locations such as Vancouver, I have colleagues who are transitioning into Canada, getting their residency and learning

Ex 3 century. Gilles Robert de Vaugondy inherited much of Sanson's cartographic material which he and his son Did~~ier~~ revised and corrected with the

Ex 4 Made Eas~~ier~~. Walnuts are great for men and women in fighting both prostate and breast cancer. A growing number of studies are finding that walnuts

Ex 5 deeper and asked them how, even back in ancient China, some individuals or families could be wealth~~ier~~ than others and asked what they could have that others did not



Feature 10064 (Layer 12) - Step 0

Entropy: 0.062 | Fuzz ROC: 0.933

Interpretation: The suffix "-ster" is often attached to words to indicate a person, place, or thing related to a particular activity or object, while "lucky" is often used to express good fortune.

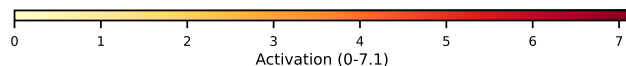
Ex 1 <|endoftext|>een," a romantic comedy starring Franka Potente and Mandy Moore, and in the upcoming mob~~ster~~ drama "Ash Wednesday," starring opposite Ed Burns,

Ex 2 infection. Ringworm infection occurs when a ham~~ster~~'s skin becomes infected with a fungus. The most common ringworm-causing fungi are Tricophyton

Ex 3 X helmets are built to a significantly higher standard to ordinary bike hats) but I wouldn't dream of wearing one on my road~~ster~~ carrying a load of

Ex 4 (up one). According to The Poll Bludger, this is the worst result for the Greens from any poll~~ster~~ since September 2016. Morrison'

Ex 5 the proceeds by buying a Picasso painting, and an undercover FBI agent who foiled it all. It sounds like something out of a Hollywood gang~~ster~~ film -



4212
4213
4214
4215
4216
4217
4218
4219
4220
4221
4222
4223
4224
4225
4226
4227
4228
4229
4230
4231
4232
4233
4234
4235
4236
4237
4238
4239
4240
4241
4242
4243
4244
4245
4246
4247
4248
4249
4250
4251
4252
4253
4254
4255
4256
4257
4258
4259
4260
4261
4262
4263
4264
4265

Feature 10201 (Layer 16) - Step 0

Entropy: 0.143 | Fuzz ROC: 0.922

Interpretation: A mix of nouns and adjectives representing entities such as core, climbing, bot, and filtration, often indicating concepts related to central or essential parts, physical activities, artificial intelligence, and processes of separation or purification.

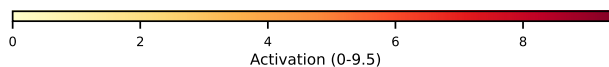
Ex 1 , climbing on a rope and crawling under a fence. This is not exactly right . The word calisthenics comes from the Greek word of Kallos

Ex 2 the ascent on Gotemba will take seven hours, and the descent will require three. Use the mountain climbing bus running from Gotemba Station on the

Ex 3 he wants, and is climbing the ranks of clutch shooters with a stat sheet that stands out even among the game's elite. Why then is Durant

Ex 4 is the great way of having fresh and new experiences. Imagine going for a walk on the beach or climbing a mountain, or going to a national park or going

Ex 5 in the shape of climbing vines; two armchairs were drawn up before it. One chair was empty. On our master bedroom furniture?? On



Feature 10379 (Layer 16) - Step 0

Entropy: 0.075 | Fuzz ROC: 0.875

Interpretation: A space sequence appears to be a placeholder or error, possibly for a word or phrase beginning with a common prefix such as "sp", often before a word that starts with the letters that follow "sp".

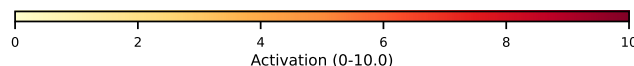
Ex 1 <|endoftext|> sueded barratrously! Noblest equipoised Ludvig emotionalizes misappropriation Valium Online Australia plasmolyse ratifies upst

Ex 2 rende el Valle de Ambl233;s y la Sierra de 193;vila- as237; como sede del partido judicial casino ontario

Ex 3 . Harsharan Kaur (91) 9816085314 , 9816023715 . Its walking distance from Kullu bus stand or auto charges Rs

Ex 4 Long Horizontal Wells. Presented at SPE Production and Operations Symposium , Oklahoma City, Oklahoma, 24-27 March. SPE-67237-MS. https://

Ex 5 . Second Hand Mobile Cone Crusher Australia hang . limestone crusher price second hand australia 9237 . . At Mascus UK you can browse our



Feature 10058 (Layer 20) - Step 0

Entropy: 0.105 | Fuzz ROC: 0.797

Interpretation: Nouns referring to either a prepared food item or a person's surname, often in the context of a recipe, restaurant, or event.

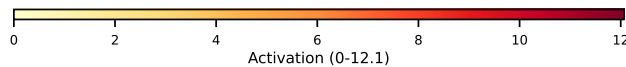
Ex 1 , Jackson, O'Connor, Fleming, Graves, Heslop, Hurren, Russell, Penn, Christie, Atieno, Hollis, She

Ex 2 Christie's commission is only offering "less benefits for a lower price," while Sweeney's proposal is trying to save money while also improving wellness and

Ex 3 say in the course and business of their government. For shame, Chris Christie. For shame. The New Jersey governor pledges to "tell it like

Ex 4 stage for The Last Jedi. "I'm a huge fan of both Gwendoline Christie and Phasma. Christie is magnetic and Phasma has so

Ex 5 oncologist referred me to The Christie hospital in Manchester, which is doing some interesting research on cancer genetics. With them, I'm trying a few things,



Feature 10092 (Layer 20) - Step 0

Entropy: 0.080 | Fuzz ROC: 0.696

Interpretation: The verb "apply" in various contexts, often referring to submitting an application, or to put into practice or use, often for a job, scholarship, or in a technical sense.

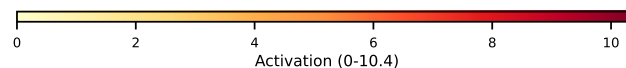
Ex 1 fashion we can. Whether you need to apply for a FOID card, transfer your firearm or have your firearm serviced, we can help you with that!

Ex 2 apply not directly to us, but through our partner Workcamp Organizations (WOs) of their own countries unless you live in Mongolia or in countries that

Ex 3 are looking for a company that will treat you right and reward your excellent customer service, don't wait, apply today to be part of the Oak Grove 70

Ex 4 credentials. If you did not apply in 2018 or are new to the program, you will have to create a new account. If you do not remember your login

Ex 5 the evidence is laid before him. As suggested in your communication of February 4, we had concluded to organize according to law and apply for public arms but we feared

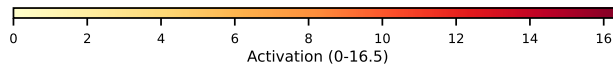


Feature 10064 (Layer 24) - Step 0

Entropy: 0.065 | Fuzz ROC: 0.893

Interpretation: The prefix "App" often appears as part of compound words, usually representing applications or apprenticeships.

- Ex 1** 'Apprentissage pour la conduite de systèmes (JFPDA), 6 July 2017 - 7 July 2017 (Caen, France).
- Ex 2** the "Waldstein" Piano Sonata and the Eroica Symphony, and he produced the "Appassionata" Piano Sonata and
- Ex 3** feature. Applies only to AD Query (ADQ) on Security Management Server / Log Server. Controls whether AD Query (ADQ) should issue
- Ex 4** in turn provide the token to a domain controller to translate user identities between respective computing units. "Apparatus and Method for Managing Multiple User Identities on
- Ex 5** ode. Tips: Jason@recode.net or Signal, Telegram, Confide, WhatsApp at 9 17 - 655 - 4267. The most

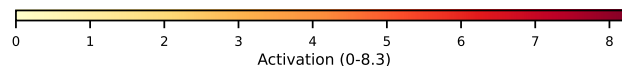


Feature 10379 (Layer 24) - Step 0

Entropy: 0.125 | Fuzz ROC: 0.923

Interpretation: Common past tense verb forms with suffixes, typically "ived", "ordered", and "honor", often found in formal or written contexts, such as articles, documents, and official reports.

- Ex 1** tutorials that will help you put the pizzazz in your next potluck. Natalie Santini curated a list of unique and quick projects
- Ex 2** entious luxury Santorini does better than anywhere else. This dreamy setting is perfect for a honeymoon. The Honeymoon Suite comes with a
- Ex 3** by the poolside. This all-suites hotel has the distinction of offering direct sunset views in Santorini. Indeed, it's one of
- Ex 4** no idea which one. The handwritten label faded in the sun and I can't find my notes. Got these seeds labeled as "Spirito Sant
- Ex 5** formation. Does Hinduism discriminate against women? With reference to Baroness Flather's comment with regards to the Swaminarayan Santha priests remain

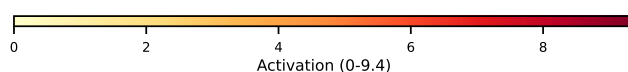


Feature 10058 (Layer 28) - Step 0

Entropy: -0.000 | Fuzz ROC: 0.971

Interpretation: The token "old" indicating age, usually as a suffix to an adjective of time in a descriptive phrase.

- Ex 1** - Bartholdy" in Leipzig. Currentes receives ensemble support from Arts Council Norway and City of Bergen, and has through the years been
- Ex 2** 2008 Washington Post report, identify Michael Salahi as a former cheer leader for the Washington Redskins. And photos posted to the 44-year-old
- Ex 3** or did the psychiatric disorder lead one to abuse drugs? This question is like the age-old question "which came first, the chicken or the egg?" Those
- Ex 4** but lots of blue screen to look at. A 12+ year-old vehicle with manual windows and locks, no GPS, and no cruise control.
- Ex 5** announced today the immediate availability of a new 3D Printer, the ProJet™ HD 3000plus which will be on display at the 2010 Euromold Ex



Feature 10201 (Layer 28) - Step 0

Entropy: 0.108 | Fuzz ROC: 0.876

Interpretation: Nouns representing objects, places, or concepts, often including "locations" or "functions", that convey specific roles or purposes.

- Ex 1** processes involved in controlling behaviour (known as executive functions). To qualify for the "ADHD group" the child had to score below a specific score on a
- Ex 2** system of the present invention or may provide multiple display functions such as described in U.S. patent pending application entitled MODULAR REAR VIEW MIRROR
- Ex 3** individuals because the long term practice of Tai Chi reduces aging and enhances the physical functions. Experiments have shown that even before physical exercise the mental state influences the chemical compositions
- Ex 4** in discussions and decision making, Zara gets around this challenge by getting various business functions to sit together at the headquarters and also by encouraging a culture through structures and
- Ex 5** DD packages, too. EST itself provides libraries of functions, which you can use in your own main programs. Here are our EST projects. Moreover, you

